Start-up Financing, Entry and Innovation

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Abstract

I develop and estimate an equilibrium model of the VC market to quantify distortions and explain cross-country differences in VC activity. The model maps directly to observable start-up funding histories and outcomes, facilitating estimation. For US start-ups first funded in 2005–2015, my estimates suggest that 40% shut down despite having positive continuation value; with continued funding, half would achieve a successful exit. In the UK, the model attributes the gap in activity to the US to financing conditions and acquisition opportunities, and counterfactuals suggest there may be gains to retargeting existing government support towards late-stage start-ups.

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1 Introduction

Entrepreneurship is a central driver of economic growth, but realising its benefits relies on the efficient allocation of capital. While most firms secure loans against pledgeable assets or predictable cash flows, high-growth start-ups with limited revenues face difficulty financing operations with debt (Hall and Lerner, 2010). These firms instead seek funding from specialist intermediaries – venture capitalists – who, unlike banks, supply equity and finance operating losses in pursuit of substantial payoffs. Venture capital (VC) has funded many of the largest and most innovative US firms; yet, concerns remain that some firms may be underfunded and that limited access to VC for European start-ups constrains innovation.[1]

Understanding the drivers of VC allocation and outcomes is challenging. Lower VC activity in Europe may reflect limited opportunities or frictions in allocating capital to entrepreneurs; and the drivers matter for policy design. However, because *ex ante* start-up quality is poorly observed, disentangling supply and demand for capital is difficult. Furthermore, even in the mature US market, financial frictions mean some start-up projects may be terminated prematurely.[2] Empirically, distinguishing poor fundamentals from premature termination is complicated because the counterfactual under continued financing is unobserved.

Motivated by these challenges, I develop and estimate a tractable equilibrium model of the VC market to study (i) how financing frictions shape start-up outcomes in the US, and (ii) why other developed economies have less VC activity. The model yields a structural interpretation of microdata on start-up funding histories, providing a way to tackle the identification challenge and a basis for quantitative counterfactual analyses that unpack start-up outcomes and simulate policy interventions. The model features frictional matching and staged financing: VCs inject capital in stages to limit losses

[1] Since 1978, approximately 50% of US IPOs were VC-backed, generating over 90% of R&D spending among these firms (Gornall and Strebulaev, 2021); 1978/1979 marks the start of the modern VC model, when the capital gains and ERISA "prudent man" reforms were enacted (Gompers, 1994).

[2] Premature termination is common in models of VC contracting (e.g. Bergemann and Hege 1998; Cornelli and Yosha 2003; Mayer 2022). Relatedly, Admati and Pfleiderer (1994) features underinvestment à la Myers (1977). Beyond VC, dynamic long-term contracting can yield inefficient liquidation (e.g. Clementi and Hopenhayn 2006).

when entrepreneurs hide failure. However, staging means the start-up must reach profitability or raise more capital before funding runs out. In turn, difficulties raising follow-on funding create financing risk, akin to rollover risk in debt markets, and mean some viable start-ups may fail to secure more funding and so shut down prematurely.[3]

For the US, I estimate that approximately 40% of VC-backed start-ups shut down prematurely, implying that substantial value is lost; with continued funding, half of these firms would achieve an acquisition or IPO. Across countries, I find that differences in VC activity, specifically US versus UK, reflect tighter financing conditions and limited acquisition opportunities, rather than the quality of funded firms. Simulating a budget-neutral policy that redirects 5% of total UK VC funding ($\approx$ £$0.5-1$Bn/year) from early to late-stage rounds raises the number of start-ups that scale independently by 15%.

The model features frictional matching between entrepreneurs and VCs and a principal-agent problem: entrepreneurs may hide failure *ex post* to reap private benefits, creating an external-finance premium. In response, VCs stage capital injections over time so only a small tranche is at risk between reviews. However, because VCs typically rely on future investors for additional rounds, initial commitments are forward looking and rise when follow-on funding is uncertain. Therefore, funding concerns at the macro level induce front-loading at the micro level, as was evident in 2008 when Sequoia advised portfolio companies to "raise as much as possible".[4] Incorporating these mechanisms, the model maps round-level funding histories (timing, size and frequency) to project characteristics and funding conditions, which enables estimation and counterfactuals.

The matching friction captures the difficulty evaluating projects, which complicates the process of securing investment, and may delay market clearing and create congestion.[5] Once a VC agrees to invest, the separation of ownership and control inherent to

[3]Nanda and Rhodes-Kropf (2017) introduce financing risk, which has parellels to rollover risk in debt markets (Acharya et al., 2011; He and Xiong, 2012), to the academic literature. Paul Graham (Y Combinator founder) frames this as being "default alive" vs "default dead": "do they make it to profitability on the money they have left?" He cautions that founders often assume raising again will be easy, but "that assumption is often false." See `https://www.paulgraham.com/aord.html`.

[4]See `https://articles.sequoiacap.com/rip-good-times`. Implicit is that Sequoia was not offering to provide follow-on funding unilaterally, instead relying on other investors.

[5]These problems arise even for successful ventures: for example, Airbnb's Brian Chesky described facing a series of rejections when introduced to prominent investors; see `https://medium.com/@bchesky/7-rejections-7d894cbaa084`. On the investor side, Bessemer Venture Partners, a leading VC firm,

the relationship exposes them to moral hazard. Entrepreneurs invest in R&D and come to learn more about the project's viability than the VC, but this information is private and can be hidden from VCs, leading to "inefficient continuation" (Gompers, 1995).[6] The resulting external finance premium depends on (i) the likelihood that the project delivers a successful outcome; (ii) the development horizon; and (iii) the size of the capital injection. Importantly, because larger capital injections increase the premium, VCs have an incentive to stage capital injections over time.

Staging capital injections means that when entrepreneurs hide failure, less capital is wasted. However, because VCs often lack the ability or will to fund a start-up to positive cash flow on their own, follow-on funding requires attracting new investors. The information and project screening issues that arose when the firm first sought funding resurface and additional funding may not arrive, even for viable projects.

These dynamics create "financing risk": viable start-ups may not secure follow-on funding (Nanda and Rhodes-Kropf, 2017). Small capital injections mitigate agency costs but increase reliance on future funding, raising exposure to financing risk. Market tightness determines the availability of follow-on funding and so financing risk is endogenous. The optimal capital commitment balances these considerations, prompting small capital injections when agency frictions are severe, and larger injections when the VC forecasts limited future funding availability.

Three key insights emerge from the theoretical analysis. First, some projects are more difficult to finance than others. This is true for more novel projects, characterised by lower success rates and higher payoffs, as in Nanda and Rhodes-Kropf (2017), but also for long-horizon projects. These projects are endogenously funded through more rounds, which increases their exposure to financing risk; *ex ante* investment incentives are then weaker – so fewer are funded – and, conditional on funding, success is less likely. Second, funding histories encode information about project characteristics and the funding environment. Specifically, moments such as the number of and duration

maintains a list of prominent companies that it passed over; see `www.bvp.com/anti-portfolio`.

[6] Entrepreneurs derive private benefits from operating their own firms (Hurst and Pugsley, 2011) and so are unlikely to want to shut them down upon the arrival of bad news.

between funding rounds depend on financing market tightness and project characteristics. Therefore, the model provides a structural interpretation of the data, which facilitates learning these primitives from observables. I formalise this through a global identification result for the baseline model. Third, there is a novel contracting externality. The choice of capital injection by a given start-up and VC affects the meeting rate in the financing market; for instance, because longer-duration funding reduces the need for follow-on funding in the future. Agents in the model do not internalise this externality, and therefore contracts are generally inefficient.

To bring the model to data and make realistic quantitative statements, I extend it in two ways. First, I include a distinct role for acquisitions, which act as a substitute for VC financing by providing an alternative route to liquidity. Most start-up exits occur through acquisitions, making their inclusion essential. Second, I introduce multiple hurdles in a start-up's journey to full development. The model can then capture differences in how early-stage and late-stage start-ups access funding, reflecting the greater information investors have as start-ups progress. Using the quantitative model, I study start-up outcomes in the US and unpack cross-country differences in VC activity.

In the US context, I estimate the model on VC funding round and exit data for all start-ups that received their first early-stage funding round between 2005 and 2015. In the quantification, start-ups are *ex ante* homogeneous. Nevertheless, the model captures the rich *ex post* heterogeneity observed in the data, including the full shape of the distributions of the number of funding rounds, the duration between rounds, exit multiples, the burn rate (i.e. rate of capital utilisation) and time-to-exit. This rich *ex post* heterogeneity is explained by shocks and information revealed after initial funding, even among start-ups that appear identical at the outset. Therefore, the analysis provides a structural counterpart to evidence on the difficulty of predicting *ex ante* which start-ups will ultimately be successful (Kerr et al., 2014).

Using the estimated model, I then analyse whether financing frictions lead to premature termination in the US market. Whereas the cause of start-up failures is unobserved empirically, simulating the model allows me to estimate the share of VC-backed start-ups that shut down before uncertainty about their projects is resolved and thus constitute inefficient closures. I find that approximately 60% of US VC-backed start-ups are either acquired, remain independent and scale, or face technological or commercial challenges that lead to closure. The remaining 40% fail due to issues in accessing capital, implying that they shut down with positive NPV projects or, equivalently, that continuation would be optimal for a cash-rich entrepreneur.

When matching and agency frictions are removed, these firms are funded until uncertainty about their potential is resolved and many go on to have successful exits. The share that scale independently – e.g. through an IPO – rises from 5% to 11% (+6pp) and pre-commercial acquisitions rise from 25% to 39% (+14pp); together, this implies successful exits rise from 30% to 50% (+20pp). These findings point to substantial losses from frictions in VC markets, but my analysis also highlights an additional margin of inefficiency. To reduce the risk that start-ups shut down prematurely in the future, initial investors extend the funding horizon beyond the level that minimises agency costs. This induces inefficient continuation and, therefore, capital is misallocated in equilibrium: premature termination (underinvestment) coexists with inefficient continuation (overinvestment). My estimates suggest that, at any point in time, 12% of start-ups with capital remaining from their previous funding round should have been shut down, whereas 23% of viable start-ups are actively seeking funding for development.

With the US benchmark in hand, I next examine cross-country differences in VC activity. Despite market imperfections, the US VC market has been instrumental in financing many of today's most R&D–intensive firms (Gornall and Strebulaev, 2021). Conversely, Europe's strong scientific base has not translated into commercial applications at the same scale, and lower VC activity has prompted policies to address a perceived "funding gap" (e.g. Draghi 2024). However, whether lower VC activity in Europe reflects fewer opportunities or frictions in allocating capital is *a priori* unclear. The lack of metrics of *ex ante* start-up quality compounds the identification challenge. I address this with the model, which distinguishes quality from financing frictions via their distinct footprints in funding histories. I focus on a US-UK comparison, where

London's status as a global financing centre, its strong universities, and comparable legal environment make the UK's lower VC intensity particularly puzzling.[7]

I find that financing conditions and acquisition opportunities, not project quality, drive the US-UK gap. UK start-ups raise fewer, more widely spaced rounds and are less likely to be acquired, consistent with tighter financing and a thinner acquisition market; conversely, I find no discernible difference in project quality among funded start-ups. Overall, financing conditions are most important, explaining more than half of the difference in start-up creation incentives.

These findings provide guidance for ongoing policy initiatives aimed at plugging funding gaps. Many programmes traditionally focus on seed and early-stage (Wilson, 2015), but high-growth start-ups often remain unprofitable for years, so late-stage capital is crucial to reach positive cash flow.[8] Furthermore, in the UK, fewer start-ups reach late-stage rounds. In a budget-neutral counterfactual, I find gains from reallocating 5% of total VC funding from early to late-stage rounds: fewer firms are initially funded, but are more likely to raise late-stage capital and scale independently, rather than be acquired at pre-commercial stages. Pre-commercial acquisitions fall (–9%), while the number of projects scaled independently (e.g. through an IPO) rises (+15%). Although total exits decline slightly, this compositional shift raises total surplus. Reassuringly, recent interventions recognise the importance of supporting late-stage start-ups.[9]

Related literature A large literature in entrepreneurial finance demonstrates that VC supply matters for innovation and start-up activity (Kortum and Lerner, 2000; Samila and Sorenson, 2011; Chen and Ewens, 2025) and studies the role that VCs play in screening and selection (Sørensen, 2007; Gompers et al., 2020), active monitoring and expertise (Hellmann and Puri, 2002; Hsu, 2004; Bernstein et al., 2016), and contracting (Kaplan and Strömberg, 2003; Ewens et al., 2022). I focus on staged financing and the

[7] Evidence from developing-country private equity shows that high enforcement and common-law jurisdictions use US-style separation of cash-flow and control rights, whereas low enforcement and civil-law countries are more likely to use debt and common stock, relying on majority ownership for control (Lerner and Schoar, 2005). The US and UK share a common-law legal origin.

[8] Figure B.1 shows that most US-headquartered, VC-backed IPOs were unprofitable at time of listing.

[9] In 2018, the British Business Bank launched British Patient Capital, a late-stage fund of funds.

need for multiple investors to coordinate to fund a firm to positive cash-flow.[10] Staging is designed as a form of monitoring to curb inefficient continuation (Gompers, 1995; Lerner, 1998; Tian, 2011) and is the primary governance tool used by VCs (Sahlman, 1990; Hall and Lerner, 2010), but exposes firms with little interim cash flow to financing risk (Nanda and Rhodes-Kropf, 2017).[11] Relative to this literature, my contribution is to embed these mechanisms in a tractable equilibrium model that can be estimated from round-level data.[12] I use the framework to quantify financing risk in the US, assess funding gaps across markets and consider policy interventions.

This paper also builds on existing models of the VC market. Among theoretical contributions, Inderst and Müller (2004) is the first to model VC as a search-and-matching market and studies how market tightness shapes contracting under double-sided moral hazard. Michelacci and Suarez (2004) and Silveira and Wright (2016) also develop search and matching models[13]; Sannino (2024) studies VC value-added in an equilibrium environment; and Jovanovic and Szentes (2013) links VC excess returns to VC scarcity. I follow Inderst and Müller (2004) in adopting a search-and-matching framework but my focus on staged financing differs from existing work. Theoretically, firms compete for capital across "cohorts" and contracting is generally inefficient because neither entrepreneurs nor VCs internalise how contracts affect the timing of future fundraising and the availability of funds for other start-ups. Empirically, my focus on staging means the model is readily mapped to data on start-up funding rounds.

Several papers incorporate VC into quantitative models.[14] Closest to my paper,

[10]The involvement of new investors can serve a contracting role. Admati and Pfleiderer (1994) show that combining new and existing investors in subsequent funding rounds disciplines insiders' incentives to overprice securities issued to outsiders and mitigates the *ex post* information monopoly that arises when firms can raise new capital only from existing investors, as in Sharpe (1990); Rajan (1992).

[11]Staging can also mitigate hold-up/renegotiation (Neher, 1999) and facilitate learning (Bergemann and Hege, 1998). Importantly, the VC retains abandonment options (see e.g. Cornelli and Yosha 2003; Bergemann and Hege 2005; Dahiya and Ray 2012).

[12]Mayer (2022) studies hidden failure; optimal contracts induce disclosure and can imply premature termination. I abstract from failure payments and study market equilibrium with many investors but retain the feature that VCs tolerate some risk of premature termination to curb inefficient continuation.

[13]They study the role of the stock market in recycling capital and the VC fund lifecycle, respectively.

[14]Ates (2018) and Akcigit et al. (2022) embed VC in endogenous-growth models where VCs provide of expertise as well as finance; Ando (2024) considers angels, in addition to VCs and banks; Ando et al. (2025) studies VC and advanced technology adoption; and Opp (2019) develops an endogenous growth model in the macro-finance tradition, focussing on VC's risk properties and cyclical nature.

Greenwood et al. (2022) also study dynamic VC contracting, but whereas they study the diversion of funds within long-term contracts, I study hidden failure under staged contracts that require firms to secure follow-on financing from new investors.

Beyond the VC literature, my cross-country application relates to work on financial and economic development (King and Levine, 1993; Rajan and Zingales, 1998) and quantitative analyses of financial frictions and entrepreneurship (Buera et al., 2011; Cavalcanti et al., 2023). A common, though not universal (Greenwood et al., 2010), feature of this literature is that entrepreneurs can self-finance and gradually relax their constraints through retained earnings. By contrast, start-ups in my model generate no interim cash flows and rely on external financing until they reach profitability. This distinction changes the implications for policy. In a model calibrated to Europe, Aragoneses and Saxena (2025) find gains from reallocating government VC funding from the late to the early stage, whereas my UK analysis finds gains from shifting funding towards the late stage because early-stage entry and investment depend on the prospect of obtaining later financing. Together, the results suggest that the appropriate stage at which to target policy depends on the ability of firms to self-finance.

The remainder of the paper is organised as follows. Section 2 specifies the baseline and quantitative models, section 3 estimates the model for the US, and section 4 conducts a cross-country comparison. Section 5 concludes.

2 Model

2.1 Economic environment

Before presenting the model, I provide an overview of the key dynamics that it attempts to capture. Three features of the venture capital market are central to the analysis. First, difficulty evaluating projects complicates the process of finding willing investors. Second, agency conflicts lead investors to be cautious of committing too much capital to any given project. Third, institutional features of the market create a need for multiple investors to coordinate to fund a start-up.

Consider an entrepreneur with a business idea. The project derives no intermediate cash flows and cannot be collateralised, so debt is unsuitable and the entrepreneur seeks equity financing from a VC firm. Obtaining this investment faces two broad challenges. First, difficulty evaluating projects prolongs the process of finding a willing investor, and funding may conceivably never arrive. Second, once investment has occurred, agency conflicts arise from the separation of ownership and control. In particular, the entrepreneur may conceal bad news and continue operating the project beyond the date that the VC would prefer to close it down. As discussed in Gompers (1995), this leads to "inefficient continuation", which staging helps to overcome. Panel A of Figure 1 shows that the typical VC-backed start-up in the US raises multiple funding rounds.

In practice, it is common for *new* investors to participate in subsequent funding rounds, while some incumbents do not reinvest (Panels B and C, Figure 1). This reflects various practical considerations, such as diversification incentives and fund-lifecycle dynamics, and the desire for external validation (Kerr et al., 2014; Nanda and Rhodes-Kropf, 2018) and expertise (Brander et al., 2002).[15] However, for the entrepreneur, the need to seek additional investors for subsequent rounds means that matching issues they faced initially reappear.

In light of these concerns, the initial investor(s) may increase upfront funding to reduce the need for subsequent rounds. At first glance, the notion that initial investor(s) could provide extra funding *ex ante*, but may be unwilling *ex post*, at least unilaterally, is puzzling. However, portfolio concerns, VC fund lifecycle dynamics and liquidity issues imply that capital now and capital later are imperfect substitutes. Negative shocks to one portfolio company can restrict the availability of follow-on funding for other firms in a VC's portfolio (Townsend, 2015), while fixed structure limits the scope for later investment (Haran Rosen et al., 2025). VCs also face liquidity risk and, anticipating potential liquidity issues, they prefer to make investments early in the fund's lifecycle (Maurin et al., 2023). When liquidity issues arise, they make fewer investments (Li,

[15]The distribution of start-up outcomes is heavily right-skewed (Hall and Woodward, 2010) and the formation of investor syndicates aids diversification across investments (Pitchbook–NVCA Yearbook, 2024). Lerner (1998) also notes that bringing on new investors can help to solve agency conflicts within the VC fund that may lead to inefficient refinancing decisions.

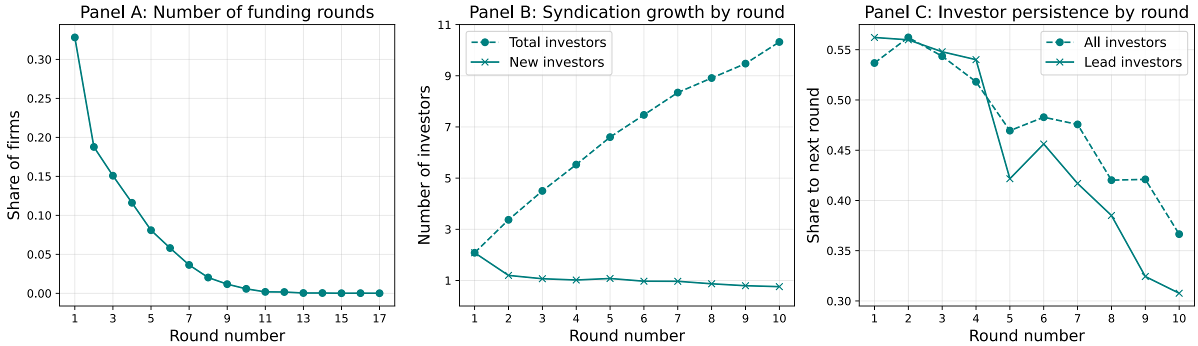

Figure 1: Staged financing and investor syndication

Source: Thomson Reuters, US VC-backed start-ups first funded between 2005–2015, including data to end-2022 (see section B.1). Data is censored at 7 years after initial funding (panel A, to ensure uniform censoring) or 12 years (panels B and C). Panel A shows the distribution of the number of rounds. Panels B and C summarise syndication patterns. Panel B shows the average total (dashed) and new (solid) investors by round; only one undisclosed investor is included (per Nanda and Rhodes-Kropf (2018)). Panel C shows the share of investors in round n that reinvest in round $n + 1$, conditional on subsequent funding; undisclosed investors are excluded.

2024). These liquidity shocks may have a common component and affect aggregate capital commitments, so that the need for new investors may arise exactly when securing funding is most difficult. This creates *financing risk*: viable projects may be forced to shut down because additional funding does not arrive (Nanda and Rhodes-Kropf, 2017).

In this setting, the financing environment influences what gets funded and whether projects scale successfully: the project may not be funded at all, or not funded to completion. Furthermore, the types of contracts written between entrepreneurs and VCs encode information about the extent of agency frictions and aggregate financing conditions, as they shape the capital injection decisions at each round.

An operational theory of the market The preceding discussion motivates a model of the venture capital market with the following structure. Entrepreneurs with projects and VCs with capital meet in a frictional search-and-matching market. On the one hand, the matching friction is a tractable representation of the difficulty evaluating projects, which implies that the market can take time to clear. Relatedly, it endogenises the availability of capital based on the relative supply from investors and demand from entrepreneurs. On the other hand, it provides a simple way to capture coordination issues among investors across funding rounds. Specifically, I require that an entrepreneur searches for a new VC each time it raises capital. During their search, they may be forced to terminate positive NPV projects, and so are subject to financing risk.

Incomplete contracting leads VCs to stage capital injections. Specifically, once investment has occurred, the arrival of news about the project is private information of the entrepreneur. Due to private benefits, the entrepreneur may conceal a negative result, wasting any remaining capital. This problem of "inefficient continuation" drives the external cost of funds above the internal cost and leads VCs to stage capital injections. To retain tractability, I summarise each commitment by a stochastic funding horizon, $T_\omega \sim \text{Exp}(\omega)$, which expires at constant hazard rate ω. This avoids carrying remaining capital as a state variable. A lower contract rate provides the entrepreneur with a longer funding commitment, but increases the VC's exposure to inefficient continuation. Conversely, a higher contract rate reduces this exposure but forces the start-up to return to the matching market more frequently. Finally, the model is closed by specifying an entry condition and a matching technology, so that meeting rates are consistent with both contracts and the entrepreneurs' entry decisions and the contracts signed.

2.2 The baseline model: specification

This section presents the baseline equilibrium model; section 2.3 presents a characterisation of the equilibrium. Time is continuous and the horizon is infinite. I focus on the steady-state equilibrium, so all transition rates and stocks are constant over time.

2.2.1 Agents and technology

There are two types of agents, entrepreneurs and venture capital firms (VCs). All agents are risk neutral and discount the future at rate ρ. At any instant, an exogenous flow Γ of entrepreneurs have a one-time chance to create a *start-up*, drawing some entry cost $c \sim U[0, \sigma]$. A start-up is created if the entrepreneur finds it profitable to pay the entry cost c; otherwise, the idea is lost.

Following entry, all start-ups are *ex ante* homogeneous. Each start-up's project requires flow capital investment k. Conditional on paying k, the results of the development process are realised at Poisson arrival rate κ and are positive (success) with probability p and negative (failure) with probability $1-p$. Success results in a payoff π, whereas a failure returns zero. In addition to any pecuniary return associated with the project, entrepreneurs also derive a non-pecuniary flow benefit $x_b > 0$ when the project is actively being funded. For technical reasons set out in section 2.2.2, I let $x_b \to 0$.

As a frictionless benchmark, suppose that the entrepreneur is wealthy and can self-finance the project. Let $T_\kappa \sim \text{Exp}(\kappa)$ denote the time at which the result of the development process is realised. The value of the project is then

$$\bar{V}_s = E\left[e^{-\rho T_\kappa} p\pi - \int_0^{T_\kappa} e^{-\rho t} k\, dt\right] = \frac{\kappa p\pi - k}{\rho + \kappa}. \tag{1}$$

The entrepreneur would finance the project if $\bar{V}_s > 0$, or equivalently if $k < \kappa p\pi$. I use $\bar{V}_s$ as the frictionless measure of project quality below.

However, in the model, entrepreneurs have no wealth and must seek external financing from VCs. In the baseline, I assume that entrepreneurs have all the bargaining power in contracting; I relax this in section A.2. There is a measure M of VCs, each of which may finance at most one project at a time. VCs face no capital constraints with respect to a single investment, but cannot provide follow-on funding to an entrepreneur that they previously invested in, should they require it. Conditional on a meeting, the entrepreneur and VC agree on the project characteristics (k, κ, p, π).

2.2.2 The financing market

Access to external finance is subject to two frictions: entrepreneurs and VCs meet in a frictional search market, and contracting is subject to moral hazard.

Search-and-matching friction Start-ups search for a VC following entry or the exhaustion of previous funding, provided they did not realise a negative result. VCs can screen out projects that have already failed, so these firms cannot obtain follow-on funding. VCs search for a start-up to invest in if they are not currently committed to another start-up. Search is costly due to discounting, but there are no explicit search costs. I denote by μ_s the measure of start-ups searching for capital and by μ_{vc} the measure of searching VCs. The measures μ_s and μ_{vc} are the aggregate state variables in the model and determine the flow of meetings via the matching function

$$m(\mu_s, \mu_{vc}) = \varepsilon\, \mu_s^{\alpha}\, \mu_{vc}^{\beta} \tag{2}$$

where $\alpha, \beta > 0$. The Poisson arrival rate of meetings for an entrepreneur and VC are

$$\nu = \frac{1}{\mu_s} m(\mu_s, \mu_{vc}) \quad \text{and} \quad \upsilon = \frac{1}{\mu_{vc}} m(\mu_s, \mu_{vc}), \tag{3}$$

respectively. During search, start-ups fail at exogenous rate λ; VCs are infinitely lived.

Contracting and moral hazard Once an entrepreneur and VC meet, they negotiate an equity contract to fund the start-ups development cost, k. Contracts specify an equity stake ς for the VC and a funding termination rule, which has two components. Firstly, the funding termination rule may condition on the arrival of news about the project at date T_κ. Secondly, the entrepreneur and VC may agree to a random date, $T_\omega \sim \text{Exp}(\omega)$, at which the funding commitment expires.

Consider first the contingent component, which matters when news arrives prior to the funding commitment expiring; that is, when $T_\kappa < T_\omega$ *ex post*. Due to the information asymmetry, in any contract that conditions on news at time T_κ, it must be incentive compatible for the entrepreneur to report to the VC.

There are two cases to consider, success or failure. Given their private benefits, the entrepreneur never reports failures but may report success.[16] Reporting success immediately at date T_κ leads the VC to terminate funding, meaning the entrepreneur forgoes any additional non-pecuniary benefit. However, concealing the news delays the pecuniary payoff by T_ω.[17] The entrepreneur must decide whether the value of the private benefit is sufficient to compensate for the loss in their pecuniary payoff due to

[16]Concealing failure allows the entrepreneur to continue to reap their private benefits, $x_b > 0$, until funding expires at date T_ω, whereas reporting failure allows the VC to shut down the venture.

[17]The environment is stationary, so if the entrepreneur finds it profitable to delay reporting at the time the result is obtained, they will never report until the contract expires, which happens at date T_ω.

discounting, which leads to an incentive compatibility constraint.[18]

Proposition A.1 in the Appendix shows that, for any set of model parameters, there always exists an upper bound $\bar{x}_b > 0$ such that for $x_b < \bar{x}_b$, then the entrepreneur always reports positive results. Intuitively, if the non-pecuniary benefit is sufficiently small, the entrepreneur would never delay their pecuniary payoff to continue to enjoy private benefits. Therefore, given $x_b < \bar{x}_b$, the entrepreneur always reports successes to the VC, but never reports failures; this appears to accord with anecdotal accounts of entrepreneur behaviour.[19] The assumption $x_b \to 0$ guarantees that this is the case.[20]

With the contingent features determined, it is simple trace the implications of the choice of non-contingent funding termination, $T_\omega \sim \text{Exp}(\omega)$. Analogous to a larger capital injection, a lower "contract rate" ω provides a longer term funding commitment. However, in this incomplete contracting environment, it increases the VC's exposure to the entrepreneur's opportunistic behaviour, because it increases both the probability that the entrepreneur will uncover negative news *and* the outstanding capital commitment when that information arrives. Specifically, the expected capital commitment associated with contract (ω, ς) is given by[21]

$$\begin{aligned} K(\omega) &= \underbrace{\frac{k}{\rho+\kappa+\omega}}_{\substack{\text{Frictionless expected} \\ \text{capital cost}}} + \underbrace{\frac{\kappa\,(1-p)}{\kappa+\omega}}_{\text{Pr(negative result)}} \underbrace{\frac{k\,(\kappa+\omega)}{(\rho+\omega)\,(\rho+\kappa+\omega)}}_{\text{Remaining commitment}} \\ &= \left(1+\frac{\kappa\,(1-p)}{\rho+\omega}\right)\frac{k}{\rho+\kappa+\omega}. \end{aligned} \tag{4}$$

where $K'(\omega) < 0$ and $K''(\omega) > 0$. The expected capital cost is composed of two terms. First, in a complete contracting environment, the entrepreneur and VC could write a contract to terminate funding at the realisation of the result, date $T_\kappa \sim Exp(\kappa)$. The

[18]See Remark A.2. The IC depends on features of the project and environment, (π, ρ, x_b), but also on the share, $1-\varsigma$, that the entrepreneur maintains in the project. This leads to the implication that the entrepreneur must retain sufficient "skin-in-the-game" to report truthfully.

[19]See, for example, Chapter 4 of Janeway (2018), "No news is never good news".

[20]The VC holds a claim that has some resemblance to participating preferred stock, in which the VC receives a liquidation preference and shares in the payoff; such features are common in VC contracts (Kaplan and Strömberg, 2003; Ewens et al., 2022). In the model, when the project has a liquidation value the VC receives a share of the payoff *and* cuts funding, which acts like a claim against the company.

[21]See section A.3.1 for the derivation.

contract rate would be $\omega = 0$, so that funding would never be withdrawn prematurely, and the VC would expect to inject $k/(\rho + \kappa)$ into the start-up before the result is realised. The first term reflects this benchmark, which is equivalent to the capital cost that would arise if the entrepreneur could invest their own funds. However, the agency conflict introduces an additional term: the entrepreneur conceals the arrival of bad news, engaging in inefficient continuation. The second line of equation (4) shows that this agency conflict introduces a wedge over the frictionless benchmark.

The optimal contract Abstracting from VC bargaining power, the optimal contract, (ω, ς), maximises the value of the project to existing shareholders (the entrepreneur and any prior investors) subject to a participation constraint (PC) of the new investor.[22]

For a given contract (ω, ς), the VC incurs expected capital costs $K(\omega)$, given by equation (4), and receives a share ς of the value of the firm gross of their investment, denoted $V_d(\omega)$.[23] Therefore, the PC is $\varsigma V_d(\omega) \geq K(\omega)$ and the optimal contract solves

$$\begin{aligned} V^M = & \sup_{\{\omega \in [0,\infty), \varsigma \in [0,1]\}} \left\{ (1-\varsigma)\, V_d(\omega) \right\} \\ \text{s.t.} \quad & \varsigma\, V_d(\omega) \geq K(\omega) \end{aligned} \tag{5}$$

where $K(\omega)$ is given by equation (4), $V_d(\omega)$ solves the HJB equation

$$\rho\, V_d = \kappa\, [p\, \pi - V_d] + \omega\, [V_s - V_d] \tag{6}$$

and V_s is the value of a start-up in search, which is taken as given in the optimisation. To see why V_d solves equation (6), recall that news about the project arrives at rate κ and is successful with probability p, yielding a payoff π. However, at rate ω, the funding commitment terminates, in which case the entrepreneur must search for new funding;

[22] I consider the extension to VC bargaining power in section A.2.
[23] This is equivalent to the "post-money" value of the firm.

its value then becomes V_s. In equilibrium, the value of search solves

$$(\rho + \lambda)\, V_s = \nu\, [V^M - V_s]. \tag{7}$$

which follows because a start-up in search fails at exogenous rate λ and meets with a VC at rate ν, in which case it has value V^M. The binding participation constraint pins down ς as a function of ω, which then implies that $V^M = V_d(\omega) - K(\omega)$. Therefore, the optimal contract rate, ω, maximises the value of the firm net of investment costs.

2.2.3 Closing the model

To close the model, I specify the entry decision and conditions that determine the two state variables, μ_s and μ_{vc}, which pin down the equilibrium meeting rates, ν and υ.

Entry Given V_s, the entry decision is a simple cutoff rule. A given start-up enters if and only if their entry cost $c \leq V_s$, where V_s is the value of search. Given $c \sim U[0, \sigma]$, the probability of drawing an entry cost below the threshold $Pr(c \leq V_s) = V_s/\sigma$. Each potential entrant faces this same probability, so that the flow of new entrants that enter endogenously is given by $\Gamma\,(V_s/\sigma)$. It will not be possible to disentangle Γ from σ, so I will substitute in $\tilde{\sigma} = \sigma/\Gamma$ going forwards. The endogenous flow of entry is then $V_s/\tilde{\sigma}$.

Steady-state conditions Finally, it remains to specify the conditions that determine μ_s and μ_{vc}. Rather than specify a condition for μ_{vc} directly, it is simpler to consider how many firms receive funding in equilibrium and to determine μ_{vc} as the residual given the fixed measure M of VCs. In this spirit, I will refer to a start-up that has already obtained a negative result but is still receiving funding as 'unproductive', as opposed to 'productive' start-ups that have funding and may still achieve success. I denote by μ_d^p the measure of start-ups in productive development and by μ_d^u the measure of start-ups in unproductive development. Since the total measure of VCs is M and a measure $\mu_d^p + \mu_d^u$ are funding start-ups, the measure of searching VCs is given by $\mu_{vc} = M - \mu_d^p - \mu_d^u$. The state-to-state transitions for a single firm are summarised by

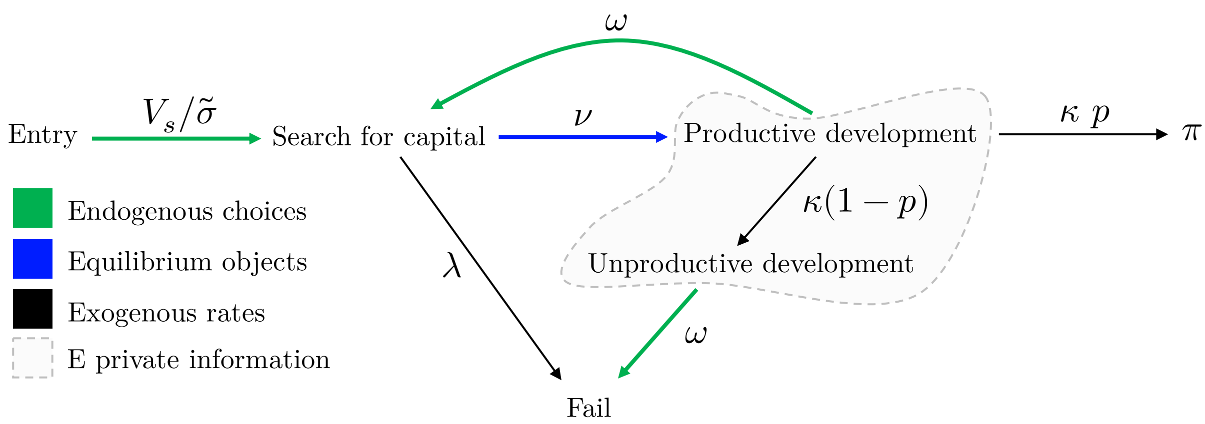

Figure 2: State transitions

Entry reflects an aggregate flow; other arrows represent individual-level transition intensities.

Figure 2. In steady-state, the inflows and outflows from each state must be equal to one another. Consider first the measure of firms in productive development. A flow $\nu\mu_s$ of start-ups enter this state from search, having signed contracts with VCs, and a flow $(\kappa+\omega)\mu_d^p$ exit this state when realising a result about their project or seeing their funding run out. Next, consider the measure of firms in unproductive development. Firms enter this state from productive development when they obtain a negative result, a flow $\kappa(1-p)\mu_d^p$, and exit this state when their funding runs out, a flow $\omega\mu_d^u$. As Figure 2 illustrates, transitions into this state are unobserved to the VC. Finally, consider the measure of firms in search. Firms return to search from productive development when their funding runs out, an inflow $\omega\mu_d^p$. Since firms in unproductive development would be screened out by subsequent VCs, they find it optimal to close down and so do not return to search. In addition to the inflow from productive development, new entrants begin their lifecycle in search of capital, a flow $V_s/\tilde{\sigma}$. Conversely, start-ups exit search when they fail during the search process or meet with VCs, a flow $(\lambda+\nu)\mu_s$. Therefore, the steady-state conditions governing μ_d^p, μ_d^u and μ_s are given by[24]

$$\mu_d^p: \qquad \nu\,\mu_s = (\kappa+\omega)\,\mu_d^p \tag{8}$$

$$\mu_d^u: \qquad \kappa(1-p)\mu_d^p = \omega\mu_d^u \tag{9}$$

$$\mu_s: \qquad \omega\mu_d^p + \frac{V_s}{\tilde{\sigma}} = (\lambda+\nu)\,\mu_s \tag{10}$$

[24]Nondegenerate stationary equilibria require $\omega > 0$; otherwise, some VCs never return to search.

These equations can be written purely in terms of μ_s and μ_{vc}.[25] Then,

Definition 1. *A steady-state equilibrium is a tuple* $(V_d, V_s, V^M, \omega, \varsigma, \mu_s, \mu_{vc})$*, where* (ω, ς) *solve maximisation problem* (5) *given the meeting rates* (ν, υ) *in equation* (3)*, and equations* (6)*,* (7) (8)*,* (9) *and* (10) *are satisfied.*

2.3 The baseline model: equilibrium characterisation

In this section, I analyse properties of the equilibrium. First, taking financing conditions in the matching market as given, I solve the contracting problem and consider comparative statics. These exercises illustrate: (i) how observable features of VC contracts respond to project characteristics and financing market conditions, thereby encoding information about them, and (ii) how frictions in VC markets shape start-up entry and outcomes. Second, I consider efficiency and discuss the feedback between the agency and matching frictions, which endogenously determines financing conditions.

2.3.1 Funding histories, outcomes, and the optimal contract

To connect the model to the data, I first characterise funding histories and outcomes for a given contract rate, ω. I then solve for the optimal contract rate and show how project characteristics and financing conditions affect these observables. The model admits a simple mapping from model primitives and the contract rate to funding histories and start-up outcomes, as outlined by the following proposition.

Proposition 1. *For a given contract rate, ω:*

(i) The expected discounted capital commitment associated with a funding round is

$$K(\omega) = \left(1 + \frac{\kappa(1-p)}{\rho + \omega}\right) \frac{k}{\rho + \kappa + \omega}.$$

[25] From equation (9), $\mu_d^u = \frac{\kappa(1-p)}{\omega}\mu_d^p$, so that $\mu_{vc} = M - \mu_d^p - \mu_d^u = M - \left(1 + \frac{\kappa(1-p)}{\omega}\right)\mu_d^p = M - \Big(1 + \frac{\kappa(1-p)}{\omega}\Big)\frac{\nu}{\kappa+\omega}\mu_s$, where the last equality follows from equation (8). Furthermore, for μ_d^p in equation (10) from equation (8) yields $\frac{\omega}{\kappa+\omega}\nu\mu_s + \frac{V_s}{\tilde{\sigma}} = (\lambda + \nu)\mu_s$.

(ii) Conditional on receiving at least one funding round, the number of rounds satisfies

$$N_f \sim \text{Geometric}(q_d), \qquad q_d = \frac{\kappa}{\kappa+\omega} + \frac{\omega}{\kappa+\omega}\frac{\lambda}{\lambda+\nu},$$

with support $N_f \in \{1, 2, \ldots\}$ *and* $E[N_f] = 1/q_d$.

(iii) Conditional on receiving a subsequent funding round, the duration between successive rounds satisfies

$$T_{br} \sim \text{Hypoexponential}(\kappa+\omega, \lambda+\nu), \qquad \textit{with} \quad E[T_{br}] = \frac{1}{\kappa+\omega} + \frac{1}{\lambda+\nu}.$$

(iv) The probabilities of eventual success for a firm in productive development, p_d, *and in search,* p_s, *are*

$$p_d = (1-\Delta_p)p, \qquad p_s = \frac{\nu}{\lambda+\nu}\, p_d, \qquad \Delta_p = \frac{\lambda\omega}{\kappa(\lambda+\nu)+\lambda\omega}.$$

where $\Delta_p \in [0, 1)$ *is the outcome distortion and, under staged financing,* $p_d, p_s < p$.

Corollary 1. *Holding the model primitives fixed, a higher contract rate implies a smaller expected capital commitment, more funding rounds and less time between funding rounds (in expectation), and a greater outcome distortion*

$$K'(\omega) < 0, \qquad \frac{\partial E[N_f]}{\partial \omega} > 0, \qquad \frac{\partial E[T_{br}]}{\partial \omega} < 0, \qquad \frac{\partial \Delta_p}{\partial \omega} > 0.$$

The expression for $K(\omega)$ follows immediately from equation (4). The remaining results follow from the model's Markovian structure. The probability of another funding round is memoryless, as it depends only on the current state and not on the number of prior funding rounds. As a result, the distribution of the number of funding rounds follows a Geometric distribution, the only discrete probability distribution with the memoryless property.[26] The term q_d is simply the probability that the current funding

[26]In practice, this memoryless property is a strong assumption and likely violated in the data. The quantitative model relaxes it.

round is the firm's last, which occurs if either project uncertainty is resolved before the current commitment expires, or the commitment expires and the firm fails before meeting another VC. Regarding the time between funding rounds, conditional on obtaining another round, the firm first spends an exponentially distributed period in development, with rate $\kappa+\omega$, and then an exponentially distributed period searching for capital, with rate $\lambda + \nu$; their sum is hypoexponential. Finally, the success probabilities satisfy

$$p_d = \frac{\kappa}{\kappa+\omega}\,p + \frac{\omega}{\kappa+\omega}\,p_s, \qquad p_s = \frac{\nu}{\lambda+\nu}\,p_d,$$

and solving these equations yields the outcome distortion in Proposition 1.

Corollary 1 provides a bridge between contracts and observables. A higher contract rate leads to smaller, more frequent capital commitments, but simultaneously lowers the success rate by increasing exposure to financing risk. I now turn to the determination of the optimal contract rate. Taking the meeting rate in the financing market, ν, the model admits closed-form solutions for value and policy functions.[27]

Proposition 2. *Given ν and for $k < \kappa p \pi$, there is a unique solution for $\{V_d, V_s, V^M, (\omega, \varsigma)\}$ in terms of model parameters and the resulting equilibrium is characterised by a (weakly) positive and finite contract rate, $\omega \in [0, \infty)$.*

Corollary 2. *Given ν and $k < \kappa p \pi$, the equilibrium contract rate is given by*

$$\omega = \max\left\{0, -\rho + \mathcal{C} + \sqrt{\mathcal{C}\mathcal{F} + \mathcal{C}^2}\right\}, \quad \textit{where} \quad \mathcal{C} = \frac{\kappa(1-p)k}{\kappa p \pi - k} \quad \textit{and} \quad \mathcal{F} = \kappa + \nu \times \frac{\rho+\kappa}{\rho+\lambda} \tag{11}$$

and $\omega \to \infty$ otherwise. $\mathcal{C}$ captures characteristics and $\mathcal{F}$ captures need for and access to follow-on funding.

The contract may feature $\omega = 0$, in which case the VC commits to fund the project to completion with certainty and corresponds to *upfront financing.* Conversely, for

[27]Under generalised Nash bargaining case, both ν and υ must be specified. VC bargaining power introduces two incentives shaping the optimal contract: (i) VCs profit from each investment but can only invest in one firm at a time, so each deal entails an opportunity cost (cf. Michelacci and Suarez 2004); and (ii) VCs anticipate dilution by future investors who also extract surplus. Effect (i) pushes towards shorter-duration contracts, whereas (ii) induces the opposite – the net effect of VC bargaining power on ω is ambiguous (see Proposition A.2).

$\omega > 0$, the entrepreneur may require additional funding for their project; this is *staged-financing* and is an endogenous feature of the model. In what follows, I analyse the model under staged financing, which is the empirically relevant case.

Towards understanding the key determinants of the optimal contract rate, it will be helpful to be able to isolate changes in contracting that are attributed to changes in the "quality" of the project, from those that arise due to changes in the project's characteristics. To this end, I define quality as the NPV of a project to a wealthy entrepreneur that can self-finance development: $\bar{V}_s$ in equation (1). Under risk-neutrality, exponential discounting, and perfect capital markets, $\bar{V}_s$ is a sufficient statistic for investment decisions and therefore constitutes the relevant frictionless benchmark. To the extent that different projects are equal in $\bar{V}_s$ but not in V_s, the comparative statics reveal how financing frictions create distortions to entry and investment across projects with equal quality, but different characteristics.

To formalise this, I define a quality-compensating comparative static (QCCS), which varies project characteristics while the payoff, π, is adjusted to hold quality, $\bar{V}_s$, fixed.[28]

Definition 2. *Consider a project with characteristics (k, κ, p, π). I define*

(a) Project quality as $\bar{V}_s = \frac{\kappa p \pi - k}{\rho + \kappa}$;

(b) A quality-compensating comparative static (QCCS) in parameter x as the derivative with respect to x while adjusting π so that quality, $\bar{V}_s$, is held constant.

With this definition in place, the following proposition considers the determinants of the optimal contract rate

Proposition 3. *The optimal contract rate ω is strictly*

- *increasing in the meeting rate (ν) and the flow investment cost (k);*
- *decreasing in the success probability (p), payoff (π), and failure rate in search (λ).*

[28]For example, a QCCS with respect to the arrival rate of R&D outcomes, κ, compares a baseline project to a project with a shorter development horizon, higher κ, but lower payoff in case of success, lower π. The compensating decrease in π means I capture the effect of κ coming purely from the timing of the payoff, rather than from changes in the underlying project NPV.

- *For the arrival rate of outcomes (κ), there is a threshold, $\bar{\kappa} > k/p\pi$, such that ω is strictly decreasing in κ below this threshold and strictly increasing above it.*

Holding project quality, $\bar{V}_s$, fixed (QCCS), the contract rate ω is strictly decreasing in the R&D success probability (p) and strictly increasing in arrival rate of outcomes (κ).

Proposition 3 reflects a trade-off between agency costs and financing risk. A lower contract rate commits more capital and insures the start-up against having to secure follow-on financing. This insurance is valuable when financing conditions are tight, when ν is low or λ is high, and when premature termination would destroy substantial project value, for instance because the project has a high payoff π relative to capital cost k, a high probability of success, p, or is expected to complete quickly, a high κ.

Longer commitments, however, exacerbate the agency friction. When p is low, negative news is more likely, increasing the scope for inefficient continuation. Holding the funding horizon fixed, shortening the development horizon, a higher κ, also raises agency costs because uncertainty is more likely to be resolved with the current capital injection, which increases the likelihood that the entrepreneur hides failure. These considerations lead the parties to shorten the commitment by increasing ω. The effect of κ is therefore ambiguous in a standard comparative static. A higher κ strengthens the incentive to align the funding and development horizons and reduces the need for follow-on financing, both of which increase ω. At the same time, it raises project quality by bringing expected payoffs forward and reducing total investment requirements, which increases the value of financing insurance and lowers ω. The QCCS removes this quality effect: holding project quality fixed, a shorter development horizon therefore results in a shorter funding commitment, so the funding and development horizons co-move positively. This is the model's notion of milestone financing.

Taken together, Proposition 3 and Corollary 1 show how funding histories and outcomes reflect project characteristics and financing conditions. The following result provides a bridge from observable funding histories to the model's outcome distortion.

Corollary 3. *Consider two start-ups $i = a, b$ facing the same financing conditions, ν and λ. If $E[N_f^a] > E[N_f^b]$, then $\Delta_p^a > \Delta_p^b$.*

Corollary 3 says that, within a common financing environment, projects whose optimal contracts imply more funding rounds also face greater outcome distortions. Both objects depend on the relationship between the funding and development horizons: more frequent returns to the financing market increase the start-up's exposure to premature termination. The result therefore connects an observable feature of funding histories to an otherwise latent distortion in start-up outcomes.

The parameters p, π, and k do not enter $E[N_f]$, $E[T_{br}]$, or Δ_p directly, so their effects on these objects operate entirely through the optimal contract. A higher probability of success, p, or payoff, π, lowers the contract rate, resulting in fewer and more widely spaced funding rounds and a smaller outcome distortion. A higher flow investment cost, k, has the opposite effect.

The arrival rate of project outcomes, κ, is more subtle because it affects funding histories and outcomes both directly and through the contract. Holding project quality fixed, a higher κ raises the contract rate and reduces the expected time between rounds, but its effects on the expected number of rounds and the outcome distortion are ambiguous because the funding and development horizons both adjust. Long-horizon projects receive more funding rounds whenever the optimal contract rate responds less than proportionally to the project completion rate. This condition holds for empirically relevant parameter values.[29] Corollary 3 then implies that, within a common financing environment, long-horizon projects also face greater outcome distortions.

Financing conditions similarly affect funding histories both directly and through the endogenous response of contracts. Easier access to follow-on financing, a higher ν, leads firms to accept smaller and shorter commitments: they raise less capital per round, complete more rounds, and spend less time between rounds. Conversely, when financing becomes difficult to obtain, firms insure themselves by securing larger and longer commitments. The resulting effect on outcomes is ambiguous. Tighter financing directly increases the risk of premature failure, but it also induces firms to raise more

[29]Under the QCCS, long-horizon projects receive more funding rounds in expectation if and only if $\kappa\,\omega'(\kappa)/\omega < 1$. A sufficient condition is $\pi < \bar{\pi}$, where $\bar{\pi}$ depends on other model parameters. In the quantitative model, the implied threshold $\bar{\pi}$ is orders of magnitude larger than the estimated value of π.

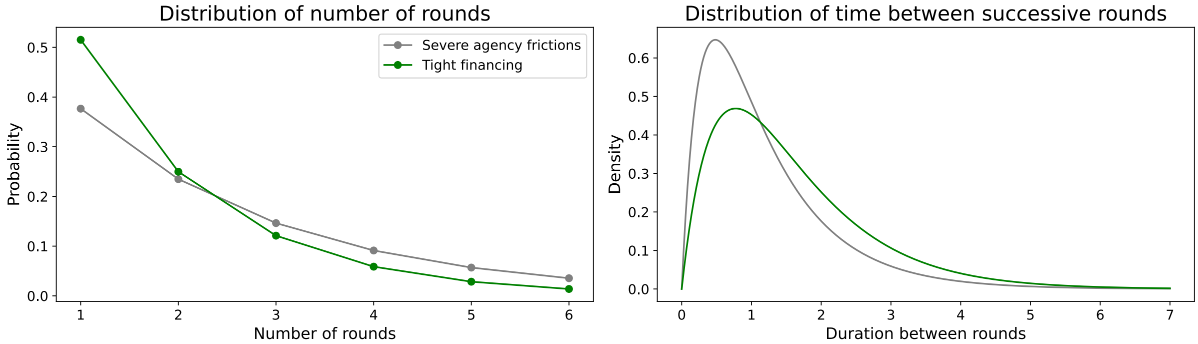

Figure 3: Funding histories under agency and financing frictions

The figure depicts the distributions of the number of funding rounds and the duration between rounds under two hypothetical parameterisations. Tight financing is generated by lowering the meeting rate, ν; severe agency frictions is generated by lowering the success probability, p.

capital per round, reducing their exposure to that risk. When financing markets are sufficiently tight, this insurance response can dominate.

Figure 3 illustrates the contrasting implications of agency and financing frictions for observable funding histories. More severe agency frictions produce more, closely spaced funding rounds, whereas tighter financing conditions produce fewer rounds with longer durations between them. Section A.3.1 reports complete comparative statics.

2.3.2 Entry

Recall that flow rate of entry is given by $V_s/\tilde{\sigma}$, where V_s is the value of search. If entrepreneurs could self-finance, this value would instead be $\bar{V}_s$, and the flow of entry would be $\bar{V}_s/\tilde{\sigma}$. To illustrate how financing frictions reduce entry incentives, consider the limiting case $\rho \to 0$, in which the timing of payments is irrelevant. Then,[30]

$$V_s = \frac{\nu}{\lambda + \nu}(1 - \Delta_p)\frac{\kappa p \pi - k\left(1 + \frac{\kappa(1-p)}{\omega}\right)}{\kappa} \tag{12}$$

which is analogous to the expression for $\bar{V}_s|_{\rho\to 0}$, with the exception of three wedges that emerge. First, only a share $\nu/(\lambda+\nu)$ of entrants ever receive funding, reducing the incentive to enter. Second, the outcome distortion attenuates entry incentives. It premultiplies the entire expression, reflecting both that $p_d = (1-\Delta_p)p$ is the relevant success

[30]Equivalently, $V_s = \frac{\nu}{\lambda+\nu}\big(p_d \times \pi - E[N_f] \times K(\omega)|_{\rho\to 0}\big)$.

probability, and that the entrepreneur may be forced to shut down before completing the development process, which reduces their total capital investment.[31] Finally, the agency friction increases the effective flow capital cost by a factor $1 + \kappa(1-p)/\omega$.

For an arbitrary ρ and with contracts chosen optimally, I summarise the combined effect of these forces by defining the *entry distortion*, Δ_s, according to

$$V_s = (1-\Delta_s)\bar{V}_s, \quad \text{where} \quad 1-\Delta_s = \frac{\mathcal{F}-\kappa}{\mathcal{F}+2\mathcal{C}+2\sqrt{\mathcal{C}\mathcal{F}+\mathcal{C}^2}} \tag{13}$$

where $\mathcal{C}, \mathcal{F}$ are defined in Corollary 2. Given unit elasticity of entry with respect to the value of search, the entry distortion, Δ_s, gives the proportionate decline in entry owing to financing frictions, analogous to the outcome distortion, Δ_p for outcomes. The key results for entry are given by the following proposition

Proposition 4. *The value of entry V_s is strictly*

- *increasing in the meeting rate* (ν)*, R&D success probability* (p)*, payoff* (π)*, and arrival rate of R&D outcomes* (κ)
- *decreasing in the flow investment cost* (k) *and failure rate in search* (λ)*.*

Holding project quality, $\bar{V}_s$, fixed (QCCS), the entry value V_s is strictly increasing in the R&D success probability (p) *and ambiguous in the arrival rate of outcomes* (κ)*.*

In most cases, the intuition for the sign is relatively straightforward. Entry is higher when financing is easier to access or projects are higher quality. For the development horizon, κ, there are various competing effects. Allowing quality to adjust, entry is higher for short-horizon projects, reflecting both the timing of payoffs and more limited capital requirements associated with a short-horizon project. Holding quality fixed, the effect is technically ambiguous but in practice the effect goes in the same direction as the outcome distortion, Δ_p.[32] Therefore, because, for given quality, longer-horizon projects tend to be funded with more rounds, they face greater entry distortions.

[31] In the frictionless case, the expected investment duration is κ^{-1} compared to $E[N_f] \times (\kappa+\omega)^{-1} = \kappa^{-1}(1-\Delta_p)$ in the frictional economy, where $(\kappa+\omega)^{-1}$ is the average development spell.

[32] In the limit as $\rho \to 0$, among start-ups facing the same financing conditions, $E[N_f^a] > E[N_f^b]$ implies $\Delta_s^a > \Delta_s^b$, mirroring Corollary 3.

2.3.3 Efficiency

The purpose of this section is threefold. First, I demonstrate that the frictionless benchmark in which the entrepreneur can self-finance can be recovered in the model in the limit as $\nu \to \infty$. This illustrates that the distortions associated with the agency friction arise through its interaction with the matching friction. Second, in the case where ν is finite, I provide a measure for the extent of capital misallocation in the market; capital is misallocated because unproductive start-ups receiving funding coexist with unfunded viable projects searching for capital. Finally, I show that, in general, the contracts and entry rate that arise in the decentralised equilibrium do not coincide with the constrained planner's solution, so the equilibrium is generally not constrained efficient.

Frictionless benchmark Were entrepreneurs able to self-finance, the entry rate would be $\bar{V}_s/\tilde{\sigma}$. The model nests this case in the limit as $\nu \to \infty$.

Lemma 1. *As search becomes frictionless, $\nu \to \infty$, the value of search, V_s, converges to the frictionless value, $\bar{V}_s$: $\lim_{\nu \to \infty} V_s = \bar{V}_s$.*

As $\nu \to \infty$, expected delays between funding rounds vanish. Funding can be split into arbitrarily small, instantaneous rounds, eliminating scope to hide failure and driving the external-finance premium to zero; in other words, staging fully overcomes the agency friction. The value of search (and thereby entry) therefore coincides with $\bar{V}_s$.

Capital misallocation For finite ν, the model features capital misallocation. Defined as the share of firms with funding that are unproductive, this is given by

$$\frac{\mu_d^u}{\mu_d^u + \mu_d^p} = \frac{\kappa\,(1-p)}{\kappa\,(1-p) + \omega} \tag{14}$$

which follows from equation (9).[33] Capital misallocation is decreasing in ω. Therefore, for parameters other than κ and p, factors that push up the contract rate, ω, reduce capital misallocation, whereas factors that lead to a lower contract rate typically increase

[33] In the baseline model where every funding round involves the same amount of capital raised, this is equivalent to the share of capital that is misallocated.

misallocation. In particular, when financing conditions for start-ups tighten, a lower ν, capital misallocation rises. Intuitively, capital is misallocated in the model because of the interaction of the agency and matching frictions; in an environment with only agency frictions, the optimal contract rate would specify $\omega \to \infty$ and no capital would be misallocated. Therefore, capital misallocation reflects insurance against financing risk and my estimates of capital misallocation in section 3.3 are indicative of the costs that firms incur to militate against it. Derivations and the complete comparative statics are provided in section A.3.4.

Constrained efficiency Some degree of capital misallocation is inevitable; a separate question is whether a planner subject to the same frictions would choose the same contract and entry rate as the decentralised equilibrium. To consider this, it is necessary to allow for VC bargaining power $1-\delta$ in negotiations, where $\delta = 1$ is the case considered in the baseline model; this general version of the model is described in section A.2 of the Appendix. The planner maximises

$$TS = \int_0^\infty e^{-\rho t}\Big([\kappa p\pi - k]\mu^p_{d,t} - k\mu^u_{d,t} - \frac{\tilde{\sigma}}{2}\Lambda_t^2\Big)\, dt \tag{15}$$

where Λ_t denotes the flow rate of entry at time t. The flow value of total surplus is equal to the flow payoff, $\kappa p\pi\mu^p_{d,t}$, net of funding, $k(\mu^p_{d,t} + \mu^u_{d,t})$, and entry costs, $\frac{\tilde{\sigma}}{2}\Lambda_t^2$.[34]

Proposition 5. *If the matching function exhibits constant returns to scale, $\beta = 1-\alpha$, and the bargaining weights in the contracting problem are equal to their respective matching function elasticities, $\alpha = \delta$ and $\beta = 1-\delta$, then the decentralised steady-state allocation is constrained efficient.*

The efficiency condition is analogous to the Hosios condition in the search literature. For a fixed contract rate, ω, matching-function bargaining weights ensure that the congestion and appropriability effects associated with start-up entry offset exactly. The contract rate introduces a similar "re-entry" margin: by determining how frequently

[34]Entry is cutoff rule, so entry Λ_t implies the cutoff $\hat{c}_t = \tilde{\sigma}\Lambda_t$ and flow entry costs $\int_0^{\hat{c}_t} c/\tilde{\sigma}\, dc = \frac{\tilde{\sigma}}{2}\Lambda_t^2$.

productive start-ups and VCs return to the matching market for follow-on financing, ω governs their rate of re-entry into search. Through the equilibrium flow-balance conditions, this re-entry rate affects the measures of investors and entrepreneurs in search and, consequently, market tightness. Thus, anticipated market tightness influences contract choice, while contract choice feeds back into the financing conditions faced by other firms. Under the Hosios condition, private incentives on both the initial-entry and re-entry margins coincide with those of the constrained planner. Outside this special case, private contracting does not generally internalise its effect on the availability of capital to other firms.

2.4 Quantitative extensions for estimation

This section adapts the baseline model for quantification by making two adjustments. First, I introduce two stages to the R&D process, allowing financing conditions to improve over the start-up's development cycle as more information is produced about its prospects. This is important for two reasons: (i) it introduces selection into premature termination *ex post*, because start-ups that show initial success may face easier financing conditions when seeking future funding; and (ii) it allows the model to capture the empirical hazard rate for successful exits, which is first increasing and then decreasing in the time since first funding (Jovanovic and Szentes, 2013). Capturing exit hazards is particularly important because I estimate the model using censored data. The two-stage structure also maps naturally to the data, where funding rounds are typically classified as either "early" or "late" stage. Second, I introduce acquisitions as a distinct form of exit for firms in the late stage. Acquisitions are empirically important but also provide an alternative path to liquidity for start-ups that struggle to obtain the capital required to scale independently, which means they substitute for financing. I permit acquisitions only in the late stage, mirroring the environment in Arora et al. (2024), where a start-up must overcome both technology and commercialisation risk but may sell to an acquirer before attempting to overcome the second hurdle.

2.4.1 Model outline

Section A.4 presents the equilibrium conditions; here, I focus on features that differ from the baseline. For clarity, Figure 4 depicts the potential paths that a start-up may take, omitting the unproductive development states for readability.

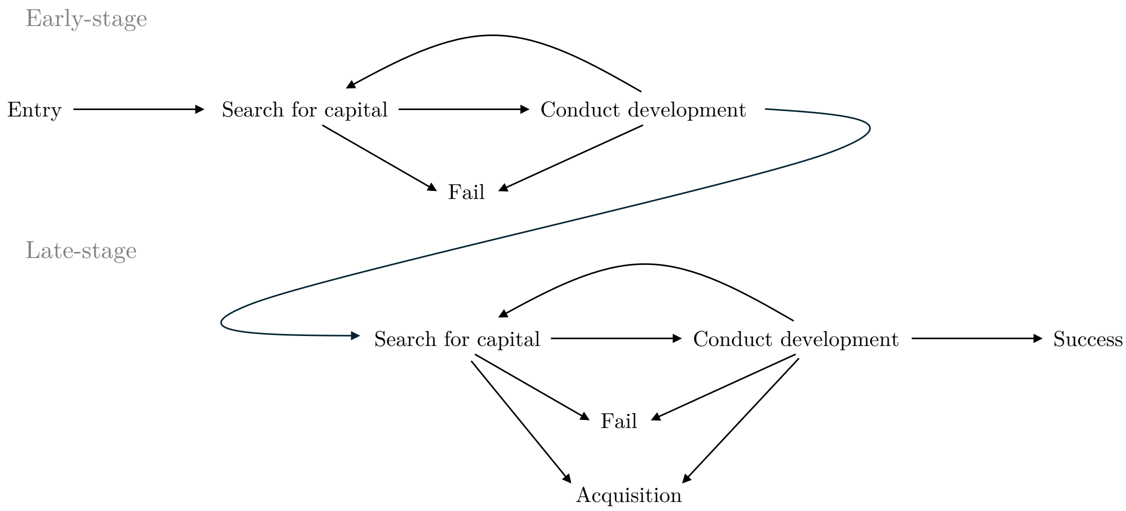

Figure 4: Start-up lifecycle

For ease of exposition, the two states "productive development" and "unproductive development" have been combined into "conduct development".

The early-stage The early-stage has the same structure as the baseline model. Start-ups enter with value $V_{s,e}$. In search, they fail at stage-agnostic rate λ, and meet a VC at stage-specific rate ν_e. Conditional on meeting with a VC, the start-up signs a contract with contract rate ω_e, which provides funding for the early-stage only. In the early-stage, investing at flow rate k_e realises a result at stage-agnostic rate κ, which is successful with probability p_e/p_l, where p_l is the probability of overcoming late-stage uncertainty, conditional on being in the late-stage and obtaining a result.[35] Success in the early-stage leads the start-up to search for late-stage funding.

The late-stage Start-ups enter with value $V_{s,l}$. During search, they fail at rate λ and finds a VC at rate ν_l. In general, $\nu_l \neq \nu_e$. Conditional on meeting with a VC, the start-up signs a contract with contract rate ω_l, which provides funding for the late-stage

[35]This convention means that, in a frictionless world, each start-up in the early-stage would be successful with probability $p_e = (p_e/p_l) \times p_l$.

only. In the late-stage, investing at flow rate k_l realises a result at stage-agnostic rate κ, which is successful with probability p_l.

The key difference is that start-ups in the late-stage may be acquired. An acquisition is the sale of a partially-developed project and may occur in search or development. In both states, potential acquirers arrive at rate ϕ, which is exogenous and the same for all firms. The acquirer values the fully-developed project possessed by the start-up at $\epsilon \times \pi$, where $\epsilon \in [0, \infty)$ is a random variable with CDF $G(\cdot)$ and that ϵ is match-specific, common knowledge and *i.i.d.* across matches. However, the acquirer purchases the start-up before it has reached full development so there is residual uncertainty. In the late-stage, a start-up's project would succeed with probability p_l if fully-funded, so the acquirer's valuation for the project is $\epsilon\, p_l\, \pi$. This is known to the start-up.

The start-up makes a take-it-or-leave-it offer and extracts the acquirer's valuation, $\epsilon p_l \pi$, whenever it exceeds the start-up's continuation value, $V_{i,l}$, for $i \in \{s, d\}$.[36] The probability that an acquisition occurs conditional on a meeting is given by

$$\text{Development:} \qquad Pr(\epsilon\, p_l\, \pi \geq V_{d,l}) = 1 - G(V_{d,l}/p_l\pi)$$

$$\text{Search:} \qquad Pr(\epsilon\, p_l\, \pi \geq V_{s,l}) = 1 - G(V_{s,l}/p_l\pi)$$

which permits defining an effective arrival rate of acquisitions $\hat{\phi}_i = \phi \cdot [1 - G(V_{i,l}/p_l\pi)]$ for $i \in \{s, d\}$. The effective arrival rate of acquisitions modifies the arrival rate of potential acquisitions by the acceptance rules outlined above and so is endogenous. In the quantification, I will impose $\epsilon \sim Exp(1/\xi)$, so that ξ is the average 'synergy'. With this in mind, the HJB equations for late-stage search and development are given by

$$\text{Development:} \qquad \rho V_{d,l} = \kappa\big[p_l\pi - V_{d,l}\big] + \omega_l\big[V_{s,l} - V_{d,l}\big] + p_l\pi\xi\phi \exp\Big\{-\frac{V_{d,l}}{p_l\pi\xi}\Big\} \tag{16}$$

$$\text{Search:} \qquad (\rho+\lambda)V_{s,l} = \nu_l\big[V_l^M - V_{s,l}\big] + p_l\pi\xi\phi \exp\Big\{-\frac{V_{s,l}}{p_l\pi\xi}\Big\} \tag{17}$$

where V_l^M is the value of a meeting in the late-stage, which is defined analogously to the baseline model. Relative to the baseline model, the final term in each equation

[36] Without a theory of acquisition demand, little is gained by allowing bargaining over the surplus.

captures the expected flow value of acquisition opportunities.

Acquisitions also alter the expected capital cost associated with a contract. With acquisitions, a firm in productive development experiences an event – a result, acquisition, or funding exhaustion – at rate $\kappa + \omega_l + \hat{\phi}_d$. I assume that the contract written between the start-up and VC is such that the VC terminates funding when the firm is acquired. For an unproductive start-up, the entrepreneur can prevent the acquirer from conducting the due-diligence to learn ϵ, because they have control rights. Therefore, the VC is left to honour their commitment, which expires at rate ω_l. The implication is that the expected capital cost associated with contract rate ω_l is

$$K_l(\omega_l) = \left(1 + \frac{\kappa\,(1 - p_l)}{\rho + \omega_l}\right) \frac{k_l}{\rho + \kappa + \omega_l + \hat{\phi}_d} \tag{18}$$

The optimal contract is then found by solving the contract problem, analogous to equation (5), where the value functions are given by equations (16) and (17) and the capital cost by (18). The full equilibrium conditions are described in section A.4.

Acquisitions and financing Introducing acquisitions delivers a simple insight: start-ups are more likely to accept offers while searching for capital than while funded.

Proposition 6. *Given that the optimal contract rate is finite, $\omega_l < \infty$, the effective arrival rate of acquisitions is higher in search than in development: $\hat{\phi}_s > \hat{\phi}_d$.*

The result follows because a start-up has a lower continuation value while searching for capital than while funded and it therefore accepts acquisition offers at lower valuations, so that a larger share of acquisition opportunities results in trade. In this sense, acquisitions substitute for financing by providing an alternative exit route, which is more valuable when financing is tight.

For the quantification, I allow success probabilities, investment costs, and meeting rates to differ across stages. I restrict the rate of uncertainty resolution, κ, and the failure rate in search, λ, to be common across stages, limiting the number of parameters that must be separately identified. Allowing $k_e \neq k_l$ accommodates the larger capital requirements of later-stage firms (Gompers, 1995), while allowing $\nu_e \neq \nu_l$ permits access to capital to improve as information about the start-up accumulates.

3 Quantification for the US VC market

The section has three objectives. First, I show how funding histories and exits identify the model's primitives. Second, I validate the model's ability to replicate key features of the data, which leads to the insight that a model with *ex ante* homogeneous start-ups can capture rich *ex post* heterogeneity in start-up outcomes and funding histories. Third, I use the estimated model to quantify distortions in the US market. The estimates imply that 40% of VC-backed start-ups shut down prematurely. Absent frictions, the share of successful exits rises from 30% to 50%. Furthermore, 12% of capital is held by unproductive start-ups while 23% of viable start-ups are searching for funding.

3.1 Identification strategy

This section discusses the identification strategy. I begin with a simple identification result for the baseline model, which shows that a set of readily observable moments is sufficient to identify the parameters of the baseline model of section 2.2. For the quantitative model, I use a richer, overidentifying set of moments, supplemented with parameters estimated from auxiliary duration models, but the identification logic carries over from the simpler setting.

3.1.1 Identification of the baseline model

In the baseline model, the objective is to recover estimates for $\theta = (p, k, \kappa, \pi, \lambda, \nu)$, and $\tilde{\sigma}$.[37] To this end, consider the following vector of population moments

$$\mathbf{m} = \big(E[\mathbb{1}\{\text{success}\}], E[N_f], E[T_{br}], E[\text{time-to-exit}], E[\text{burn rate}], E[\text{exit multiple}]\big) \tag{19}$$

where $E[\mathbb{1}\{\text{success}\}] = p_d$ is the probability of success for funded firms, $E[N_f]$ and $E[T_{br}]$ are, respectively, the expected number of funding rounds and the expected duration

[37]Unlike labour-market data, VC data do not reveal the stocks searching on either side; I therefore estimate partial-equilibrium meeting rates rather than the matching function.

between funding rounds, and E[time-to-exit], E[burn rate] and E[exit multiple] are, respectively, the expected time from first funding to a successful exit, the expected burn rate and the expected exit multiple; see section A.3.6 for closed-form expressions.

Let $\Theta = [0,1] \times \mathbb{R}^5_+$ denote the parameter space, where $\theta \in \Theta$, and denote by $\Theta^{adm} \subset \Theta$ the subset of the parameter space for which the optimal contract rate, ω, is positive and finite, and so all moments $\mathbf{m}$ are well-defined. Then, for $\theta \in \Theta^{adm}$, denote by $\mathbf{m} \in [0,1] \times \mathbb{R}^5_+$ the vector of population moments implied by the model. Letting $\mathcal{M}^{\text{adm}} = \mathbf{m}(\Theta^{\text{adm}})$ denote the corresponding set of admissible moment vectors, and defining $G : \Theta^{\text{adm}} \to \mathcal{M}^{\text{adm}}$ by $G(\theta) = \mathbf{m}(\theta)$, the following proposition demonstrates that the set of moments $\mathbf{m}$ is sufficient to recover the parameters θ uniquely.

Proposition 7. *The mapping $G : \Theta^{\text{adm}} \to \mathcal{M}^{\text{adm}}$ is injective. In particular, for any admissible moment vector $\mathbf{m} \in \mathcal{M}^{\text{adm}}$ there exists a unique $\theta \in \Theta^{\text{adm}}$ such that $G(\theta) = \mathbf{m}$.*

The parameters (λ, ν, κ) affect the timing of events directly, whereas (p, π, k) influence timing only indirectly via contract rate, (p, π, k). Information on the former comes from moments that encode the frequency and timing of events, which include the expected time-to-exit, time between rounds, and number of rounds. However, these moments only allow for recovering the composite transition rates, $\kappa + \omega$ and $\lambda + \nu$, and the implied cycling intensity between search and productive development, $\nu\omega$.[38]

To isolate the role of primitives, (λ, ν, κ), from contracts, ω, in driving these moments, it is necessary to choose moments that allow for the optimal contract to be recovered, which requires choosing moments that load explicitly on (p, k, π). For instance, many rounds in quick succession – high $E[N_f]$ and low $E[T_{br}]$ – could reflect a high meeting rate, ν, or a high contract rate, ω. Distinguishing between these explanations requires information on the primitives that determine the optimal contract: if (p, k, π) indicate a high-quality project for which the optimal contract rate is low, then observing many rounds in quick succession must reflect a high meeting rate, not a high

[38]Other moments that depend purely on the transient block of the continuous-time Markov chain governing movement between productive development and search before project resolution or closure could also be used. However, because (p, π, k) enter this block only through ω, such moments provide no direct information about these primitives.

contract rate. The success probability, expected burn rate, and expected exit multiple are natural candidates to inform (p, k, π).[39]

The particular set of moments used here is sufficient but not necessary. More generally, identification requires moments that recover the transition structure governing funding histories, together with moments that provide direct information on project success, costs, and payoffs.

With θ identified, V_s is given by equation (13). The entry rate is $V_s/\tilde{\sigma}$, which suggests that $\tilde{\sigma}$ can be backed out given the empirical entry rate. In practice, the model's notion of entry is unobserved, but the condition determining the flow of first funding rounds can be used instead. In steady state, it follows that

$$\tilde{\sigma} = \Big(\frac{\nu}{\lambda + \nu} \times \frac{1}{\text{flow first-time rounds}}\Big) \times V_s \tag{20}$$

which enables recovery of $\tilde{\sigma}$. See section A.3.6 for the derivation.

3.1.2 Identification in the quantitative model

The baseline result provides a guide to identifying the quantitative model, which adds two development stages and acquisitions and is estimated on censored data. As a result, I do not provide a formal identification result but instead carry forward the central insight that funding histories discipline the transition dynamics, while outcome and capital moments provide direct information on project success, costs, and payoffs. I adapt the moment set to the richer environment and move to an overidentified specification.

Acquisitions Acquisitions introduce the parameters (ϕ, ξ) into the late-stage. The acquisition block itself depends jointly on (p, π, ϕ, ξ). As a result, many targeted moments respond similarly to changes in any of (p, π, ϕ, ξ); see Figure C.1. Three moments

[39] One could use observed capital injections for k, rather than the burn rate. In practice, the burn rate is cleaner: the capital injection itself is non-monotonic (Corollary A.1), making it less than ideal (see e.g. Strebulaev and Whited 2012). The issue is that, although k directly raises the capital injection, $\bar{K}$, it causes a reduction in the funding horizon, which may dominate. The burn rate is preferable because it normalises by the time between rounds. For π, using the exit-multiple, rather than the exit valuation directly, is more appropriate when estimating an *ex ante* homogeneous firm model because it better compensates for the implicit pooling of firms that may differ in their levels of both k and π.

help to disentangle these parameters. First, exit mode separates (p, π) from (ϕ, ξ): higher (p, π) increase the success rate, while higher (ϕ, ξ) mainly shifts mass toward acquisitions. The acquisition-to-success ratio therefore helps distinguish the two blocks. Second, with acquisitions the overall exit multiple is less useful for separating p from π because acquisition prices are anchored to $p \times \pi$, not π alone. I therefore target the mean exit multiple for successes only. Finally, separating ϕ from ξ is difficult because both raise the incidence of "quality-adjusted" offers. Using information on the shape of the distribution of acquisition exit multiples provides a route forward: the share of high-exit-multiple acquisitions rises with ξ, but is relatively insensitive or decreasing in ϕ; see Figure C.1. I include the share of acquisition multiples that exceed ten (10X).

Development stages In the early stage, (ν_e, k_e, p_e) are stage-specific, while (κ, λ) are shared across stages. Early-stage "success" is progression to the late stage: conditional on completing early-stage R&D, firms progress with probability p_e/p_l. I discipline p_e/p_l using progression moments; with p_l pinned down by late-stage moments, these targets discipline p_e. Specifically, I target (i) the share of firms that reach the late stage and (ii) the share of firms with exactly one round.[40] I also include the mean number of early-stage rounds and the early-stage mean burn rate (winsorised at 5% in both tails). To sharpen identification, I also add an auxiliary duration target implied by the model. Within stage, inter-round durations are hypoexponential; for two successive early-stage rounds, $T_{br}^{e \to e} \sim \text{Hypo}(\kappa + \omega_e, \lambda + \nu_e)$. Fitting this distribution by MLE to observed early-stage inter-round durations provides direct information on $\lambda + \nu_e$.[41]

Late-stage targets are analogous. I include the mean number of late-stage rounds among firms that progress to the late-stage, the late-stage mean burn rate (5% winsorised), and a late-stage auxiliary duration target: with acquisitions, $T_{br}^{l \to l} \sim \text{Hypo}(\kappa + \omega_l + \hat{\phi}_d, \lambda + \nu_l + \hat{\phi}_s)$; I therefore target the MLE-based estimate of $\lambda + \nu_l + \hat{\phi}_s$.

Finally, following section 3.1.1, I also target the mean time-to-exit. I also target the

[40]In the two-stage model, the total (i.e. across stages) number of rounds is no longer geometric; it responds directly to p_e/p_l, beyond its indirect effect via ω_e.

[41]This auxiliary target is a convenient way to use within-stage timing information. Because the two hypoexponential rate parameters are unlabelled in the likelihood, I obtain (h_1, h_2) and target $\max\{h_1, h_2\}$ as the counterpart to $\lambda + \nu_e$. See section C.1.

share of firms with a funding round more than five years after the first, which helps to ensure that the firm attrition in the model aligns with the data. This is important given that I estimate the model on censored data. Table C.2 reports moment elasticities around the estimates. I leave the share of successful exits untargeted for validation.

3.2 Data and estimation

3.2.1 Dataset construction

The sample includes all firms in the Thomson Reuters venture capital dataset from the US that received their first round of venture capital funding between 2005 and 2015, inclusive. Raw data is censored based on the date of first funding, so to maintain consistency across the data, I exclude any firm observations that occur more than seven years after their first funding round, following Ewens and Farre-Mensa (2020).

I construct the dataset as follows. I keep financing rounds labelled as "seed", "early stage", "expansion", or "later stage", and denote seed and early-stage rounds as "Early" and expansion and later-stage rounds as "Late". I exclude any firms that only receive "Late" funding. I complement the data with market capitalisation data from CRSP for IPOs and M&A price data from Crunchbase when price data from Thomson Reuters is unavailable. All values are in 2015 USD. I impute missing investment amounts, following Jagannathan et al. (2022), and set missing deal values equal to 1.5 times the total investment, following Kerr et al. (2014).[42]

Finally, I need to map modes of exit in the data to the notion of "success" and "acquisition" in the model. I define an acquisition in the data as an M&A transaction by a non-financial buyer and a success as an IPO or an M&A transaction by a financial (including SPAC) buyer, reflecting the idea that non-financial buyers typically have the capabilities to acquire start-ups at pre-commercial stages, whereas financial buyers do not. The final sample includes 11,030 firms and 33,158 funding rounds. Among these firms, 400 were successful and 2,034 were acquired within the same period. Section B.1 provides further details on data construction and Table B.3 includes summary statistics.

[42]Imputation applies to about 10% of US rounds (and slightly more for the UK); see section B.1.

3.2.2 Estimation

To maintain consistency with the data construction procedure, I censor the simulated data in the same way before computing moments. I fix the discount rate $\rho = 0.08$. A unit of time in the model corresponds to a year. The remaining parameters are estimated in two steps. First, I jointly estimate all parameters except for $\tilde{\sigma}$, denoted by θ, by minimising the criterion function

$$C(\theta) = \sum_j \left(\frac{\tilde{m}_j - m_j(\theta)}{\frac{1}{2}|\tilde{m}_j| + \frac{1}{2}|m_j(\theta)|} \right)^2 \tag{21}$$

where $\tilde{m}_j$ is the empirical moment (or auxiliary parameter) and $m_j(\theta)$ its model counterpart computed at θ.[43] Second, I recover $\tilde{\sigma}$ from the counterpart to equation (20) for the full model, where I set the flow rate of initial rounds to 1.00 for the US.

Parameter estimates Table 1 reports the parameter estimates and standard errors. Parameter estimates should be interpreted as the values for the average start-up.

The average start-up requires approximately eight years of funding to fully commercialise ($\approx 2 \times 1/\kappa$), consistent with typical VC timelines. It requires annual investment of $k_e =$ \$4.16Mn for the first half and $k_l =$ \$16.15Mn for the second half. If successful in scaling independently, its value is $\pi =$ \$218.18Mn. However, the start-up may fail due to technological or commercial challenges: conditional on uncertainty resolution, this happens with probability $1 - p_e/p_l = 23\%$ in the early-stage and $1 - p_l = 70\%$ in the late-stage. In the late-stage, acquisition opportunities arrive at the rate of one every 18 months ($\approx 12/\phi$) and the average offer price is approximately \$50Mn ($\approx p_l \pi \xi$).

However, financing frictions hinder start-ups' development. In the early-stage, the probability that a viable firm in search fails before securing funding is $\lambda/(\lambda + \nu_e) = 20\%$ and the average duration spent searching is $12/(\lambda + \nu_e) = 3.8$ months. In the late-stage,

[43] The criterion function sums over squared percentage deviations between the empirical and simulated objects with equal weighting. The choice of equal, rather than optimal, weighting scheme reflects the need to utilise variation in certain moments, specifically coming from exit payoffs, in order to identify the model's parameters. For instance, the sample contains over 30,000 funding rounds but only about 400 successes. Because the exit multiple is key for identifying π, equal weighting ensures this low-frequency variation is not dominated by high-frequency moments.

Parameter		Estimate	Std. Error
Discount rate	ρ	0.08	-
R&D milestone arrival rate	κ	0.26	0.01
Failure rate in search	λ	0.62	0.04
Early-stage development level	p_e	0.23	0.01
Early-stage flow investment cost ($Mn USD/year)	k_e	4.16	0.18
Early-stage meeting rate	ν_e	2.54	0.13
Late-stage development level	p_l	0.30	0.01
Late-stage flow investment cost ($Mn USD/year)	k_l	16.15	0.25
Late-stage meeting rate	ν_l	5.20	0.26
Success payoff ($Mn USD)	π	218.18	15.40
Acquisition offer arrival rate	ϕ	0.65	0.07
Mean acquisition 'synergy' ($\varepsilon \sim \text{Exp}(1/\xi)$)	ξ	0.74	0.08
Entry-cost parameter	$\tilde{\sigma}$	2.68	0.37

Table 1: US parameter estimates

The table reports parameter estimates from the model in section 2.4. Estimation uses a simulated sample of 220,600 firms, $S = 20$ times the size of the empirical sample. $\rho = 0.08$ is fixed externally. Parameters (except $\tilde{\sigma}$) are estimated by minimising (21) and I report asymptotic standard errors, where the covariance matrix of moment conditions is computed from 1,000 bootstrap samples, clustered at the firm level (see section C.1 for details). To avoid local minima in the estimation, I initialise a simulated annealing algorithm with 200 (randomly selected) starting points. With the resulting candidate parameter estimates, I run a local minimisation (Nelder-Mead) using the results from the first step as initial guesses. With $S = 20$, simulation noise is miniscule, so I report standard errors that simply reflect sampling variability (applying an adjustment raises standard errors by $< 1\%$). With these estimates, the entry cost parameter, $\tilde{\sigma}$, is recovered from the condition determining the flow rate of first-time funding rounds. To compute its standard error, I take 1,000 samples from the asymptotic distribution of parameter estimates, solve the model, and recover the implied value for $\tilde{\sigma}$. The standard error is the standard deviation of $\tilde{\sigma}$ among this sample. All amounts are in 2015 USD.

financing conditions are more favourable: the probability that a viable firm in search fails before securing funding falls to 10% and securing funding takes an average of just 2 months. The improvement in conditions is consistent with greater information as the start-up develops. Finally, the estimates suggest that the average start-up can survive 19 months ($\approx 12/\lambda$) after exhausting its previous capital injection before shutting down.

The estimated financing-market meeting rates, ν_e and ν_l, benefit from external validation. To the best of my knowledge, Ewens et al. (2022) provide the only comparable estimates. Entrepreneurs and VCs meet only for first-round funding in their model and the relevant comparison is their estimated deal frequency, which averages 1.57 per year and ranges from 0.16 to 3.56 across entrepreneur-quality deciles (Table 6, Ewens et al. 2022). My estimate of $\nu_e = 2.54$ lies within this range and above the average, which

Panel A: Targeted moments		
Moment	Data	Model
Mean # early rounds	1.98	1.91
Mean # late rounds | progressed to late	2.46	2.43
Share one round (%)	32.81	26.47
Mean duration b/w rounds (years)	1.16	1.25
Estimate of $\lambda + \nu_e$	3.30	3.16
Estimate of $\lambda + \nu_l + \hat{\phi}_s$	5.94	6.15
Share funded after year 5 (%)	19.68	18.25
Mean burn rate | early round ($Mn/year)	7.91	7.49
Mean burn rate | late round ($Mn/year)	23.78	24.89
Share to late-stage (%)	41.56	41.86
Mean time-to-exit (years)	3.67	3.33
Acquisition-to-success ratio	5.08	5.09
Mean exit multiple | success	9.24	9.06
Share acquisition multiples > 10X (%)	8.01	8.17
# first funding rounds	1.00	1.00
Panel B: Untargeted moments		
Share successful (%)	3.63	3.78
Share acquired (%)	18.44	19.24
Mean time-to-success (years)	4.44	3.40
Mean time-to-acquisition (years)	3.52	3.31
Mean # funding rounds†	3.01	2.93
Mean exit multiple	5.79	5.33

Table 2: Targeted and untargeted moments

Panel A shows 15 moments (including two auxiliary parameter estimates) used in the estimation procedure. Empirical moments are computed from the sample of 11,030 firms. Other than for the estimates of $\lambda + \nu_e$ and $\lambda + \nu_l + \hat{\phi}_s$, simulated moments are computed using the parameter values given in Table 1 to simulate a sample of 220,600 ($= 20 \times 11,030$) firms. For $\lambda + \nu_e$ and $\lambda + \nu_l + \hat{\phi}_s$, the parameter values are taken directly to reduce computational burden. In computing the empirical and simulated moments, data on a given firm is censored seven years after its first funding round. Panel B shows a set of untargeted moments. † The overall mean number of rounds is not targeted directly, but is implied by the targeted stage-specific round counts and late-stage progression share.

directionally sensible: their estimates concern first-round funding, whereas my estimate is disciplined by re-matching for follow-on capital, which may be easier to secure once a start-up has obtained initial VC backing. The comparison supports the magnitude of my estimate and suggests that I am not overstating early-stage matching frictions.

Model fit and validation Panel A of Table 2 reports the targeted moments, which the model captures well. Panel B reports untargeted moments. Although the acquisition-to-success ratio is targeted, their levels are not, and the model closely matches both shares. It also captures the mean exit multiple and correctly predicts that successes occur later than acquisitions, although successes do arrive too early.

Figure 5 assesses model fit across funding histories, capital intensity, and exits. Panel A reports information on funding rounds. Three points are noteworthy. First, Panel A (i) shows that the model replicates the declining distribution of the number of rounds. This is not mechanical in the two-stage model: if most project uncertainty were concentrated in the late stage, firms would typically progress and raise at least two rounds. Matching the declining distribution therefore disciplines how uncertainty is distributed across stages. Second, Panel A (iii) shows that the model closely replicates the average time from the first funding round to each subsequent round. In the model, this profile must be upward sloping because firms are *ex ante* homogeneous, but it need not be upward sloping empirically when firms differ *ex ante*. The upward slope in the data therefore suggests that such unmodelled heterogeneity is not excessive. Third, Panel A (iv) shows that the model replicates funding-continuation dynamics, which is important because the model is estimated using censored data.

Panel B assesses the model's ability to capture capital intensity. Because capital injections are constant within stage, the model cannot reproduce the substantial within-stage variation in round sizes observed in the data. Accordingly, Panel B (i) shows that the model misses the widening dispersion in cumulative capital raised across rounds, but closely tracks the median and its slope. Panel B (ii) shows that the model nevertheless captures the skewness of the burn-rate distribution. Because variation in model-implied burn rates is driven primarily by differences in the duration between rounds rather than round sizes, this fit suggests that *ex post* heterogeneity – shocks occurring after firms raise capital – is an important driver of funding dynamics.

Finally, Panel C (i) shows that the model reasonably captures the skewness in start-up outcomes, a key feature of the market (Hall and Woodward, 2010). Panel C (ii) shows that it also captures the distribution of exit timing, although this fit is driven primarily by acquisitions (see Table 2, Panel B).

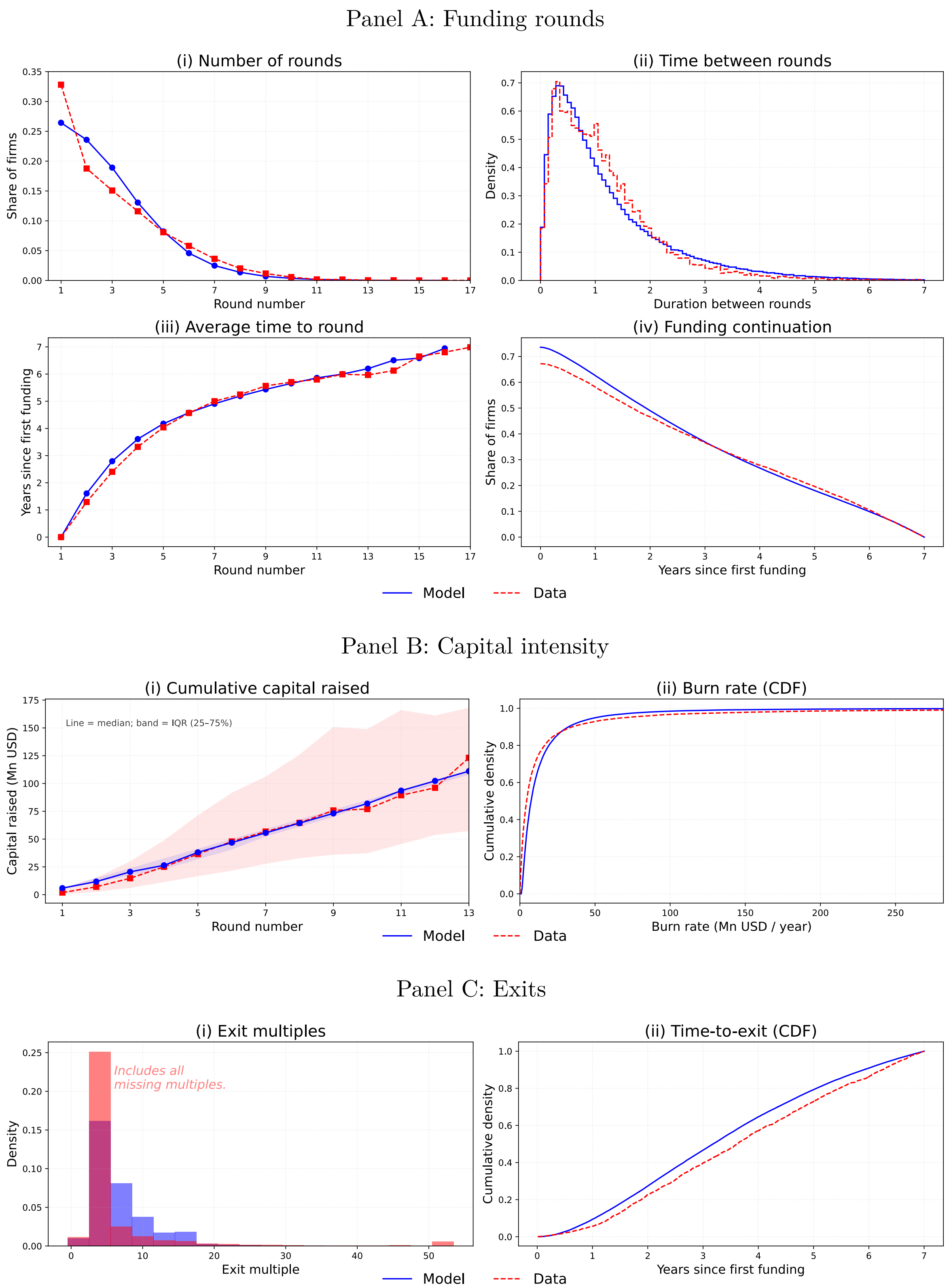

Figure 5: Model validation

The figure shows outcomes in the model and data across funding rounds (panel A), capital intensity (panel B) and exits (panel C). All empirical and simulated data is censored at seven years following a firm's first funding round. "Funding continuation" in Panel A (iv) reports the share of firms that raise capital more than x years after their first funding round. To aid visualisation, cumulative capital raised is shown up to the 13th round; the burn rate CDF is reported up to the 99th percentile; and exit multiples are winsorised at 50X. Missing empirical exit multiples are set to 1.5X, following Kerr et al. (2014).

3.3 Analysis of the US venture capital market

In this section, I use the estimated model to assess distortions in the US venture capital market. I first decompose start-up outcomes to measure how often viable projects shut down before completion. I then compare entry incentives and outcomes in the frictional equilibrium with a frictionless benchmark in which wealthy entrepreneurs can self-finance.[44] Finally, I quantify capital misallocation in the US market.

Financing distortions Panel A of Table 3 reports a decomposition of start-up outcomes. Column two reports model-implied outcomes from simulated data, censored in the same way as the estimation sample. The estimates suggest that 17.2% of VC-backed start-ups remain active seven years after their first funding round, without having had a successful exit. Column three reports uncensored results. Approximately 70% of funded firms fail, but the estimates suggest that approximately 40% ($\approx 25.7 + 15.1$) are forced to shut down prematurely, before uncertainty is resolved. Most of these failures occur in the early-stage, consistent with more severe matching frictions at this stage.

These estimates suggest there are substantial issues in financing young, innovative firms. However, caution is required in interpreting them. Premature closure does not imply that these firms would necessarily have succeeded with additional funding. Rather, it indicates that positive-NPV projects shut down before uncertainty is resolved. Panel B evaluates the associated losses using the self-financing counterfactual. In column two of Panel B, I report the success and acquisition distortions, defined analogously to those in section 2.3 of the baseline model. Specifically, the success distortion, Δ_p, is the wedge between the actual success rate and the success rate when a wealthy entrepreneur can self-finance.[45] Analogously, the acquisition distortion, Δ_a, is the wedge between the actual acquisition rate and the acquisition rate under self-financing. Both of these distortions condition on receiving at least one round of funding, mirroring my empirical sample. The probability of success is approximately $\Delta_p = 56\%$ lower when the

[44] See section A.4.1 for the relevant equations. Equivalently, the frictionless benchmark obtains when $\nu_e \to \infty$ and $\nu_l \to \infty$; see Lemma 1.

[45] This is analogous to the "outcome distortion" defined in section 2.3. I refer to it here as a "success distortion" because the full model also includes a distinct role for acquisitions.

Panel A: Outcomes and modes of failure	Censored	Uncensored
Share successful (%)	3.8	4.9
Share acquired (%)	19.2	24.5
Share no outcome (%)	17.2	0.0
Share failed (%)	59.7	70.6
... in early-stage development (%)	16.1	18.2
... in early-stage search (%)	22.5	25.7
... in late-stage development (%)	9.0	11.6
... in late-stage search (%)	12.2	15.1
Panel B: Distortions	**Agency & Matching**	**Matching only**
Success distortion, Δ_p (%)	55.5	14.9
Acquisition distortion, Δ_a (%)	37.0	5.7
Entry distortion, Δ_s (%)	71.5	31.7
Panel C: Capital misallocation		
Share unproductive | early-stage (%)		12.3
Share unproductive | late-stage (%)		11.1
Share unproductive (%)		12.0
Share capital misallocated (%)		11.7
Share viable firms in search (%)		23.0

Table 3: US market assessment

The table reports start-up outcomes, measures of distortions induced by matching and agency frictions, and capital misallocation for the US estimation. Panel A reports modes of success and failure for firms from data simulated given estimated parameter values. Panel B reports distortions: for an outcome x in the frictional equilibrium, distortions are defined such that $x = (1 - \Delta_x)\bar{x}$, where $\bar{x}$ is the value in the economy without frictions. Distortions relate to the success share, Δ_p, the acquisition share, Δ_a, and the entry value, Δ_s. Column "Agency & Matching" measures the combined effect of both frictions; column "Matching only" removes agency frictions (no hidden failure) while retaining matching frictions. Panel C reports figures on capital misallocation. Within each stage, this is defined as in equation (14). Across stages, the "share unproductive" reports the share of all firms receiving funding that are unproductive, whereas the "share capital misallocated" reports the share of capital allocated to unproductive firms, $(k_e\mu^u_{d,e} + k_l\mu^u_{d,l})/\big((k_e(\mu^u_{d,e} + \mu^p_{d,e}) + k_l(\mu^u_{d,l} + \mu^p_{d,l})\big)$. The share of viable firms in search is computed as $(\mu_{s,e} + \mu_{s,l})/(\mu_{s,e} + \mu^p_{d,e} + \mu_{s,l} + \mu^p_{d,l})$.

entrepreneur has to secure external funding, a 6 percentage point reduction. In other words, if the entrepreneur were wealthy and could self-finance, the share of successful outcomes would rise from 4.9% to 11.0%. Acquisition outcomes are also adversely affected: the $\Delta_a = 37\%$ distortion is equivalent to a 14 percentage point reduction in the probability of acquisition, from 39% to 25%. The positive acquisition distortion implies that the effect of premature closure dominates the tendency for financing constraints to induce entrepreneurs to accept acquisition offers.

Together, the results suggest that financial frictions both inhibit entrepreneurs' ability to scale independently and reduce transfers of technology to other firms. Column three of Panel B reports the same distortions in an economy where entrepreneurs can commit not to engage in inefficient continuation, but must still search for funding for each stage.[46] Abstracting from moral hazard in contracting cuts the success distortion $\Delta_p = 56\%$ to 15% and the acquisition distortion $\Delta_a = 37\%$ to 6%. This suggests that the interaction between agency and matching frictions is quantitatively important.

The success and acquisition distortions allow for understanding the *ex post* effects of financing. To investigate the *ex ante* effects, Panel B also reports the entry distortion, Δ_s, defined, as in equation 13, as the wedge between the actual entry value, $V_{s,e}$, and the entry value under self-financing, $\bar{V}_{s,e}$. The value of the average entrepreneurial project is approximately $\Delta_s = 72\%$ lower when the entrepreneur relies on venture capital financing, compared to a wealthy entrepreneur that can self-finance. Therefore, frictions in access to external finance have large detrimental effects on start-up creation. Converting this into a measure of lost entry requires an estimate of the supply curve of entrepreneurial projects, which is beyond the scope of this paper. Nevertheless, the magnitude of the distortion points to substantial effects on start-up creation.

Capital misallocation In addition to losses from premature terminations, the economy features misallocation: some viable start-ups seek funding, while other start-ups are still having capital from their previous funding round but should have shut down. Panel C of Table 3 reports the main results on capital misallocation. In the aggregate, misallocation can be computed based on the share of unproductive firms receiving capital or the share of capital allocated to unproductive firms. In practice, both figures are similar: 11.7% of capital is misallocated, whereas 12% of firms are unproductive, reflecting the similar degrees of misallocation across stages.

Taken together, these results highlight a key tension in the VC market: premature termination (underinvestment) coexists with inefficient continuation (overinvestment).

[46] In this economy, the entrepreneur and VC can write a state-contingent contract that terminates funding when news about the project arises, which drives the effective development costs back to k_i for $i \in \{e, l\}$.

On the one hand, roughly 40% of projects are closed down prematurely while 23% of viable projects are actively searching for capital. On the other hand, a substantial share ($\approx 12\%$) of capital is tied up in firms that have already ceased to create value.

3.4 Robustness

This section discusses robustness to the estimation procedure. The first set of concerns regard the sample period and censoring horizon. The baseline sample covers first financings in 2005–2015 with seven-year censoring, so events are observed through end-2022, which includes the sharp run-up in US VC activity culminating in 2021 and easing in 2022. To ensure this is not driving parameter estimates, I re-estimate the model on a 2005–2010 subsample, maintaining seven-year censoring. Table C.3 reports the results and shows that parameter estimates are of comparable magnitude. The earlier time-period also permits a test of the effect of censoring and I re-estimate the model for the 2005–2010 subsample with twelve-year censoring. Table C.3 shows that parameter estimates are very similar. Figure C.2 provides a further diagnostic: using the two estimated parameter vectors, I simulate data, apply censoring at $h \in \{3, \ldots, 12\}$, and recompute the targeted moments. The simulated moments line up closely across h, including for $h > 7$, which is out of sample for the seven-year censored estimation.

A second concern is the decision to winsorise burn rates. Empirical variation in burn rates reflects both the size of capital injections and the duration bewteen rounds, whereas only the duration between rounds varies in the model within each stage. Consequently, extremely large deal values add mass to the right tail of the burn rate distribution that the model does not explicitly capture. I winsorise in the estimation to prevent tail observations, which would primarily identify the *variance* of k_i for $i \in \{e, l\}$ if it were modelled, from having undue influence on average *level* of capital costs across firms. Figures C.3 and C.4 report estimates and model fit across alternative winsorisation thresholds. Overall, parameter estimates are stable across thresholds; only (k_e, k_l) and π adjust mechanically as less winsorisation retains more right-tail mass, raising mean burn rates and the payoff required to match the mean success multiple.

4 Venture capital across countries: US versus UK

In this section, I use the model to diagnose the sources of UK–US disparities and evaluate policy interventions. I find that access to capital and a thin acquisition market are quantitatively important constraints on UK start-up activity, rather than differences in the characteristics of funded projects. I then conduct a budget-neutral counterfactual that reallocates funding across stages and show that reallocating capital from the early- to late-stage can improve outcomes on the margin.

4.1 Market failure or limited investment opportunities?

Table 4 provides an overview of key differences between the US and UK markets for the period 2005-2015; see Table B.3 for further summary statistics.[47] On a per-capita basis, there are twice as many US VC-backed start-ups, and they have superior outcomes. In what follows, I use the model to separate competing explanations for these differences.

Country	Firms	Successes	Acquisitions
US	11,030 (1.00)	400 (3.6%)	2,034 (18.4%)
UK	1,066 (0.48)	26 (2.4%)	97 (9.1%)

Table 4: US–UK Comparison

The table provides an overview of key aggregate statistics for the US and UK. Counts shown first; values in parentheses are normalised figures. For 'Firms' the value is the population-adjusted ratio to the US (US = 1.00). For 'Successes' and 'Acquisitions' it is the percentage of firms. All data is censored at 7 years following a firm's first funding round.

4.1.1 Estimation

Identification The model separates financing frictions from other factors because each mechanism leaves distinct footprints in funding histories. Funding patterns are particularly indicative: all else equal, if UK underperformance reflected riskier or lower-quality projects, we should expect to see more frequent, shorter funding rounds.[48] Conversely, if the funding environment is the constraint, funded start-ups secure longer-

[47] Axelson and Martinovic (2016) analyse exit patterns at the European and the US level.

[48] Riskier projects benefit from a wait-and-see approach, due to increased agency costs, leading to frequent capital injections. Conversely, lower-quality projects face a lower marginal benefit of insurance against financing risk, so optimally lean towards shorter-term funding commitments.

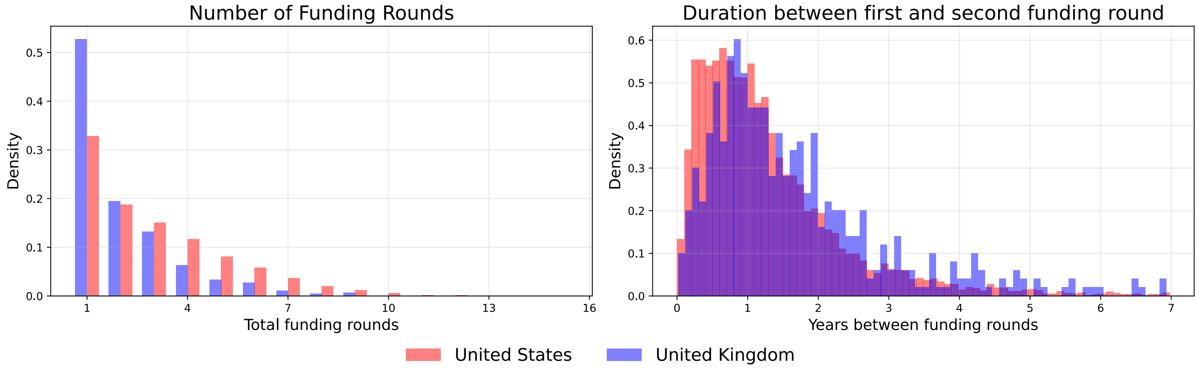

Figure 6: US versus UK funding patterns

The figure shows funding patterns for the UK and US samples. The left plot shows the distribution for the number of funding rounds for US firms (red) and UK firms (blue). The right plot shows the distribution of the duration of time between the first and second funding round for firms in each country. Focussing on the first-to-second round implies that data is uniformly censored; this choice is otherwise immaterial. All data is censored at 7 years following a firm's first funding round.

duration commitments to insure against returning to the market.

Figure 6 compares the number of funding rounds and duration between rounds; see Table C.6 for the full set of UK moments. On average, UK VC-backed start-ups complete fewer rounds within the first seven years of funding (2.1 vs 3.0) with longer intervals between rounds (17.5 months vs 13.9 months). Greater project quality in the UK or longer-horizon projects could also generate this pattern, so funding histories alone cannot distinguish these possibilities from tighter financing conditions. The separate estimation below uses the full set of targeted moments to do so.

Nevertheless, the comparison is suggestive of tighter financing conditions. Consistent with this interpretation, when US capital supply increased after the 1979 ERISA "prudent man" clarification – associated with a large inflow of VC funding (Gompers and Lerner, 1999) – cohorts first funded thereafter exhibited more funding rounds and shorter intervals between rounds than earlier cohorts (Figure B.2). Seen through the model, this is exactly the footprint of easing financing conditions and aligns with my interpretation of US–UK differences.

Returning to the UK, tighter financing conditions coincide with substantially worse acquisition outcomes. Tighter financing could reduce acquisitions mechanically by preventing firms from reaching the late stage. However, UK start-ups are around one-fifth

Parameter	US	UK	Δ(US–UK)
Early-stage meeting rate, ν_e	2.54 (0.13)	1.09 (0.19)	1.45^{***}
Late-stage meeting rate, ν_l	5.20 (0.26)	2.28 (0.41)	2.92^{***}
Acquisition-offer arrival rate, ϕ	0.65 (0.07)	0.33 (0.07)	0.32^{***}

Table 5: Key US–UK parameter estimates

The table reports the key parameter differences between the US and UK estimates. Full parameter estimates and details of the UK estimation are reported in section C.2. Significance levels are: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

less likely to raise late-stage funding than US start-ups (33% versus 42%), but only half as likely to be acquired (9.1% versus 18.4%). Moreover, conditional on reaching the late stage, acquisitions substitute for financing (Proposition 6), so tighter financing increases firms' willingness to accept acquisition offers. Taken together, these patterns point to a thinner acquisition market in the UK.

Results The parameter estimates in Table C.4 for the UK confirm this interpretation; Table 5 provides the parameters where the differences are statistically significant.[49] The early and late-stage meeting rates, ν_e and ν_l, and the arrival rate of acquisition offers, ϕ, are lower in the UK and the differences are statistically significant. By contrast, funded-project characteristics do not differ significantly across markets.[50]

The differences in access to financing are economically large. The average time spent searching for early-stage funding is 6.3 months in the UK, compared with 3.8 months in the US; in the late stage, the corresponding figures are 3.6 and 2 months. Each search is also more than twice as likely to end in failure in the UK.[51] The difference in acquisition markets is also economically important. Although there is no discernible difference in offer quality, parameterised by ξ, acquisition offers arrive approximately twice as frequently in the US.

Relative to the US, a clear issue in the UK market is difficulty securing follow-on funding. 53% of UK VC-backed firms do not raise additional funding within seven years

[49] The estimation procedure mirrors that for the US in section 3. Section C.2 replicates the model validation exercises for the UK.

[50] My estimation strategy does not identify start-up characteristics for unfunded projects.

[51] 43% of searches end in failure in the early-stage, and 25% in the late-stage ($=\lambda/(\lambda + \nu_l + \hat{\phi}_s)$); see Tables C.4 and C.6. These numbers compare to 20% and 10% in the US, respectively.

of their first round (Table C.6), compared to 33% in the US (Table 2), and just 33% of UK start-ups raise late-stage funding, compared to 42% in the US. These issues feed into start-up outcomes. UK start-ups are less able to scale independently and therefore place greater value on acquisition opportunities. In the UK estimation, start-ups accept 66% of acquisition offers in search and 49% in development, compared to 50% and 41% in the US, respectively.[52] Tentatively, this suggests that capital may be misallocated between stages: funding fewer projects at the outset but ensuring that follow-on funding is available to those that demonstrate intermediate success may improve outcomes. I consider this counterfactual exercise in section 4.2.

US projects under UK financing To isolate the role of financing conditions, I replace the US early- and late-stage meeting rates with their UK estimates, holding the remaining US parameters fixed; Table C.5 reports the results. This reduces the US entry value from \$3.3Mn to \$1.6Mn and the share of funded firms that succeed from 4.9% to 3.0%, indicating that financing conditions account for a substantial part of the US–UK gap. The stage-specific counterfactuals show that early-stage financing conditions have a larger effect on entry incentives, whereas late-stage conditions have a larger effect on whether funded firms scale independently.

4.2 Policy evaluation

Motivated by the finding that UK start-ups particularly struggle to raise follow-on and late-stage financing, I now conduct the following experiment. Starting from the estimated parameter values from Table C.4, I adjust the meeting rates (ν_e, ν_l) to induce a 5% reallocation of capital from the early- to late-stage, while holding total (flow) investment in the market fixed.[53] This reflects the sort of marginal adjustment that policymakers could induce given existing budgets.[54] I consider two cases regarding the

[52]This is computed as $\hat{\phi}_i/\phi$ for $i \in \{s, d\}$. For the UK, $\hat{\phi}_s = 0.22$ and $\hat{\phi}_d = 0.16$. For the US, $\hat{\phi}_s = 0.33$ and $\hat{\phi}_d = 0.26$. The estimates of ϕ are shown in Table 5.

[53]Total flow investment is defined as the sum of capital investment into start-ups of all stages, $k_e \times (\mu^p_{d,e} + \mu^u_{d,e}) + k_l \times (\mu^p_{d,l} + \mu^u_{d,l})$. This measure captures both "productive" and "unproductive" start-ups and is analogous to the term that enters equation (15), adapted for the two-stage model.

[54]Given recent trends in VC activity, it amounts to between £0.5-1Bn annually; see PitchBook (2025).

Panel A: Policy lever		
Outcome	Fixed Entry	Endogenous Entry
ν_e	0.63	0.48
ν_l	4.91	5.15

Panel B: Outcomes		
Outcome	Fixed Entry	Endogenous Entry
Shares		
Share success (%Δ)	50.94	51.54
Share acquired (%Δ)	20.11	19.68
Share to late (%Δ)	33.25	33.05
Counts		
No. funded firms (%Δ)	-23.76	-23.99
No. successful firms (%Δ)	15.08	15.19
No. acquired firms (%Δ)	-8.43	-9.03
No. firm exits (%Δ)	-3.33	-3.78
Aggregate		
Total flow surplus (%Δ)	29.71	16.09

Table 6: UK budget-neutral reallocation across stages (5% of total investment)

The table reports results from a policy that shifts 5% of total VC investment from the early to late-stage, holding total investment constant. "Shares" are conditional on receiving funding, whereas "counts" also reflect changes in the flow of firms receiving such funding. The intervention is implemented by adjusting the meeting rates, (ν_e, ν_l), while holding total (flow) investment, K^{total} fixed, $K^{\text{total}} = K^{\text{e}} + K^{\text{l}}$ and $K^{\text{i}} = k_i(\mu^p_{d,i} + \mu^u_{d,i})$ for $i \in \{e, l\}$. The intervention is conducted under two assumptions on start-up creation: (i) holding start-up creation constant at its initial value; (ii) allowing start-up creation to adjust endogenously, according to $V_{s,e}/\tilde{\sigma}$. The share reallocated is defined as $(K^l_{\text{new}} - K^l_{\text{init}})/K^{\text{total}}$. Flow surplus is defined analogously to equation (15), but augmented to account for acquisitions and two stages. Specifically, it is given by $\kappa p_l \pi \times \mu^p_{d,l} + \hat{\phi}_d p_l\, \pi E[\epsilon | \epsilon > V_{d,l}/p_l \pi] \times \mu^p_{d,l} + \hat{\phi}_s p_l\, \pi E[\epsilon | \epsilon > V_{s,l}/p_l \pi] \times \mu_{s,l} - K^{\text{total}} - (\tilde{\sigma}/2) \times \Lambda^2$, where Λ is the flow rate of start-up creation. In case (i), this is held constant at the initial value of $V_{s,e}/\tilde{\sigma}$. In case (ii), it equals $V_{s,e}/\tilde{\sigma}$ as $V_{s,e}$ adjusts endogenously to changes in (ν_e, ν_l).

optimal response of start-up creation, either (i) holding start-up creation constant, or (ii) allowing start-up creation to adjust endogenously. These two cases correspond, respectively, to an assumption of a fixed supply of entrepreneurs and to an assumption that start-up creation responds to entry incentives with unit elasticity. Note that in both cases, the total number of firms that receive funding still adjusts because the share of entrants that receives funding depends on ν_e.

Table 6 reports the results of this exercise. The reallocation increases the number of successes by 15%, but leads to a fall in pre-commercial acquisitions by 8-9%, depending on the start-up creation adjustment. Although fewer firms receive initial funding (-24%), the share that reaches the late stage rises (+33%), reflecting both longer early-stage commitments, because reliable late-stage funding raises the value of early-stage experimentation (Kerr et al., 2014; Ewens et al., 2018), and easier access to capital once this intermediate milestone is reached. With more favourable late-stage funding conditions, firms are more likely to scale independently, rather than be acquired at pre-commercial stages. Policymakers then face an extensive-intensive margin trade-off: holding budgets fixed, targeting the early-stage means backing many firms, often with insufficient capital to scale, whereas targeting the late-stage means fewer firms are funded, but for those that show early promise, capital is available to scale-up. In the model, targeting the late stage dominates: total exits fall (−3–4%), but the compositional shift towards successes, which correspond to fully developed projects rather than pre-commercial transfers, increases total flow surplus by 16–30%, depending on assumptions about the entry response.[55]

These results suggest there are gains to a budget-neutral policy that reallocates capital from the early- to late-stage. To ensure these results are not driven by the particular choice of capital reallocation, Figure C.6 reports the effects across steady states as capital is shifted between stages in both directions. The results demonstrate that gains arise from shifting capital from the early- to late-stage, not vice versa, and that the gains are monotonic in the amount of capital reallocated. I focus on the 5% policy because it is economically relevant and close to the practical limit on reallocation: without sufficient early-stage funding, there are too few late-stage firms to absorb additional capital. It is worth noting that the key conclusion – that reallocating existing support to late-stage start-ups improves surplus – relies on start-ups remaining dependent on external finance. If firms could self-finance after receiving a favourable early signal, supporting more early-stage experimentation might instead be optimal, as Aragoneses and Saxena (2025) show in a setting where entrepreneurs can save out of retained earnings. My analysis is complementary and highlights the extreme case in which firms remain unprofitable for many years, which is particularly relevant for the subset of firms that access VC financing; since 2000 in the US, the only year in which a majority of VC-backed IPOs were profitable was 2009 (Figure B.1).

[55] The difference between the two cases is almost entirely accounted for by the change in total entry costs when start-up creation responds.

5 Conclusion

This paper develops a tractable equilibrium model of start-up financing and uses it to assess the US market and unpack the differences in VC activity between the US and UK. In the model, entrepreneurs seek funding from VC, but financing is subject to a classical principal-agent problem: following investment, the entrepreneur learns more about the idea's potential and has an incentive to conceal bad news. To mitigate this problem, the VC limits the size of their capital injection, but these smaller rounds force the entrepreneur back to the financing market repeatedly, and follow-on funding is not always easily forthcoming. The optimal contract balances these two concerns, prompting small, frequent capital injections when the agency friction is severe, and larger upfront commitments when future funding is more uncertain.

The main contribution of the theoretical analysis is to formulate this dynamic problem in a tractable model that admits closed-form solutions. I then show that the feedback from contracting choices into market tightness implies that the VC market is not constrained efficient in general. Furthermore, a key insight is that funding histories and start-up outcomes can be used to learn about *both* the characteristics of the start-ups receiving funding, and the financing conditions they face when raising capital.

This insight allows me to apply the model to assess the US market, and unpack the drivers of the differences between the US and UK markets. In the US, my estimates suggest that approximately 40% of VC-backed start-ups shut down prematurely, while viable firms searching for capital coexist with unproductive firms that continue to receive funding. Financing frictions therefore generate both underinvestment and inefficient continuation.

In the UK, the model attributes lower VC activity to tighter financing conditions and a thinner acquisition market, rather than differences in the characteristics of funded projects. UK start-ups face particular constraints in securing follow-on funding, and I show that a 5% budget-neutral reallocation of funding from the early to late-stage can raise steady-state flow surplus by 16–30%.

During the preparation of this work, the author used ChatGPT/Codex for coding and copy editing. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

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Appendix

A Theory appendix

A.1 The contract

Consider the environment of section 2.2. As discussed in the main text, the entrepreneur (E) will never report negative results. The contract in the body of the paper makes assumptions such that the E always reports positive results. Specifically, the private benefit is taken to zero in the limit, $x_b \to 0$. The proposition below demonstrates that this is sufficient for the E to always report positive results and never report negative results.

Proposition A.1. *For any set of parameters* $(\rho > 0, k > 0, \kappa > 0, 0 < p < 1, \pi > 0, V_s \geq 0)$ *with* $k < \kappa p \pi$*, there exists a* $\bar{x}_b > 0$ *such that if* $x_b < \bar{x}_b$*, then the E reports positive results but does not report negative results.*

Before stating the proof, I give a brief outline of the argument. In part 1, I derive the payoffs to the E given a contract (ς, ω) under two cases: (i) the E never reports results and (ii) the E reports only positive results. I show that given (ς, ω), the E always prefers the payoff associated with case (ii). In part 2, I derive a condition under which case (ii) is not incentive compatible; that is, a condition under which a commitment to report positive results is not credible. In part 3, I demonstrate that for any set of parameters, case (ii) can always be made incentive compatible by choosing a suitably small non-pecuniary payoff to the E, x_b. Part 4 concludes the proof.

Proof of Proposition A.1.

Part 1. Without loss of generality, assume $x_b < k$.[56] The E never reports negative results, since $x_b > 0$. Suppose a positive result arrives at time t and call the decision to

[56] This assumption is not restrictive since the proof attempts only to obtain an upper bound on x_b.

report a positive result $\chi = 1$, where $\chi = 0$ reflects the choice not to report a positive result. $\chi = 0$ implies that the payoff, π, from success is delayed until time $t + T_\omega$, for $T_\omega \sim Exp(\omega)$, so has time-$t$ expected value $\frac{\omega}{\rho+\omega}\pi$.[57] If $\chi = 0$, the contract ends at date T_ω, regardless of the result.

The value of the firm in development, V_d, is determined analogously to equation (6)

$$\rho\, V_d = \begin{cases} \kappa\,[p\,\pi - V_d] + \omega\,[V_s - V_d] & \text{if } \chi = 1 \\ \kappa\,[p\,\pi\,\frac{\omega}{\rho+\omega} - V_d] + \omega\,[V_s - V_d] & \text{if } \chi = 0 \end{cases} \tag{A.1}$$

Denote by $V_d^{np}(\chi)$ the non-pecuniary value of the contract to the E. The value to the E of contract (ς, ω) is $V_E(\omega, \varsigma, \chi) = (1 - \varsigma)\, V_d(\omega, \chi) + V_d^{np}(\omega, \chi)$, where

$$V_d^{np} = \begin{cases} \left(1 + \frac{\kappa\,(1-p)}{\rho+\omega}\right) \frac{x_b}{\rho+\kappa+\omega} & \text{if } \chi = 1 \\ \frac{x_b}{\rho+\omega} & \text{if } \chi = 0 \end{cases} \tag{A.2}$$

where the expression for $V_d^{np}(\omega, \chi)$ is analogous to $K(\omega, \chi = 1)$ in equation (4). Similarly, $K(\omega, \chi = 0) = \frac{k}{\rho+\omega}$ because the contract ends at time T_ω regardless of any potential result. Since the VC has no bargaining power, their participation constraint binds, $\varsigma V_d(\omega, \chi) = K(\omega, \chi)$. The value of the contract to the E is then $V_E = (1 - \varsigma)V_d + V_d^{np} = V_d + V_d^{np} - K(\omega)$, or

$$V_E(\omega, \chi) = \begin{cases} \frac{\kappa\,p\,\pi + \omega\,V_s}{\rho+\kappa+\omega} - \left(1 + \frac{\kappa\,(1-p)}{\rho+\omega}\right) \frac{k - x_b}{(\rho+\kappa+\omega)} & \text{for } \chi = 1 \\ \frac{\kappa\,p\,\pi\,\frac{\omega}{\rho+\omega} + \omega\,V_s}{\rho+\kappa+\omega} - \frac{k - x_b}{\rho+\omega} & \text{for } \chi = 0 \end{cases} \tag{A.3}$$

This leads to the following remark, which states that the E prefers ex-ante the contract in which they report positive results to the VC. Intuitively, given $x_b < k$, the cost of compensating the VC exceeds the private benefit they attain from continuation, so they would prefer to commit to stop development at time T_κ.

Remark A.1. *Given* $0 < x_b < k$, *for any contract rate* ω *and endogenously priced*

[57] $E[e^{-\rho T_\omega} \times \pi] = \frac{\omega}{\rho+\omega}\pi$, where expectations are taken over T_ω.

equity stake, $V_E(\omega, \chi = 1) > V_E(\omega, \chi = 0)$.

Part 2. The benefit to the E of delay, $\chi = 0$, upon observing a positive result is $\frac{x_b}{\rho+\omega}$. The cost to the E is to delay the pecuniary payoff, of which they obtain share $1-\varsigma$. The delay therefore costs the E $E[(1-e^{-\rho T_\omega})(1-\varsigma)\pi] = \frac{\rho}{\rho+\omega}(1-\varsigma)\pi$, where expectations are taken over T_ω.[58] This leads to the following incentive compatibility constraint, which states that the E's share of the pecuniary payoff must be sufficiently large for them to optimally report positive results.

Remark A.2. $\chi = 1$ *is incentive compatible iff* $1 - \varsigma \geq \frac{x_b}{\rho\pi}$.

Part 3. Consider the contract problem when the E commits to report positive results, $\chi = 1$

$$\sup_{\{\omega\in[0,\infty),\varsigma\in[0,1]\}} \left\{ \frac{\kappa\, p\, \pi + \omega\, V_s}{\rho+\kappa+\omega} - \left(1 + \frac{\kappa\,(1-p)}{\rho+\omega}\right) \frac{k - x_b}{(\rho+\kappa+\omega)} \right\}$$

$$\text{s.t.} \qquad 1 - \varsigma \geq \frac{x_b}{\rho\,\pi}, \qquad \varsigma\, \frac{\kappa\, p\, \pi + \omega\, V_s}{\rho+\kappa+\omega} = \left(1 + \frac{\kappa\,(1-p)}{\rho+\omega}\right) \frac{k}{(\rho+\kappa+\omega)}$$

where V_s is taken as given and the IC constraint from Remark A.2 is imposed. I begin by solving this problem without consideration for the incentive compatibility and feasibility constraint for ς; thereafter, I show that x_b can be chosen such that they are slack.

Define $\bar{V}_s = \frac{\kappa p\pi-(k-x_b)}{\rho+\kappa}$. For $V_s \geq \bar{V}_s$, the objective function is everywhere increasing in $\omega \in [0, \infty)$ given $x_b < k$, so the solution is $\omega \to \infty$. Next, restrict attention to $V_s \in [0, \bar{V}_s)$. The first-order condition for an interior optimum is

$$\frac{(k - x_b)(1-p)}{(\rho+\omega)^2} + \frac{(k - x_b - \kappa\pi)\, p + V_s(\rho+\kappa)}{(\rho+\kappa+\omega)^2} = 0$$

[58]Alternatively, one could consider the incentives to delay/report over some finite interval Δt. The environment is stationary, so these two approaches are equivalent.

which delivers two candidate solutions for ω

$$\omega = \frac{\rho\kappa p\pi - \rho V_s(\rho+\kappa) - (k-x_b)(\kappa(1-p)+\rho) \pm \sqrt{\kappa^2(k-x_b)(1-p)\big(\kappa p\pi - p(k-x_b) - V_s(\rho+\kappa)\big)}}{k - x_b - \kappa p\pi + V_s(\rho+\kappa)}$$

The 'positive' solution is strictly negative on $V_s \in [0, \bar{V}_s)$. Define $\bar{p} = \frac{(k-x_b)(\rho+\kappa)^2}{\kappa(\pi\rho^2+(k-x_b)(\kappa+2\rho))}$ and $\underline{V_s} = \frac{\kappa p\pi - p(k-x_b)}{\rho+\kappa} - \frac{(k-x_b)(1-p)(\rho+\kappa)}{\rho^2}$. When $p \leq \bar{p}$, the 'negative' solution is positive for $V_s \in [0, \bar{V}_s)$; when $p > \bar{p}$, it is positive for $V_s \in [\underline{V_s}, \bar{V}_s)$. The SOC is satisfied in both cases. Conversely, when $p > \bar{p}$, the solution is negative for $V_s \in [0, \underline{V_s})$. In this case, the optimand is decreasing on $\omega \in [0, \infty)$. The solution is then $\omega = 0$.

With candidate solutions in hand, it remains to check the feasibility and incentive compatibility constraints. To this end, note the following lemma.

Lemma A.1. *If there are parameters such that $\varsigma \in (0,1)$ (i.e. strictly), then $\exists \bar{x}_b$ such that $\forall x_b \in [0, \bar{x}_b)$, $1 - \varsigma \geq \frac{x_b}{\rho\pi}$.*

Proof of Lemma A.1. The PC gives $\varsigma(\omega) = \frac{k(\kappa(1-p)+\rho+\omega)}{(\rho+\omega)(\kappa p\pi+\omega V s)}$, so that $\varsigma'(\omega) < 0$ on $V_s \in [0, \bar{V}_s)$. Furthermore, from the solution to the contract problem, $\omega'(x_b) \leq 0$, with strict inequality when $\omega > 0$ is optimal. Together, $\varsigma'(x_b) \geq 0$. Then, suppose the parameters are such that $\varsigma \in (0,1)$ and consider reducing $x_b \to 0$. The E's share, $(1-\varsigma)$, increases weakly from a positive value, whereas $x_b/(\rho\pi) \to 0$. Therefore, there exists a region $[0, \bar{x}_b)$ such that the IC is satisfied, i.e. $1 - \varsigma \geq \frac{x_b}{\rho\pi}$.

The remainder of the proof provides sufficient conditions on x_b such that Lemma A.1 can be applied. Suppose first that $\omega = 0$ is optimal. Then $\varsigma(0) = \frac{k(\kappa(1-p)+\rho)}{\kappa p\pi\rho}$. Clearly, $\varsigma(0) > 0$ since $p \in (0,1)$. Note that $\varsigma(0) < 1 \iff \kappa p\pi\rho > k(\kappa(1-p)+\rho)$. Furthermore, if $\omega = 0$ is optimal, then (from above) $p > \bar{p} \implies p\kappa\pi\rho > \rho^{-1}\big((k-x_b)(\rho+\kappa)^2 - p\kappa(k-x_b)(\kappa+2\rho)\big)$. Combining these inequalities, a sufficient condition for $\varsigma(0) < 1$ is $\rho^{-1}\big((k-x_b)(\rho+\kappa)^2 - p\kappa(k-x_b)(\kappa+2\rho)\big) > k(\kappa(1-p)+\rho)$. For $x_b < \frac{k\kappa(1-p)(\rho+\kappa)}{\rho^2+(1-p)(\kappa^2+2\rho\kappa)}$, this condition is satisfied, so $\varsigma(0) \in (0,1)$. Then, by Lemma A.1, it is possible to choose a positive x_b sufficiently small such that the IC is satisfied.

Suppose next that $\omega \in (0, \infty)$ is optimal. Note $\varsigma(\omega) > 0$ $\forall \omega \in (0, \infty)$. It remains to check $\varsigma(\omega) < 1$, which is equivalent to requiring that the pecuniary return to the

E is positive, $(1-\varsigma)\,V_d(\omega,\chi=1)=V_d(\omega,\chi=1)-K(\omega,\chi=1)>0$. Evaluating this expression

$$V_d(\chi=1)-\left(1+\frac{\kappa\,(1-p)}{\rho+\omega}\right)\frac{k}{(\rho+\kappa+\omega)}\Bigg|_{\omega=\omega^*\in(0,\infty)}=$$

$$p\pi-\frac{1}{\kappa^2}\left[\frac{(2k-x_b)\sqrt{\kappa^2(k-x_b)(1-p)\big(p\kappa\pi-p(k-x_b)-V_s(\rho+\kappa)\big)}}{k-x_b}\right.$$
$$\left.+\kappa\left(2kp-k+\rho V_s-\frac{(1-p)p(k-x_b)x_b\kappa}{\sqrt{\kappa^2(k-x_b)(1-p)\big(p\kappa\pi-p(k-x_b)-V_s(\rho+\kappa)\big)}}\right)\right]$$

This expression is decreasing in x_b on $x_b>0$ given $V_s<\frac{\kappa p\pi-p(k-x_b)}{\rho+\kappa}$, which is satisfied whenever $V_s<\bar{V}_s$ since $p\in(0,1)$. Therefore, if $V_d(\omega,\chi=1)-K(\omega,\chi=1)>0$ at $x_b=0$, it is positive over some region $x_b\in[0,\bar{x}_b)$ where $\bar{x}_b>0$. Evaluating $V_d(\omega,\chi=1)-K(\omega,\chi=1)>0$ at $x_b=0$ obtains the condition

$$p\pi-\frac{1}{\kappa}\Big(k(2p-1)+\rho V_s+2\sqrt{k(1-p)(\kappa p\pi-kp-V_s(\rho+\kappa)}\Big)>0$$

which is equivalent to

$$\kappa p\pi-k(2p-1)-\rho V_s>2\sqrt{k(1-p)(\kappa p\pi-kp-V_s(\rho+\kappa))}$$

Note that the left-hand side is positive if $V_s<\frac{\kappa p\pi-k}{\rho}+\frac{2k(1-p)}{\rho}$, which is valid on $V_s\leq\bar{V}_s$ when $x_b=0$. Therefore, squaring both sides, subtracting the right-hand side from the left-hand side and some algebra obtains

$$4kV_s(1-p)\kappa+(\kappa p\pi-k-\rho V_s)^2>0$$

which is satisfied given the initial restrictions on parameters (i.e. positivity constraints and $p\in(0,1)$). This implies that the E obtains a positive pecuniary payoff when $x_b=0$ and, by extension, over some region $x_b\in[0,\bar{x}_b)$. This in turn implies the conditions to apply Lemma A.1 are satisfied and there exists a (potentially different) upper bound

$\bar{x}_b$ such that $\forall x_b \in [0, \bar{x}_b)$, the IC is slack.

Suppose finally that $\omega \to \infty$ is optimal. Then $\lim_{\omega \to \infty} \varsigma(\omega) = 0$ and so satisfying the IC requires only that $x_b \leq \rho\pi$, which is easily satisfied.

Part 4. By Remark A.1, the E prefers the payoff associated with commitment. Furthermore, for the candidate optimal contract derived in Part 3, it is possible to find a strictly positive upper bound on x_b such that the feasibility constraint is slack, i.e. $\varsigma \in (0, 1)$, if x_b is less than this upper bound. Then, by Lemma A.1, there is a (potentially different) upper bound on x_b such that the IC constraint is also slack if x_b does not exceed this upper bound. Together, there is an interval $[0, \bar{x}_b)$ such that both the feasibility and incentive compatibility constraints are slack at the optimal contract in which $\chi = 1$. Since the contract with $\chi = 1$ is incentive compatible and preferred ex ante, the optimal contract involves $\chi = 1$ on this interval $x_b \in [0, \bar{x}_b)$. Furthermore, the expected capital cost is given by $K(\omega, \chi = 1)$ and the optimal contract rate is determined as in part 3. ■

A.2 VC bargaining power

In this section, I generalise the baseline model to allow for VC bargaining power. VC bargaining power influences the choice of contract, (ς, ω), in two ways. First, current investors face a hold-up problem vis-à-vis future investors because future investors extract surplus in their own negotiations with the start-up. Therefore, VCs see their shareholding in the firm facing greater dilution in future funding rounds relative to the setting where VCs have no bargaining power. The implication is that VCs have an incentive to provide greater funding upfront to reduce the need for the firm to return to the capital market. Second, a VC's outside option in the case that they choose not to invest in a start-up now has positive value. Investing in a given start-up involves forgoing this outside option and, therefore, incurs an opportunity cost that raises the effective flow cost of investing in a start-up above k.[59] This second effect has the oppo-

[59]A similar effect appears in Michelacci and Suarez (2004): VCs may facilitate an early-stage IPO to "recycle" their capital into new ventures. In my model, the same motive leads to earlier termination by

site implication to the first, pushing instead in the direction of a higher contract rate. In the remainder of this section, I set up the contracting problem when the VC has bargaining power and consider these effects more formally.

The VC It is now necessary to derive the value of a VC that is searching for investment opportunities or funding a start-up. Consider the value of a VC that is funding a productive start-up, which I denote by U_d^p. The VC incurs capital flow cost k and receives a share ς of the firm. At rate κ, the start-up realises a result, which is a success with probability p and returns the VC their share of the payoff π *and* the value of search for a VC, U_s, since the VC can now seek out new investment opportunities. However, with probability $1-p$, the start-up obtains a negative result and so the VC gets the value of funding an unproductive start-up, U_p^u. Finally, at rate ω, the contract expires and the VC obtains their share in the company, a value ςV_s, and returns to search. With this in mind, the HJB for a VC funding a productive start-up is given by

$$\rho U_d^p = -k + \kappa[p(\varsigma\pi + U_s) + (1-p)U_d^u - U_d^p] + \omega[\varsigma V_s + U_s - U_d^p] \tag{A.4}$$

Consider next the value of funding an unproductive start-up, U_p^u. The VC must continue to pay the flow capital cost k because they are unaware that the start-up is unproductive. In addition, at rate ω, the commitment expires and the VC returns to search. The HJB for a VC funding an unproductive start-up is then given by

$$\rho U_d^u = -k + \omega[U_s - U_d^u] \tag{A.5}$$

Finally, the value of search for a VC depends on the rate at which they meet with start-ups, which I denote by υ. At this rate the VC meets with a start-up and obtains the value of funding an productive start-up, U_p^d. The HJB for a VC in search is given

inducing a smaller up-front commitment, after which the start-up must raise additional capital.

by

$$\rho U_s = \upsilon[U_d^p - U_s] \tag{A.6}$$

The contract. Upon meeting, the VC and entrepreneur engage in Nash bargaining to determine the contract rate ω and share ς obtained by the VC. The value of the match to the VC is U_d^p and the value of their outside option is U_s. The value of the match to existing shareholders is $(1-\varsigma)V_d$, because they retain a share $(1-\varsigma)$ of the gross value of the firm, and the value of their outside option is V_s. If the entrepreneur has bargaining weight δ in the negotiations, then the optimal contract solves

$$\max_{\{\varsigma\in[0,1],\omega_i\in[0,\infty)\}} \left\{ \big((1-\varsigma)V_d - V_s\big)^{\delta} \big(U_d^p - U_s\big)^{1-\delta} \right\}$$

subject to the participation constraints, $(1-\varsigma)V_d \geq V_s$ and $U_d^p \geq U_s$, where V_s and U_s are taken as given in the optimisation. The first-order condition for the equity stake, ς, yields the following conditions

$$(1-\varsigma)V_d = V_s + \delta S \tag{A.7}$$

$$U_d^p = U_s + (1-\delta)S \tag{A.8}$$

where $S = (1-\varsigma)V_d + U_d^p - V_s - U_s$ is the surplus of the contract. As is standard in these settings, the surplus is split according to the Nash bargaining weights. Solving equations (A.4), (A.5) and (A.6) for U_d^p in terms of U_s allows for writing the surplus as

$$S = V_d - V_s - \frac{k + \rho U_s}{\rho + \kappa + \omega}\left(1 + \frac{\kappa(1-p)}{\rho + \omega}\right) \tag{A.9}$$

Since the surplus is independent of the share that goes to the VC, the contract rate simply maximises the joint surplus, $\omega = \arg\max_{\omega\in[0,\infty)}\{S\}$. The first-order condition for ω is then

$$V_d'(\omega) = -(k + \rho U_s)\left(\frac{1-p}{(\rho+\omega)^2} + \frac{p}{(\rho+\kappa+\omega)^2}\right) \tag{A.10}$$

The first-order condition for the optimal contract rate is identical to the case without bargaining power, except that k has been replaced by $k+\rho U_s$. Intuitively, every instant that the VC pays for the investment, they incur capital cost k and *opportunity cost* $\rho \times U_s$, because they cannot engage in other investment opportunities. This induces shorter duration capital commitments. However, in equilibrium, V_s will also adjust due to hold-up vis-a-vis future investors.

As I show in the proof to Proposition A.3, the optimal contract rate when their is bargaining power is

$$\omega = \max\left\{0, -\rho + \mathcal{C} + \sqrt{\mathcal{C}\mathcal{F} + \mathcal{C}^2}\right\}, \tag{A.11}$$

where

$$\mathcal{C} = \frac{\kappa(1-p)k}{\kappa p\pi - k} \quad \text{and} \quad \mathcal{F} = \kappa + \delta\nu \times \frac{\rho+\kappa}{\rho+\lambda} + (1-\delta)\upsilon\frac{\kappa p\pi}{k}$$

Clearly, setting $\delta = 1$ recovers equation (11). However, more generally, the bargaining weight has two implications. One the one hand, it affects the collective forward-looking incentives of the current E and VC. Once a VC has invested, it becomes an existing shareholder and has bargaining power δ vis-à-vis future investors. Therefore, δ reflects the size of the potential hold-up problem that the E and current VC face in future investments, which induces dilution. As δ increases, the E and VC worry less about future dilution and so reduce the size of the current capital injection.[60] Conversely, the bargaining weight affects how the VC prioritises the current investment vis-à-vis other potential investment opportunities. As δ increases, the VCs bargaining power falls in any new investment, which lowers their opportunity cost of funds. This leads them to extend more funding to the current entrepreneur.

Comparative statics Here, I provide comparative statics for the optimal contract rate in the model with VC bargaining power.

Proposition A.2. *Conditional on staged financing, the optimal contract rate, ω, is strictly increasing in k, strictly decreasing in p and π, and ambiguous with respect to*

[60]Note that from the perspective of the current investment, the need to raise follow-on funding and the associated hold-up acts analogously to financing risk: in the limit where $\delta \to 0$, current investors would be indifferent between meeting with a new investor and failing due to a lack of follow-on funding.

κ and δ. It is strictly increasing in υ for $\delta < 1$ and independent of υ for $\delta = 1$. It is strictly increasing in ν and strictly decreasing in λ for $\delta > 0$, and independent of ν and λ for $\delta = 0$. The QCCS for p is strictly negative and for κ strictly positive.

Comparative statics are preserved for all parameters considered in the $\delta = 1$ case. For $\delta < 1$, the contract rate is additionally increasing in υ, the meeting rate for VCs; this reflects the opportunity cost motive. Furthermore, the effect of bargaining power on the optimal contract rate is ambiguous. In fact,

$$\frac{\partial \omega}{\partial \delta} > 0 \iff \theta < \frac{k}{\kappa p \pi} \frac{\rho + \kappa}{\rho + \lambda}$$

Whether increases in the bargaining power of the entrepreneur, δ, increase the contract rate, ω, depends on market tightness, $\theta = \mu_s/\mu_{vc} = \upsilon/\nu$. When the market is tight – start-ups are numerous relative to VCs, θ is high – the value of the VC's outside option is high, which strengthens their motive to keep funding commitments short. In this case, marginal changes to their bargaining power have large effects on their opportunity cost of funds and, therefore, reductions in their bargaining power induce larger capital commitments, $\partial\omega/\partial\delta < 0$. Otherwise, increases in the bargaining power of insiders, δ, allows them to worry less about future dilution, and lead to shorter commitments, $\partial\omega/\partial\delta > 0$.

A.3 Proofs and supplementary results

A.3.1 Funding histories and outcomes

This section proves the results in the main text regarding funding histories and outcomes, and provides additional comparative statics.

Proof of Proposition 1 Consider each part in turn.

Part (i). Suppose the contract is signed at date $t = 0$. Until time $T = \min\{T_\kappa, T_\omega\} \sim Exp(\kappa + \omega)$, the E has not obtained a result and funding has not been withdrawn, so the VC pays k. Once time T arrives, there are three potential outcomes: (i) funding runs out, $T = T_\omega$, with probability $\frac{\omega}{\kappa+\omega}$; (ii) $T = T_\kappa$ and the result is a success, with probability $\frac{\kappa p}{\kappa+\omega}$; or (iii) $T = T_\kappa$ and the result is a failure, with probability $\frac{\kappa(1-p)}{\kappa+\omega}$. In cases (i) and (ii), funding ceases. In case (iii), the VC continues to pay k until time $T + T_\omega = T_\kappa + T_\omega$. Therefore, from time $t = 0$ to $t = T$, the flow cost to the VC is $e^{-\rho t}\, k$ from the perspective of time $t = 0$. At time T, the probability that the VC continues to pay k is $\frac{\kappa(1-p)}{\kappa+\omega}$, in which case they pay k from date T to $T + T_\omega$. The expected capital cost is then computed by taking expectations over $T_\omega \sim Exp(\omega)$ and $T \sim Exp(\kappa+\omega)$

$$\begin{aligned} K(\omega) &= \int_0^\infty \Bigg(\int_0^T e^{-\rho t}\, k\, dt \\ &\qquad + \frac{\kappa(1-p)}{\kappa+\omega} \int_0^\infty \left(\int_T^{T+T_\omega} e^{-\rho s}\, k\, ds \right) \omega\, e^{-\omega T_\omega}\, dT_\omega \Bigg) (\kappa+\omega)\, e^{-(\kappa+\omega)T}\, dT \\ &= \left(1 + \frac{\kappa(1-p)}{\rho+\omega} \right) \frac{k}{\rho+\kappa+\omega} \end{aligned} \tag{A.12}$$

Part (ii). Consider a firm in productive development. It may not raise capital again because: (i) it completes R&D in the current financing round, or (ii) its financing is withdrawn and it cannot raise new capital. Together, the probability that a firm in productive development does not raise more capital is given by

$$\begin{aligned} Pr(\text{do not raise again}) &= Pr(\text{result in current round}) + Pr(\text{return to search and fail}) \\ &= \frac{\kappa}{\kappa+\omega} + \frac{\omega}{\kappa+\omega} \frac{\lambda}{\lambda+\nu} \end{aligned}$$

For notational convenience, I denote this probability by q_d. Then

$$Pr(N_f = 1) = q_d, \quad Pr(N_f = 2) = (1-q_d)\, q_d, \quad \ldots \quad Pr(N_f = n) = (1-q_d)^{n-1}\, q_d$$

which is the probability mass function of the Geometric distribution.

Part (iii). Conditional on receiving another round, the firm first spends an exponentially distributed period in development with rate $\kappa + \omega$, followed by an exponentially distributed period in search with rate $\lambda + \nu$. The sum of these independent durations is hypoexponential.

Part (iv). The success probabilities satisfy

$$p_d = \frac{\kappa}{\kappa+\omega}p + \frac{\omega}{\kappa+\omega}p_s, \qquad p_s = \frac{\nu}{\lambda+\nu}p_d.$$

Solving this system gives the expressions in Proposition 1. ■

Proof of Corollary 1 Follows immediately from differentiation of the key objects in Proposition 1. ■

Comparative statics under optimal contracting

Corollary A.1. *Under optimal staged financing, $\omega^* > 0$:*

(i) The capital injection $\bar{K} = K(\omega^*)$ is strictly increasing in the payoff (π) and failure rate in search (λ); decreasing in the meeting rate (ν); and ambiguous with respect to the R&D success probability (p), arrival rate of R&D outcomes (κ), and flow investment cost (k). Holding project quality, $\bar{V}_s$, fixed (QCCS), the capital injection $\bar{K}$ is ambiguous with respect to the R&D success probability (p) and strictly decreasing in the arrival rate of R&D outcomes (κ).

(ii) The expected number of funding rounds $E[N_f] = 1/q_d$ is strictly increasing in the meeting rate (ν) and flow investment cost (k); decreasing in the R&D success probability (p), payoff (π), and failure rate in search (λ); and ambiguous with respect to the arrival rate of R&D outcomes (κ). Holding project quality, $\bar{V}_s$, fixed (QCCS), the expected number of funding rounds $E[N_f]$ is strictly decreasing in the R&D success probability (p) and ambiguous with respect to the arrival rate of R&D outcomes (κ).

(iii) The expected duration between successive funding rounds for a given firm $E[T_{br}] = \frac{1}{\kappa+\omega} + \frac{1}{\lambda+\nu}$ is strictly increasing in the R&D success probability (p) and payoff (π); decreasing in the meeting rate (ν) and flow investment cost (k); and ambiguous with respect to the failure rate in search (λ) and arrival rate of R&D outcomes (κ). Holding project quality, $\bar{V}_s$, fixed (QCCS), the expected duration between funding

rounds $E[T_{br}]$ is strictly increasing in the R&D success probability (p) and strictly decreasing in the arrival rate of R&D outcomes (κ).

(iv) The success probability for firms in productive development, p_d is: strictly increasing in p and π; strictly decreasing in k; and ambiguous with respect to ν, λ, and κ.

Proof of Corollary A.1 Consider each part in turn

Part (i). For $x \in \{\nu, \pi, \lambda\}$, $\frac{dK}{dx} = \frac{\partial K}{\partial \omega}\frac{\partial \omega}{\partial x} \implies \text{sign}\big(\frac{dK}{dx}\big) = -\text{sign}\big(\frac{\partial \omega}{\partial x}\big)$ since $\frac{\partial K}{\partial \omega} < 0$. The result follows immediately. For $x \in \{k, p, \kappa\}$, $\frac{dK}{dx} = \frac{\partial K}{\partial x} + \frac{\partial K}{\partial \omega}\frac{\partial \omega}{\partial x}$, where $\frac{\partial K}{\partial \omega} < 0$. For p, $\frac{\partial K}{\partial p} < 0$ and $\frac{\partial \omega}{\partial p} < 0$; for k, $\frac{\partial K}{\partial k} > 0$ and $\frac{\partial \omega}{\partial k} > 0$; and for κ, $\frac{\partial K}{\partial \kappa} < 0$ and the sign of $\frac{\partial \omega}{\partial \kappa}$ is ambiguous. It is not possible to sign these effects in general. The same logic applies for the QCCS for p. However, the QCCS for κ is strictly negative since $\frac{\partial \omega}{\partial \kappa} > 0$ in this case.

Part (ii). Recall $E[N_f] = \big(\frac{\kappa}{\kappa+\omega} + \frac{\omega}{\kappa+\omega}\frac{\lambda}{\lambda+\nu}\big)^{-1}$. Then

- $\frac{d}{d\nu}E[N_f] = \frac{\kappa\nu(\lambda+\nu)\omega'(\nu)+\lambda\omega(\nu)(\kappa+\omega(\nu))}{(\kappa(\lambda+\nu)+\lambda\omega(\nu))^2} > 0$ since $\omega'(\nu) > 0$.

- $\frac{d}{d\lambda}E[N_f] = \frac{\kappa\nu(\lambda+\nu)\omega'(\lambda)-\nu\omega(\lambda)(\kappa+\omega(\lambda))}{(\kappa(\lambda+\nu)+\lambda\omega(\lambda))^2} < 0$ since $\omega'(\lambda) < 0$.

- For $x \in \{p, \pi, k\}$, including the QCCS for p, $\frac{d}{dx}E[N_f] = \frac{\kappa\nu(\lambda+\nu)\omega'(x)}{(\kappa(\lambda+\nu)+\lambda\omega(x))^2} \implies \text{sign}\big(\frac{d}{dx}E[N_f]\big) = \text{sign}\big(\omega'(x)\big)$. The result then follows from Proposition 3.

- $\frac{d}{d\kappa}E[N_f] = \frac{(\lambda+\nu)\nu}{(\kappa(\lambda+\nu)+\lambda\omega(\kappa))^2}\,(\kappa\omega'(\kappa) - \omega(\kappa))$. This is ambiguous, the QCCS. Note $\frac{d}{d\kappa}E[N_f] < 0$ if $\frac{\partial \omega}{\partial \kappa}\frac{\kappa}{\omega} < 1$; i.e. ω the elasticity of ω with respect to κ is less than one.

Part (iii). For all parameters $x \in \{k, p, \pi\}$, $\text{sign}\big(\frac{d}{dx}E[T_{br}]\big) = -\text{sign}\big(\omega'(x)\big)$. The result then follows from Proposition 3. For κ, $\text{sign}\big(\frac{d}{d\kappa}E[T_{br}]\big) = -\text{sign}\big(1+\omega'(\kappa)\big)$, which is ambiguous in the general case, but the QCCS is negative, since $\omega'(\kappa) > 0$ in this case. For ν, $\frac{d}{d\nu}E[T_{br}] = -\frac{\omega'(\nu)}{(\kappa+\omega(\nu))^2} - \frac{1}{(\lambda+\nu)^2} < 0$, where the inequality follows from Proposition 3, and $\frac{d}{d\lambda}E[T_{br}] = -\frac{\omega'(\lambda)}{(\kappa+\omega(\lambda))^2} - \frac{1}{(\lambda+\nu)^2}$, which is ambiguous since $\omega'(\lambda) < 0$ by Proposition 3.

Part (iv) The result for p, π and k follows directly from differentiation of p_d given the results from Proposition 3. For κ, $\text{sign}\big(\frac{dp_d}{d\kappa}\big) = \text{sign}\big(\omega(\kappa) - \kappa\omega'(\kappa)\big)$, which is ambiguous.

Next, consider $\frac{dp_d}{d\nu} = \frac{\kappa p\lambda(\omega-(\lambda+\nu)\omega'(\nu))}{(\kappa(\lambda+\nu)+\lambda\omega)^2} \implies \text{sign}\big(\frac{dp_d}{d\nu}\big) = \text{sign}\big(\omega - (\lambda+\nu)\omega'(\nu)\big)$, which can equivalently be cast as a statement about the semi-elasticity of ω with respect to ν; that is, if $\frac{\omega'(\nu)}{\omega} < \frac{1}{\lambda+\nu}$, then $\frac{dp_d}{d\nu} > 0$. To see that $\frac{dp_d}{d\nu} < 0$ is possible, note that there are parameter values such that $\omega = 0$ but $\omega'(\nu) > 0$ (see the proof to Proposition A.3). In this case $\omega - (\lambda+\nu)\omega'(\nu) = -(\lambda+\nu)\omega'(\nu) < 0$, so $\frac{dp_d}{d\nu} < 0$. To see that $\frac{dp_d}{d\nu} > 0$ is possible, note that $\lim_{\nu\to\infty} p_d = p$ and $p_d \leq p$, so p_d must approach p from below as $\nu \to \infty$. This means there must be some ν for which $\frac{dp_d}{d\nu} > 0$. Finally, to see that $\exists\tilde{\nu}_d \geq 0$ such that if $\nu \geq \tilde{\nu}_d$, $\frac{dp_d}{d\nu} > 0$ and otherwise $\frac{dp_d}{d\nu} < 0$, note that $\frac{d}{d\nu}(\omega - (\lambda+\nu)\omega'(\nu)) = -(\lambda+\nu)\omega''(\nu) > 0$ where the inequality follows since $\omega''(\nu) < 0$, so there at most one root for $\omega-(\lambda+\nu)\omega'(\nu) = 0$. Furthermore, $\frac{d}{d\nu}(\omega-(\lambda+\nu)\omega'(\nu)) > 0$ implies that if $\frac{dp_d}{d\nu} \geq 0$ for some $\nu = \hat{\nu}$, then $\frac{dp_d}{d\nu} > 0 \ \forall\nu > \hat{\nu}$. Therefore, we can define $\tilde{\nu}_d > 0$ s.t. $\omega - (\lambda+\nu)\omega'(\nu) = 0$ or $\tilde{\nu}_d = 0$ if there is no root and the statement holds.

For λ, $\text{sign}\big(\frac{dp_d}{d\lambda}\big) = \text{sign}\big(-\nu\omega - \lambda(\lambda+\nu)\omega'(\lambda)\big)$. To see that this can be be positive, note that there exists a λ such that $\omega = 0$ and $\omega'(\lambda) < 0$; at this point, small reductions in λ imply a movement from upfront financing to staged-financing and so lower the success probability. To see that $\frac{dp_d}{d\lambda} < 0$ is possible, note that $\lim_{\lambda\to 0} p_d = p$ and $p_d \leq p$, so p_d must approach p from below as $\lambda \to 0$, implying there is some region for which $\frac{dp_d}{d\lambda} < 0$. For p_s, clearly $\frac{dp_d}{d\lambda} < 0 \implies \frac{dp_s}{d\lambda} < 0$.

Finally, $\frac{dp_d}{dp} = \frac{\kappa(\lambda+\nu)(\kappa(\lambda+\nu)-\lambda p\omega'(p)+\lambda\omega)}{(\kappa(\lambda+\nu)+\lambda\omega)^2}$ and $\frac{dp_s}{dp} = \frac{\nu}{\lambda+\nu}\frac{dp_d}{dp}$. By Proposition 3, $\omega'(p) < 0$, so $\frac{dp_d}{dp} > 0$ and $\frac{dp_s}{dp} > 0$. Furthermore, $\frac{d}{dp}\frac{\kappa\,(\lambda+\nu)}{\omega\,\lambda+\kappa\,(\lambda+\nu)} = -\frac{\kappa\lambda(\lambda+\nu)\omega'(p)}{(\kappa(\lambda+\nu)+\lambda\omega)^2} > 0$ since $\omega'(p) < 0$.
■

Proof of Corollary 3. Consider two start-ups $i = a, b$ facing the same funding conditions, ν and λ. Suppose $E[N_f^a] > E[N_f^b]$. Then,

$$\left(\frac{\kappa_a}{\omega_a+\kappa_a} + \frac{\omega_a}{\kappa_a+\omega_a}\frac{\lambda}{\lambda+\nu}\right)^{-1} > \left(\frac{\kappa_b}{\omega_b+\kappa_b} + \frac{\omega_b}{\kappa_b+\omega_b}\frac{\lambda}{\lambda+\nu}\right)^{-1} \implies \frac{\kappa_a}{\omega_a} < \frac{\kappa_b}{\omega_b}$$

The result the follows immediately from the expression for $\Delta_p = \frac{\lambda\omega}{\kappa(\lambda+\nu)+\lambda\omega} = \frac{\lambda}{\frac{\kappa}{\omega}(\lambda+\nu)+\lambda}$.

■

A.3.2 Equilibrium and the optimal contract

In this section, I show that for given meeting rates in the financing market, the value functions and contract are unique. I do so in the general case in which the VC has bargaining power, as described in section A.2. The roof of Proposition 2 then follows by setting the VC's bargaining power to zero.

Proposition A.3. *Given ν and υ and for $\kappa p\pi > k$, there is a unique solution for $\{V_d, V_s, U_d^p, U_d^u, U_s, S, \omega, \varsigma\}$ in terms of the model parameters and the resulting equilibrium is characterised by a (weakly) positive and finite contract rate, $\omega \in [0, \infty)$.*

Proof of Proposition A.3. Given ν, υ, an equilibrium is a solution for $\{V_d, V_s, U_d^p, U_d^u, U_s, S, \omega, \varsigma\}$ to the following set of equations

$$\rho\, V_d = \kappa\,[p\,\pi - V_d] + \omega\,[V_s - V_d] \tag{A.13}$$

$$(\rho+\lambda)V_s = \nu[(1-\varsigma)V_d - V_s] \tag{A.14}$$

$$\rho U_d^p = -k + \kappa[p(\varsigma\,\pi + U_s) + (1-p)U_d^u - U_d^p] + \omega[\varsigma\, V_s + U_s - U_d^p] \tag{A.15}$$

$$\rho U_d^u = -k + \omega[U_s - U_d^u] \tag{A.16}$$

$$\rho U_s = \upsilon[U_d^p - U_s] \tag{A.17}$$

$$S = V_d - V_s - \frac{k + \rho U_s}{\rho + \kappa + \omega}\left(1 + \frac{\kappa(1-p)}{\rho + \omega}\right) \tag{A.18}$$

$$(1-\varsigma)V_d = V_s + \delta S \tag{A.19}$$

$$U_d^p = U_s + (1-\delta)S \tag{A.20}$$

$$\omega = \arg\max_{\omega \in [0,\infty]}\{S\} \tag{A.21}$$

The first two equations come from the start-up's problem, where I have substituted $V^M = (1-\varsigma)V_d$. The next three equations come from the VC's problem, as set out in section A.2 of the Appendix. The final four equations relate to the optimal contract: they define, respectively, the surplus, the optimality conditions for the equity share in the contract, and the optimal contract rate.[61]

Towards a proof, note that given $(\omega, S, \nu, \upsilon)$, all remaining equilibrium objects are pinned down uniquely. Specifically, combining equations (A.14) and (A.19), $(\rho+\lambda)V_s = \nu\delta S$, so V_s is known given (ν, S). V_d then follows immediately from equation (A.13) given (ω, V_s). The equity stake then follows from equation (A.19). Finally, the VC value functions can then be recovered from equations (A.15), (A.16) and (A.17), given knowledge of $(\omega, \upsilon, \varsigma, V_s)$.

In light of this, to show that the equilibrium, given (ν, υ), is unique, it is sufficient to demonstrate that there is a unique (ω, S) consistent with the equilibrium conditions. To this end, rather than taking the first-order condition as a function of V_s and U_s, as in equation (A.10), and *then* solving the resulting system of equations in (ω, V_s, U_s), it is simpler to first eliminate the search values from the optimisation problem and then solve for the optimal contract. Although the contract is derived taking as given the search values, (V_s, U_s), this approach is valid because the equilibrium relationships for V_s and U_s are strictly linear in S. Therefore, differentiating the reduced surplus fixed point – while holding the meeting rates ν and υ constant – yields the exact same critical points as the private first-order condition. Towards an expression for S, combining the above expressions allows to solve for V_s in terms of model parameters and the equilibrium

[61]There are more equations than unknowns, but technically one of the equations (A.19) or (A.20) is redundant.

contract rate, ω

$$
\begin{aligned}
(\rho+\lambda)V_s &= \nu\delta S \\
&= \nu\delta\left[V_d - V_s - \frac{k+\rho U_s}{\rho+\kappa+\omega}\left(1+\frac{\kappa(1-p)}{\rho+\omega}\right)\right] \\
&= \nu\delta\left[\frac{\kappa p\pi+\omega V_s}{\rho+\kappa+\omega} - V_s - \frac{k+\rho U_s}{\rho+\kappa+\omega}\left(1+\frac{\kappa(1-p)}{\rho+\omega}\right)\right] \\
&= \nu\delta\left[\frac{\kappa p\pi+\omega V_s}{\rho+\kappa+\omega} - V_s - \frac{k+\upsilon(1-\delta)\frac{\rho+\lambda}{\nu\delta}V_s}{\rho+\kappa+\omega}\left(1+\frac{\kappa(1-p)}{\rho+\omega}\right)\right]
\end{aligned}
$$

where the first line follow from combining equations (A.14) and (A.19), the second line uses the definition of S, the third line substitutes for V_d from equation (A.13), and the fourth line uses the relationship between U_s and V_s, which comes from writing these values both in terms of the surplus and then eliminating the surplus from the resulting equations.[62] Then, solving for V_s and using $\nu\delta S = (\rho+\lambda)V_s$ yields an expression for S in terms of model parameters and the contract rate, ω

$$
S(\omega) = \frac{(\rho+\lambda)\left[\kappa p\pi - \left(1+\frac{\kappa(1-p)}{\rho+\omega}\right)k\right]}{(1-\delta)\upsilon\,(\rho+\lambda)\left(1+\frac{\kappa(1-p)}{\rho+\omega}\right) - \nu(1-\delta)\,(\rho+\kappa) + (\rho+\kappa)\,(\rho+\lambda+\nu) + (\rho+\lambda)\,\omega} \tag{A.22}
$$

The optimal contract rate maximises $S(\omega)$, taking the meeting rates ν and υ as given. Differentiating with respect to ω obtains

$$
\begin{aligned}
S'(\omega) = &\frac{1}{\frac{1}{\rho+\lambda}(\rho+\omega)^2\left((\rho+\kappa)(\delta\nu+\lambda+\rho)+\omega(\rho+\lambda)+(1-\delta)\upsilon(\rho+\lambda)\left(1+\frac{\kappa(1-p)}{\rho+\omega}\right)\right)^2} \\
&\times\Bigg((\rho+\lambda)(k-\kappa p\pi)(\rho+\omega)^2 + 2\kappa k(1-p)(\rho+\lambda)(\rho+\omega) \\
&\quad + k\left(\kappa^2(1-p)(\rho+\lambda+\delta\nu)+\delta\kappa\nu(1-p)\rho\right) + (1-\delta)\kappa^2(1-p)p\pi\upsilon(\rho+\lambda)\Bigg)
\end{aligned}
$$

The denominator of $S'(\omega)$ is strictly positive, so it is sufficient to focus on the numerator. Note firstly that the numerator is a quadratic function of ω. There are two cases to consider. Suppose firstly that $k \geq \kappa p\pi$, in which case $S'(\omega) > 0$ on $\omega \in [0,\infty)$.

[62] Specifically, $\rho U_s = \upsilon(1-\delta)S$ and $(\rho+\lambda)V_s = \nu\delta S$, so that $\rho U_s = (\rho+\lambda)\frac{1-\delta}{\delta}\frac{\upsilon}{\nu}V_s$

In this case, the unique solution is $(\omega \to \infty, S = 0)$. Next, suppose that $k < \kappa p\pi$, in which case the numerator is quadratic and concave in ω, and so may omit an interior solution. Setting $S'(\omega) = 0$ and solving for ω obtains

$$\omega = -\rho + \mathcal{C} \pm \sqrt{\mathcal{C}\mathcal{F} + \mathcal{C}^2}$$

where

$$\mathcal{C} = \frac{\kappa(1-p)k}{\kappa p\pi - k} \quad \text{and} \quad \mathcal{F} = \kappa + \delta\nu \times \frac{\rho + \kappa}{\rho + \lambda} + (1-\delta)\upsilon\frac{\kappa p\pi}{k}$$

Among the two solutions, it is simple to see that the 'negative' solution is strictly negative whenever $\kappa p\pi > k$.[63] Therefore, there can be at most one solution on $\omega \in [0, \infty)$. Specifically, since the sign of $S'(\omega)$ is pinned down by a quadratic function in ω that is concave, the greater of the two solutions is a maximum and so is the optimum when it is positive. However, if the greater of the two solution is negative, then the optimum must be at $\omega = 0$, because $S'(\omega)$ is negative on $\omega \in [0, \infty)$. Combining these insights, given ν and υ, is it clear that there is a unique optimum characterised by a finite contract rate when $k < \kappa p\pi$ and so surplus. Given the preceding arguments, the equilibrium objects are uniquely determined. ■

Proof of Corollary 2. Again, I consider the general case with VC bargaining power. Solving the optimisation problem as described, the optimal contract rate is given by

$$\omega = \begin{cases} \max\left\{0, -\rho + \mathcal{C} + \sqrt{\mathcal{C}\mathcal{F} + \mathcal{C}^2}\right\} & \text{if } k < \kappa p\pi \\ \infty & \text{else} \end{cases} \tag{A.23}$$

where

$$\mathcal{C} = \frac{\kappa(1-p)k}{\kappa p\pi - k} \quad \text{and} \quad \mathcal{F} = \kappa + \delta\nu \times \frac{\rho + \kappa}{\rho + \lambda} + (1-\delta)\upsilon\frac{\kappa p\pi}{k}$$

To determine when the contract rate is set to zero, corresponding to upfront financing, consider when the max-operator is active. To this end, upfront financing occurs

[63] $-\rho + \mathcal{C} - \sqrt{\mathcal{C}\mathcal{F} + \mathcal{C}^2} < -\rho + \mathcal{C} - \sqrt{\mathcal{C}^2} = -\rho$, where the first inequality follows since $\mathcal{C} > 0$ given $k < \kappa p\pi$.

when any of the following conditions are met

$$p \in \left(\frac{k(\rho+\kappa)^2}{\kappa\left(\rho^2\pi + k(\kappa+2\rho)\right)}, 1 \right] \quad and \quad \begin{cases} \upsilon \leq \frac{1}{p\kappa^2\pi}\Psi & and \quad \delta = 0, \quad or \\ \nu \frac{(\rho+\lambda)}{k\kappa\delta(\rho+\kappa)}\Psi & and \quad \upsilon \leq f(\nu) \quad \text{and} \quad \delta \in (0,1), \quad or \\ \nu \leq \frac{(\rho+\lambda)}{k\kappa(\rho+\kappa)}\Psi & and \quad \delta = 1 \end{cases}$$

with

$$\begin{aligned} \Psi &= \frac{1}{1-p}\left(\rho^2(\kappa p\pi - k) - k(1-p)\kappa^2 - 2k(1-p)\rho\kappa\right) \\ f(\nu) &= \frac{\rho^2(\rho+\lambda)(\kappa p\pi - k) - k\kappa(1-p)\left[\kappa(\rho+\lambda+\delta\nu) + \rho(2\rho+2\lambda+\delta\nu)\right]}{(\rho+\lambda)(1-p)(1-\delta)\kappa^2 p\pi} \end{aligned}$$

with $\Psi > 0$ under the restriction on p and $f'(\nu) < 0$. The case given in the main body of the paper simply corresponds to the case of $\delta = 1$. ■

Proof of Propositions 3 and A.2 I consider first the standard comparative statics, then the QCCS (see Definition 2). Consider the general case of $\delta \in [0,1]$; Proposition 3 considers the special case, $\delta = 1$. Let $k < \kappa p\pi$ and consider the case of a positive contract rate, $\omega > 0$. Then,

$$\omega = -\rho + \mathcal{C}(z) + \sqrt{C(z)\mathcal{F}(z) + \mathcal{C}(z)^2}$$

where $\mathcal{C}(z), \mathcal{F}(z)$ makes explicit the dependence on some parameter, z. Differentiating ω with respect to some generic parameter z obtains

$$\frac{\partial\omega}{\partial z} = \left(1 + \frac{2\mathcal{C}(z) + \mathcal{F}(z)}{2\sqrt{\mathcal{C}(z)\mathcal{F}(z) + \mathcal{C}(z)^2}}\right)\mathcal{C}'(z) + \frac{\mathcal{C}(z)}{2\sqrt{\mathcal{C}(z)\mathcal{F}(z) + \mathcal{C}(z)^2}}\mathcal{F}'(z)$$

Now, consider each parameter in turn. For $z = \upsilon, \nu, \lambda, \delta$, $\mathcal{C}'(z) = 0 \implies \text{sign}(\frac{\partial\omega}{\partial z}) = \text{sign}(\mathcal{F}'(z))$. Then

- $\mathcal{F}'(\upsilon) \geq 0$, with strict inequality for $\delta < 1$.
- $\mathcal{F}'(\nu) \geq 0$, with strict inequality for $\delta > 0$.

- $\mathcal{F}'(\lambda) \leq 0$, with strict inequality for $\delta > 0$.

- $\mathcal{F}'(\delta) = \nu\frac{\rho+\kappa}{\rho+\lambda} - \upsilon\frac{\kappa p\pi}{k}$, which has ambiguous sign.

For $z = p, \pi$, $\mathcal{C}'(z) < 0$ (given $k < \kappa p\pi$) but $\mathcal{F}'(z) > 0$. Nevertheless, given $k < \kappa p\pi$, we have $\frac{\partial\omega}{\partial z} < 0$. For example, in the case of $z = \pi$

$$\frac{\partial\omega}{\partial\pi} = -\frac{1}{\kappa p\pi - k}\left(\kappa p\,\mathcal{C} + \mathcal{C}\frac{\kappa p\Big(\kappa + \delta\nu \times \frac{\rho+\kappa}{\rho+\lambda} + (1-\delta)\upsilon + 2\mathcal{C}\Big)}{2\sqrt{\mathcal{C}\mathcal{F} + \mathcal{C}^2}}\right)$$

which is negative for $k < \kappa p\pi$. Analogous arguments show that $\frac{\partial\omega}{\partial k} > 0$.

For $z = \kappa$, the expression is ambiguous. Indeed, $\mathcal{C}'(\kappa) < 0$, tends to negative infinity as κ approaches its lower bound, $k/(p\pi)$, and tends to zero as $\kappa \to \infty$. Conversely, $\mathcal{F}$ is increasing and linear in κ. Therefore, for small κ, $\mathcal{C}$ drives the behaviour and for large κ, $\mathcal{F}$ drives the behaviour. More formally, note that $\frac{\partial\omega}{\partial\kappa} > 0$ can be written

$$\mathcal{F}'(\kappa) > -\mathcal{C}'(\kappa)\left(2 + 2\sqrt{1 + \frac{\mathcal{F}(\kappa)}{\mathcal{C}(\kappa)}} + \frac{\mathcal{F}(\kappa)}{\mathcal{C}(\kappa)}\right)$$

The left-hand side is a constant. Given, $k < \kappa p\pi$, the right-hand side is strictly decreasing in κ. Therefore, there can be at most one solution to $\frac{\partial\omega}{\partial\kappa} = 0$. Furthermore, as κ approach $k/(p\pi)$ from above, it is simple to verify that $\frac{\partial\omega}{\partial\kappa} < 0$. Finally, for large κ, note that $\lim_{\kappa\to\infty}\mathcal{C}(\kappa)$ is a constant and, recalling that $F(\kappa)$ is linear, it is simple to show that $\lim_{\kappa\to\infty}\frac{\omega(\kappa)}{\sqrt{\kappa}} = C_0$ for some constant C_0. Therefore, $\omega(\kappa)$ grows like $\sqrt{\kappa}$, such that $\omega'(\kappa) \to 0$ from above. Therefore, there exists some threshold $\bar{\kappa}$ such that, for $\kappa < \bar{\kappa}$, $\omega'(\kappa) < 0$ and otherwise $\omega'(\kappa) > 0$.

For the QCCSs, replace $\pi = \frac{1}{p\kappa}\big[(\rho+\kappa)\bar{V}_s + k\big]$. Then,

$$\mathcal{C}(z;\bar{V}_s) = \frac{\kappa}{\rho+\kappa}\frac{k(1-p)}{\bar{V}_s}, \qquad F(z;\bar{V}_s) = \kappa + \delta\nu \times \frac{\rho+\kappa}{\rho+\lambda} + (1-\delta)\upsilon\frac{(\rho+\kappa)\bar{V}_s + k}{k}$$

Then, $\mathcal{C}'(p;\bar{V}_s) < 0$ and $\mathcal{F}'(p;\bar{V}_s) = 0$. It follows immediately that the QCCS for p is strictly negative. Furthermore, $\mathcal{C}'(\kappa;\bar{V}_s) > 0$ and $\mathcal{F}'(\kappa;\bar{V}_s) > 0$, so that the QCCS for κ is strictly positive. ■

A.3.3 Entry

Proof of Proposition 4. The standard comparative statics follow immediately from differentiating equation (13). For the QCCSs, note that equation (13) can be written

$$V_s = \frac{\mathcal{F} - \kappa}{\mathcal{F} + 2(\rho + \omega)} \bar{V}_s$$

To compute the QCCS, note first that $\bar{V}_s$ is held constant, so it is sufficient to consider the first term, $\frac{\mathcal{F}-\kappa}{\mathcal{F}+2(\rho+\omega)}$. p appears only via ω; then, differentiating and using the QCCS for $\omega'(p) < 0$, it follows that the QCCS for entry with respect to p is positive. For κ, we have

$$\frac{\partial}{\partial \kappa}\left(\frac{\mathcal{F} - \kappa}{\mathcal{F} + 2(\rho + \omega)}\right) = \frac{\nu}{(\rho + \lambda)\big(\mathcal{F} + 2(\rho + \omega)\big)^2} \times \big(\rho + 2[\omega - (\rho + \kappa)\omega'(\kappa)]\big)$$

Therefore, QCCS for κ is positive iff $\rho + 2(\omega - (\rho + \kappa)\omega'(\kappa)) > 0$, but ambiguous in general. For intuition, note that as $\rho \to 0$, this is equivalent to

$$\frac{\kappa\omega'(\kappa)}{\omega} < 1$$

which is the necessary condition for $\frac{\partial E[N_f]}{\partial \kappa} < 0$; in words, as $\rho \to 0$, long-horizon projects, $\kappa \downarrow$, face greater distortions when they require more funding rounds. More generally, $\rho + 2(\omega - (\rho + \kappa)\omega'(\kappa)) > 0$ is equivalent to

$$\frac{\kappa\omega'(\kappa)}{\omega} < 1 + \frac{\rho}{\rho + \kappa}\Big(\frac{\kappa}{2\omega} - 1\Big)$$

The right-hand side can be less than or exceed one, so it is not possible, in general, to state whether $\rho > 0$ loosens or tightens the elasticity condition under $\rho \to 0$. ■

The value of search To obtain equation (13), solve the following system for $\{V_d, V_s\}$

$$\rho V_d = \kappa[p\pi - V_d] + \omega[V_s - V_d]$$
$$(\rho+\lambda)V_s = \nu[V_d - K(\omega) - V_s]$$

which substitutes $V^M = V_d - K(\omega)$. This yields

$$V_s = \frac{\nu\Big[\kappa p\pi - k\Big(1+\frac{\kappa(1-p)}{\rho+\omega}\Big)\Big]}{(\rho+\lambda+\nu)(\rho+\kappa+\omega)-\nu\omega}$$

After some algebra, this is equal to

$$V_s = \frac{\mathcal{F}-\kappa}{\mathcal{F}+\frac{\mathcal{CF}+(\rho+\omega)^2}{\rho+\omega-\mathcal{C}}}\bar{V}_s = \frac{\mathcal{F}-\kappa}{\mathcal{F}+\frac{2\mathcal{CF}+2\mathcal{C}(\rho+\omega)}{\rho+\omega-\mathcal{C}}}\bar{V}_s = \frac{\mathcal{F}-\kappa}{\mathcal{F}+2(\mathcal{C}+\sqrt{\mathcal{CF}+\mathcal{C}^2})}\bar{V}_s$$

The first value obtains after factoring out $\bar{V}_s$; the second after multiplying out $(\rho+\omega)^2$ and simplifying; the third follows by factoring $\sqrt{\mathcal{CF}+\mathcal{C}^2}$ from $\frac{2\mathcal{CF}+2\mathcal{C}(\rho+\omega)}{\rho+\omega-\mathcal{C}}$ and simplifying.

A.3.4 Efficiency

This sections contains proof for results in section 2.3.3

Proof of Lemma 1 From equation (13), V_s can be expressed as

$$V_s = \frac{\kappa p\pi - k\Big(1+\frac{\kappa(1-p)}{\rho+\omega}\Big)}{(\rho+\kappa)+(\rho+\lambda)\big(\frac{\rho+\kappa+\omega}{\nu}\big)}$$

Under $\kappa p\pi > k$, $\lim_{\nu\to\infty}\omega(\nu) = \infty \implies \lim_{\nu\to\infty}\kappa p\pi - k\Big(1+\frac{\kappa(1-p)}{\rho+\omega(\nu)}\Big) = \kappa p\pi - k$. Furthermore, from equation (11)

$$\lim_{\nu\to\infty}\omega(\nu)/\sqrt{\nu} = \sqrt{\mathcal{C}\times\frac{\rho+\kappa}{\rho+\lambda}} \implies \lim_{\nu\to\infty}\omega(\nu)/\nu = 0$$
$$\implies \lim_{\nu\to\infty}\Big((\rho+\kappa)+(\rho+\lambda)\big(\frac{\rho+\kappa+\omega(\nu)}{\nu}\big)\Big) = \rho+\kappa$$

where $\mathcal{C} = \frac{\kappa(1-p)k}{\kappa p\pi - k}$. Therefore, $\lim_{\nu\to\infty} V_s = \frac{\kappa p\pi - k}{\rho+\kappa}$. ■

Corollary A.2. *Capital misallocation (equation* (14)*) is strictly*

- *increasing in the payoff* (π) *and failure rate in search* (λ)*;*
- *decreasing in the meeting rate* (ν) *and flow investment cost* (k)*;*
- *and ambiguous with respect to the R&D success probability* (p) *and arrival rate of R&D outcomes* (κ)*.*

Proof of Corollary A.2 The result follows immediately from differentiating the expression and applying the results of Proposition 3. ■

Planner's problem In the main body of the paper, only the steady state conditions are considered. Towards solving the planner's problem, it is necessary to consider the behaviour of the system out of steady state. To this end, note that at time t, there may be some firms in productive development with different contract rates $\omega \in [0, \infty)$. The same is true for the unproductive development state. I will use $\mu^p_{d,t}(\omega)$ and $\mu^u_{d,t}(\omega)$ to refer to the measures of firms at time t in productive and unproductive development, respectively, with contract rate ω. The total measure of firms in productive and unproductive development are then given by

$$\mu^p_{d,t} = \int_0^\infty \mu^p_{d,t}(\omega)\, d\omega, \quad \mu^u_{d,t} = \int_0^\infty \mu^u_{d,t}(\omega)\, d\omega \tag{A.24}$$

and the measure of VCs not financing start-ups is $\mu_{vc,t} = M - \mu^p_{d,t} - \mu^u_{d,t}$. Firm in search have no contract and so are homogeneous–they have measure $\mu_{s,t}$, such that the total flow of matches is $m(\mu_{s,t}, \mu_{vc,t})$.

To determine the state transition equations, it is necessary to know which contracts are being signed at each instant. In principle, each firm that signs a contract at time t could choose a different contract rate. In this spirit, I let χ_t denote the distribution of contract rates among contracts signed at time t. Using generalised-density notation, so that point masses are permitted, write $\chi_t(\omega)$, with

$$\chi_t(\omega) \geq 0, \qquad \int_0^\infty \chi_t(\omega)\, d\omega = 1.$$

The same convention applies to $\mu^p_{d,t}(\omega)$ and $\mu^u_{d,t}(\omega)$. The total inflow into state $\mu^p_{d,t}(\omega)$ from new contracts is then $\chi_t(\omega)m(\mu_{s,t}, \mu_{vc,t})$ and the total outflow of firms from search to development is $\int_0^\infty \chi_t(\omega)m(\mu_{s,t}, \mu_{vc,t})\, d\omega$. With this notation in hand, the state transition equations are given by

$$\dot{\mu}^p_{d,t}(\omega) = \chi_t(\omega)m(\mu_{s,t}, \mu_{vc,t}) - (\kappa + \omega)\mu^p_{d,t}(\omega) \tag{A.25}$$

$$\dot{\mu}^u_{d,t}(\omega) = \kappa(1 - p)\mu^p_{d,t}(\omega) - \omega\mu^u_{d,t}(\omega) \tag{A.26}$$

$$\dot{\mu}_{s,t} = \int_0^\infty \omega\mu^p_{d,t}(\omega)\, d\omega + \Lambda_t - \lambda\mu_{s,t} - \int_0^\infty \chi_t(\omega)m(\mu_{s,t}, \mu_{vc,t})\, d\omega \tag{A.27}$$

where Λ_t is the flow rate of entry at time t, which is equal to $V_{s,t}/\tilde{\sigma}$ in the decentralised equilibrium.

The planner's problem is to choose entry, Λ_t, and the contract rate for each firm at time t in order to maximise total surplus, given by equation (15). In practice, the choice of contract boils down to choosing $\chi_t(\omega)$ for each $\omega \in [0, \infty)$ subject to the aforementioned constraints on $\chi_t(\omega)$. The current value Hamiltonian is given by

$$\begin{aligned} H = \int_0^\infty \Big(&[\kappa p\pi - k]\mu^p_{d,t}(\omega) - k\mu^u_{d,t}(\omega) \\ &+ \gamma^p_{d,t}(\omega)[\chi_t(\omega)m(\mu_{s,t}, \mu_{vc,t}) - (\kappa + \omega)\mu^p_{d,t}(\omega)] \\ &+ \gamma^u_{d,t}(\omega)[\mu^p_{d,t}(\omega)\kappa(1 - p) - \omega\mu^u_{d,t}(\omega)]\Big)\, d\omega \\ &+ \gamma_{s,t}\big[\int_0^\infty \omega\mu^p_{d,t}(\omega)\, d\omega + \Lambda_t - \lambda\mu_{s,t} - \int_0^\infty \chi_t(\omega)m(\mu_{s,t}, \mu_{vc,t})\, d\omega] - \frac{\tilde{\sigma}}{2}\Lambda_t^2 \end{aligned} \tag{A.28}$$

The optimality condition for Λ_t is simply $\frac{\partial H}{\partial \Lambda_t} = \gamma_{s,t} - \tilde{\sigma}\Lambda_t$. For the contracts, the functional derivative of the Hamiltonian is

$$\frac{\delta H}{\delta \chi_t(\omega)} = \big[\gamma^p_{d,t}(\omega) - \gamma_{s,t}\big]m(\mu_{s,t}, \mu_{vc,t}).$$

Because the Hamiltonian is linear in χ_t subject to the unit-mass constraint, an optimal stationary plan can be chosen to place unit mass on some

$$\hat{\omega} \in \underset{\omega \in [0,\infty)}{\arg\max} \left\{ \gamma_d^p(\omega) - \gamma_s \right\}.$$

Imposing the steady-state conditions $\dot{\gamma}_{s,t} = 0$, $\dot{\gamma}_{d,t}^p(\omega) = 0$ and $\dot{\gamma}_{d,t}^u(\omega) = 0$, the shadow-values are given by

$$\begin{aligned}
\gamma_d^p(\hat{\omega}) &= \frac{\left(\rho + \lambda + \frac{\partial m}{\partial \mu_s}\right)\left(\kappa p \pi - \left(1 + \frac{\kappa(1-p)}{\rho + \hat{\omega}}\right) k\right)}{(\rho + \lambda)\left(1 + \frac{\kappa(1-p)}{\rho + \hat{\omega}}\right) \frac{\partial m}{\partial \mu_{vc}} + (\rho + \kappa) \frac{\partial m}{\partial \mu_s} + (\rho + \lambda)(\rho + \kappa + \hat{\omega})} \\
&= \frac{(\rho + \lambda + \alpha\, \nu)\left[\kappa p \pi - \left(1 + \frac{\kappa(1-p)}{\rho + \hat{\omega}}\right) k\right]}{\beta\, \upsilon\, (\rho + \lambda)\left(1 + \frac{\kappa(1-p)}{\rho + \hat{\omega}}\right) - \nu\,(1 - \alpha)\,(\rho + \kappa) + (\rho + \kappa)\,(\rho + \lambda + \nu) + (\rho + \lambda)\,\hat{\omega}}
\end{aligned} \tag{A.29}$$

and

$$\gamma_s = \frac{\alpha \nu}{\rho + \lambda + \alpha \nu} \gamma_d^p(\hat{\omega}) \tag{A.30}$$

$$\gamma_d^u(\hat{\omega}) = -\frac{k}{\rho + \hat{\omega}} - \frac{\beta \upsilon (\gamma_d^p(\hat{\omega}) - \gamma_s)}{\rho + \hat{\omega}} \tag{A.31}$$

where $\hat{\omega}$ is the time-invariant maximising contract rate defined above.

Proof of Proposition 5. The proof follows immediately from the optimality conditions. First, consider the optimal contract rate. In the decentralised equilibrium, the optimal contract rate ω maximises the surplus, S, which is given by

$$S = \frac{(\rho + \lambda)\left[\kappa p \pi - \left(1 + \frac{\kappa(1-p)}{\rho + \omega}\right) k\right]}{(1 - \delta)\upsilon\,(\rho + \lambda)\left(1 + \frac{\kappa(1-p)}{\rho + \omega}\right) - \nu(1 - \delta)\,(\rho + \kappa) + (\rho + \kappa)\,(\rho + \lambda + \nu) + (\rho + \lambda)\,\omega}$$

In the planner's problem, the optimal contract rate maximises $\gamma_d^p(\hat{\omega}) - \gamma_s$, which is given by

$$\gamma_d^p(\hat{\omega}) - \gamma_s = \frac{(\rho + \lambda)\left[\kappa p \pi - \left(1 + \frac{\kappa(1-p)}{\rho + \hat{\omega}}\right) k\right]}{\beta\, \upsilon\, (\rho + \lambda)\left(1 + \frac{\kappa(1-p)}{\rho + \hat{\omega}}\right) - \nu\,(1 - \alpha)\,(\rho + \kappa) + (\rho + \kappa)\,(\rho + \lambda + \nu) + (\rho + \lambda)\,\hat{\omega}}$$

Setting $\beta = 1 - \delta$ and $\alpha = \delta$ creates an identity, so that the contract decentralised equilibrium equals that chosen by the planner. Because the two expressions coincide pointwise for every candidate contract rate, the planner and private parties have the same argmax.

Second, consider entry. In the decentralised equilibrium, entry is given by $V_s/\tilde{\sigma}$. In the planner's problem, the optimality condition yields $\Lambda = \gamma_s/\tilde{\sigma}$. Noting that $(\rho+\lambda)V_s = \delta\nu S$ and comparing V_s to γ_s, it is clear, for given optimal contract rate, ω, that entry is efficient if $\beta = 1 - \delta$ and $\alpha = \delta$. Finally, since the contract in the decentralised equilibrium coincides with the planner's solution when the same condition is met, entry must also be efficient. ■

A.3.5 Acquisitions

Proof of Proposition 6 An equivalent statement is that if $\omega < \infty$, then $V_d > V_s$. To see that this must be the case, note that setting $\omega \to \infty$ implies $V_d = V_s$ and all the value is retained by the existing shareholders. In contrast, a finite contract rate implies that existing shareholders retain value $(1-\varsigma)V_d|_{\{\omega<\infty\}}$. If this is to be optimal, it must be that $\omega < \infty$ leaves the existing shareholders better off than $\omega \to \infty$, i.e. $(1-\varsigma)V_d|_{\{\omega<\infty\}} > V_s \implies V_d|_{\{\omega<\infty\}} > V_s$. Since $\varsigma|_{\{\omega<\infty\}} \in (0,1)$, this implies $V_d|_{\{\omega<\infty\}} > V_s$. ■

A.3.6 Identification

This section provides the proof to Proposition 7. Before stating the proof, note that the E[time-to-exit], E[burn rate] and E[exit multiple] – the expected time from first funding to a successful exit, the expected burn rate and the expected exit multiple, respectively – are given by[64]

[64] E[time-to-exit] may also be written, $E[\text{time-to-exit}] = E[N_f](\kappa+\omega)^{-1} + (E[N_f]-1)(\lambda+\nu)^{-1}$, which adds up the time in development plus time in search (i.e. between rounds). Noting $E[N_f](\kappa+\omega)^{-1} = \kappa^{-1}(1-\Delta_p)$, financing risk brings forward the time-to-exit, because only firms that complete development before funding runs out can be successful. However, the time spent in search delays completion.

$$E[\text{time-to-exit}] = \frac{(\lambda+\nu)^2+\nu\omega}{(\lambda+\nu)(\kappa(\lambda+\nu)+\lambda\omega)} \tag{A.32}$$

$$E[\text{burn rate}] = E\Big[\frac{\bar{K}}{T_{br}}\Big] = \bar{K} \times \left(\frac{(\kappa+\omega)(\lambda+\nu)\ln\left(\frac{\kappa+\omega}{\lambda+\nu}\right)}{(\kappa+\omega)-(\lambda+\nu)}\right) \tag{A.33}$$

$$E[\text{exit multiple}] = E\Big[\frac{\pi}{\bar{K}N_f}\Big] = \frac{\pi}{\bar{K}}\frac{E[N_f]^{-1}}{1-E[N_f]^{-1}}\ln\big(E[N_f]\big) \tag{A.34}$$

which uses $T_{br} \sim Hypo(\kappa+\omega, \lambda+\nu)$ and $N_f \sim Geo(q_d)$, and where $\bar{K}$ is the equilibrium expected capital commitment, defined in Corollary A.1.

Proof of Proposition 7. Recall equation (19)

$$\mathbf{m} = \big(E[\mathbb{1}\{\text{success}\}], E[N_f], E[T_{br}], E[\text{time-to-exit}], E[\text{burn rate}], E[\text{exit multiple}]\big)$$

I refer to element i of $\mathbf{m}$ by m_i. The proof consists of two steps. First, I show how to recover expressions for $\alpha_0 = \kappa+\omega$, $\alpha_1 = \lambda+\nu$, $\alpha_2 = \nu\omega$, $\alpha_3 = \kappa p$, $\alpha_4 = \pi$, and $\alpha_5 = k/(\rho+\omega)$ as functions of observable moments, $\mathbf{m}$. Then, I use the optimality condition for the contract to eliminate ω; since this equation is quadratic in ω, it suffices to show that there is at most one positive solution for ω.

Fix some $\mathbf{m} \in \mathcal{M}^{adm}$. Then, using the model-implied expressions for $m_1, ..., m_6$, it is simple to show

$$\alpha_0 = \kappa+\omega = \big(m_4 - m_3(m_2-1)\big)^{-1}$$
$$\alpha_1 = \lambda+\nu = \big(m_2 m_3 - m_4\big)^{-1}$$
$$\alpha_2 = \nu\omega = \alpha_0\alpha_1(1-m_2^{-1})$$
$$\alpha_3 = \kappa p = \alpha_0 m_1 m_2^{-1}$$

For $\alpha_4 = \pi$, note from equation (A.33)

$$\bar{K} = m_5\left(\frac{\alpha_0\alpha_1\ln\left(\frac{\alpha_0}{\alpha_1}\right)}{\alpha_0-\alpha_1}\right)^{-1}$$

then from equation (A.34)

$$\alpha_4 = \pi = \frac{m_5 m_6}{\frac{\alpha_0\alpha_1 \ln\left(\frac{\alpha_0}{\alpha_1}\right)}{\alpha_0-\alpha_1}\frac{m_2^{-1}}{1-m_2^{-1}}\log(m_2)}$$

Finally, for $\alpha_5 = \frac{k}{\rho+\omega}$, write

$$\bar{K} = \frac{k}{\rho+\kappa+\omega}\left(1+\frac{\kappa(1-p)}{\rho+\omega}\right) = \frac{k}{\rho+\omega}\left(1-\frac{\kappa p}{\rho+\kappa+\omega}\right)$$

then, using the expression for $\bar{K}$

$$\alpha_5 = \frac{k}{\rho+\omega} = m_5\left(\frac{\alpha_0\alpha_1\ln\left(\frac{\alpha_0}{\alpha_1}\right)}{\alpha_0-\alpha_1}\right)^{-1}\left(1-\frac{\alpha_3}{\rho+\alpha_0}\right)^{-1}$$

Then, the second step eliminates ω from the above relationships. To do so, I use the first-order condition and solve directly for ω in terms of moments, $\mathbf{m}$. The first order condition for ω can be written

$$\frac{k}{\rho+\omega}\kappa(1-p) = (\rho+\lambda)(\rho+\omega)\frac{V_s(\omega,\theta)}{\nu}$$

where $V_s(\omega,\theta)$ is the value of search, and

$$\begin{aligned}\frac{V_s(\omega,\theta)}{\nu} &= \frac{\kappa p\pi - k\left(1+\frac{\kappa(1-p)}{\rho+\omega}\right)}{(\rho+\lambda+\nu)(\rho+\kappa+\omega)-\nu\omega} = \frac{\kappa p\pi-(\rho+\kappa+\omega)\bar{K}}{(\rho+\lambda+\nu)(\rho+\kappa+\omega)-\nu\omega} \\ &= \frac{\alpha_3\alpha_4-(\rho+\alpha_0)m_5\left(\frac{\alpha_0\alpha_1\ln\left(\frac{\alpha_0}{\alpha_1}\right)}{\alpha_0-\alpha_1}\right)^{-1}}{(\rho+\alpha_1)(\rho+\alpha_0)-\alpha_2} > 0\end{aligned}$$

which depends only on observables, and where the inequality follows since $\theta \in \Theta^{adm} \implies k < \kappa p \pi$. Denoting $V_s(\omega,\theta)/\nu = \tilde{V}_s(\mathbf{m})$, using $\alpha_5 = k/(\rho+\omega)$ and $\alpha_3 = \kappa p$, and solving for ω, the first-order condition implies

$$\omega = -\rho + \frac{\kappa-\alpha_3}{\rho+\lambda}\frac{\alpha_5}{\tilde{V}_s(\mathbf{m})}$$

Then, to solve for ω, substitute in $\kappa = \alpha_0 - \omega$ and $\lambda = \alpha_1 - \nu = \alpha_1 - \alpha_2/\omega$

$$\omega = -\rho + \frac{\alpha_0 - \omega - \alpha_3}{\rho + \alpha_1 - \alpha_2/\omega} \frac{\alpha_5}{\tilde{V}_s(\mathbf{m})} = -\rho + \frac{(\alpha_0 - \alpha_3 - \omega)\omega}{(\rho + \alpha_1)\omega - \alpha_2} \frac{\alpha_5}{\tilde{V}_s(\mathbf{m})}$$

Rearranging as a quadratic in ω yields

$$\left[\rho + \alpha_1 + \alpha_5/\tilde{V}_s(\mathbf{m})\right]\omega^2 + \left[\rho(\rho + \alpha_1) - \alpha_2 - (\alpha_0 - \alpha_3)\alpha_5/\tilde{V}_s(\mathbf{m})\right]\omega - \rho\alpha_2 = 0$$

Finally, since $\rho > 0, \alpha_1, \alpha_2, \alpha_5 > 0$ and $\tilde{V}_s(\mathbf{m}) > 0$, the coefficient on ω^2 is positive while the constant term is negative; thus the two roots have opposite signs, and exactly one satisfies $\omega > 0$. Then, given $\alpha_0 = \kappa + \omega$, $\alpha_1 = \lambda + \nu$, $\alpha_2 = \nu\omega$, $\alpha_3 = \kappa p$, $\alpha_4 = \pi$, and $\alpha_5 = k/(\rho + \omega)$ and $\omega = \omega(\mathbf{m})$, it follows immediately that there is a unique mapping from empirical moments to model parameters. ■

Flow of first funding rounds To obtain equation (20), note that the measure of firms that have entered but not yet received funding evolves as

$$\dot{\mu}^0_{s,t} = \frac{V_{s,t}}{\tilde{\sigma}} - (\lambda + \nu_e)\,\mu^0_{s,t}$$

which implies the steady-state condition

$$\tilde{\sigma} = \left(\frac{1}{\lambda + \nu_e} \times \frac{1}{\mu^0_s}\right) V_s$$

where μ^0_s is the steady-state measure of firms that have paid the entry cost but not yet obtained VC funding. μ^0_s is unobserved in the data, but the empirical counterpart to $\nu_e \times \mu^0_s$ is the flow of first funding rounds, which is observed. Substituting in for μ^0_s then yields equation (20).

A.4 Quantitative model

A.4.1 Frictionless benchmark

In the frictionless economy without agency and matching frictions, the value functions $\{\bar{V}_{s,e}, \bar{V}_{d,e}, \bar{V}_{s,l}, \bar{V}_{d,l}\}$ solve

$$\rho\bar{V}_{d,l} = -k_l + \kappa\big[p_l\pi - \bar{V}_{d,l}\big] + p_l\pi\xi\phi\exp\Big\{-\frac{\bar{V}_{d,l}}{p_l\pi\xi}\Big\} \tag{A.35}$$

$$(\rho+\lambda)\bar{V}_{s,l} = p_l\pi\xi\phi\exp\Big\{-\frac{\bar{V}_{s,l}}{p_l\pi\xi}\Big\} \tag{A.36}$$

$$\rho\bar{V}_{d,e} = -k_e + \kappa\Big[\Big(\frac{p_e}{p_l}\Big)\max\{\bar{V}_{s,l}, \bar{V}_{d,l}\} - \bar{V}_{d,e}\Big] \tag{A.37}$$

$$\bar{V}_{s,e} = \max\{0, \bar{V}_{d,e}\} \tag{A.38}$$

These equations mirror the conditions in the model of section 2.4, but are adjusted for a wealthy entrepreneur that can self-finance the project.[65] Relative to that benchmark, note that an entrepreneur that enters the late-stage may choose not to invest, or decide not to and simply wait for an acquisition offer. Therefore, the relevant continuation value from the early to late-stage is $\max\{\bar{V}_{s,l}, \bar{V}_{d,l}\}$. An entrepreneur that does not invest in the late-stage fails at rate λ, and so I retain the subscripts s and d; although now they refer to no late-stage investment and late-stage investment, respectively. Conversely, in the early-stage the entrepreneur either invests, or the project has zero value.

Matching-only benchmark When the agency friction is removed, each development stage is financed until uncertainty is resolved, but start-ups must still search for early-

[65] Alternatively, by taking the numerical limit $\nu_e \to \infty$ and $\nu_l \to \infty$ and solving the model of section 2.4, one can obtain an approximation to the solution of this system of equations to desired accuracy. See Lemma 1.

and late-stage funding. The value functions solve

$$(\rho+\lambda)V_{s,e} = \nu_e\big(V_{d,e} - V_{s,e}\big), \tag{A.39}$$

$$\rho V_{d,e} = -k_e + \kappa\big(\hat{p}_e V_{s,l} - V_{d,e}\big), \tag{A.40}$$

$$(\rho+\lambda)V_{s,l} = \nu_l\big(V_{d,l} - V_{s,l}\big) + p_l\pi\xi\phi \exp\left\{-\frac{V_{s,l}}{p_l\pi\xi}\right\}, \tag{A.41}$$

$$\rho V_{d,l} = -k_l + \kappa\big(p_l\pi - V_{d,l}\big) + p_l\pi\xi\phi \exp\left\{-\frac{V_{d,l}}{p_l\pi\xi}\right\}, \tag{A.42}$$

where $\hat{p}_e = p_e/p_l$. Defining

$$\hat{\phi}_s = \phi \exp\left\{-\frac{V_{s,l}}{p_l\pi\xi}\right\}, \qquad \hat{\phi}_d = \phi \exp\left\{-\frac{V_{d,l}}{p_l\pi\xi}\right\}.$$

Conditional on receiving early-stage financing, the corresponding outcome probabilities are

$$P(\text{success}) = \hat{p}_e \frac{\nu_l}{\nu_l+\lambda+\hat{\phi}_s}\frac{\kappa p_l}{\kappa+\hat{\phi}_d}, \tag{A.43}$$

$$P(\text{acquisition}) = \hat{p}_e\left[\frac{\hat{\phi}_s}{\nu_l+\lambda+\hat{\phi}_s} + \frac{\nu_l}{\nu_l+\lambda+\hat{\phi}_s}\frac{\hat{\phi}_d}{\kappa+\hat{\phi}_d}\right]. \tag{A.44}$$

A.4.2 Equilibrium conditions

In this section, I provide the full set of equilibrium conditions for the model that I outline in section 2.4 and estimate in section 3. I first provide the full set of conditions taking as given the equilibrium in the capital market, and then provide details on the full equilibrium.

Taking the meeting rates in the early and late-stage markets, ν_e and ν_l as given, the following conditions characterise the equilibrium

$$\rho V_{d,e}(\omega_e) = \kappa\left[\hat{p}_e V_{s,l} - V_{d,e}(\omega_e)\right] + \omega_e\left[V_{s,e} - V_{d,e}(\omega_e)\right] \tag{A.45}$$

$$(\rho+\lambda)V_{s,e} = \nu_e\left[V_e^M - V_{s,e}\right] \tag{A.46}$$

$$V_e^M = \sup_{\{\omega_e\in[0,\infty),\varsigma_e\in[0,1]\}} \left\{(1-\varsigma_e)\, V_{d,e}(\omega_e)\right\} \quad \text{s.t.} \quad \varsigma_e\, V_{d,e}(\omega_e) \geq K_e(\omega_e),\ V_{s,e} \text{ given} \tag{A.47}$$

$$K_e(\omega_e) = \left(1 + \frac{\kappa\,(1-\hat{p}_e)}{\rho+\omega_e}\right)\frac{k_e}{\rho+\kappa+\omega_e} \tag{A.48}$$

$$\rho V_{d,l}(\omega_l) = \kappa\big[p_l\pi - V_{d,l}(\omega_l)\big] + \omega_l\big[V_{s,l} - V_{d,l}(\omega_l)\big] + \hat{\phi}_d\left[p_l\pi E[\epsilon|\epsilon > V_{d,l}(\omega_l)/(p_l\pi)] - V_{d,l}\right] \tag{A.49}$$

$$(\rho+\lambda)V_{s,l} = \nu_l\big[V_l^M - V_{s,l}\big] + \hat{\phi}_s\left[p_l\pi E[\epsilon|\epsilon > V_{s,l}/(p_l\pi)] - V_{s,l}\right] \tag{A.50}$$

$$V_l^M = \sup_{\{\omega_l\in[0,\infty),\varsigma_l\in[0,1]\}} \left\{(1-\varsigma_l)\, V_{d,l}(\omega_l)\right\} \quad \text{s.t.} \quad \varsigma_l\, V_{d,l}(\omega_l) \geq K_l(\omega_l),\ V_{s,l} \text{ given} \tag{A.51}$$

$$K_l(\omega_l) = \left(1 + \frac{\kappa\,(1-p_l)}{\rho+\omega_l}\right)\frac{k_l}{\rho+\kappa+\omega_l+\hat{\phi}_d} \tag{A.52}$$

where $\hat{p}_e = p_e/p_l$, $\epsilon \sim Exp(1/\xi)$, $\hat{\phi}_i = \phi\left[1 - F(V_{i,l}/(p_l\pi))\right]$ for $i \in \{s,d\}$, and (ω_e, ς_e) and (ω_l, ς_l) are chosen optimally in the sense of equations (A.47) and (A.51), respectively.

The meeting rates, ν_e and ν_l, are determined in equilibrium based on the supply and demand for capital in each stage. I assume that VCs specialise within each stage: a measure M_e serve the early stage and measure M_l serve the late-stage. I denote by $\mu_{s,i}$, $\mu^p_{d,i}$ and $\mu^u_{d,i}$ the measures of firms in search, productive development and unproductive development for stage $i \in \{e,l\}$, respectively. Then, the measure of VCs in each stage that are not funding start-ups is given by $\mu_{vc,i} = M_i - \mu^p_{d,i} - \mu^u_{d,i}$ for $i \in \{e,l\}$ and the flow rate of matches in stage $i = e,l$ is $m_i = m(\mu_{s,i}, \mu_{vc,i})$. Then $\nu_i = \mu^{-1}_{s,i} m(\mu_{s,i}, \mu_{vc,i})$ for $i = e,l$.

Finally, $\mu_{s,i}$, $\mu^p_{d,i}$ and $\mu^u_{d,i}$ for $i = e,l$ are determined in equilibrium by the following steady-state conditions.

$$
\begin{aligned}
\text{Early-stage productive development}: &\quad \nu_e \mu_{s,e} = (\kappa + \omega_e)\mu_{d,e}^p \\
\text{Early-stage unproductive development}: &\quad \kappa(1-\hat{p}_e)\mu_{d,e}^p = \omega_e \mu_{d,e}^u \\
\text{Early-stage search}: &\quad \omega_e \mu_{d,e}^p + \frac{V_{s,e}}{\tilde{\sigma}} = (\lambda + \nu_e)\mu_{s,e} \\
\text{Late-stage productive development}: &\quad \nu_l \mu_{s,l} = (\kappa + \omega_l + \hat{\phi}_d)\mu_{d,l}^p \\
\text{Late-stage unproductive development}: &\quad \kappa(1-p_l)\mu_{d,l}^p = \omega_l \mu_{d,l}^u \\
\text{Late-stage search}: &\quad \omega_l \mu_{d,l}^p + \kappa \hat{p}_e \mu_{d,e}^p = (\lambda + \nu_l + \hat{\phi}_s)\mu_{s,l}
\end{aligned}
$$

where $\nu_e = m_e/\mu_{s,e}$ and $\nu_l = m_l/\mu_{s,l}$. Relative to the baseline model, there are two tangible differences. Firstly, the inflow into early-stage search includes new entrants, whereas the inflow into late-stage search includes firms that are successful in overcoming the early-stage uncertainty, a flow $\kappa\hat{p}_e\mu_{d,e}^p$. Secondly, in the late-stage, firms may be acquired from productive development or search.

B Empirical appendix

B.1 Dataset construction

This section discusses dataset construction. Table B.1 shows an illustrative dataset. To create a dataset of this form, my US and UK samples are drawn from Thomson Reuters venture capital dataset; I consider all firms that raised a first "early-stage" funding round between 2005-2015. Before processing the data, I make a small number of manual corrections where there are obvious errors. The majority of errors occur when a funding round coincides with an exit event; I perform manual searches to clarify the true nature of the event and drop the false entry. I also drop duplicated funding rounds.

Among all funding rounds, I keep those tagged as one of "Seed", "Early Stage", "Expansion", "Later Stage", which reflect increasing levels of progression. If a firm has

Firm ID	Date	Event	Round Type	Amount raised	Valuation
1	28/01/2013	funding round	Early	6.0	-
1	24/09/2013	funding round	Early	6.5	-
1	04/02/2015	funding round	Late	30.0	-
1	17/04/2019	IPO	-	80.5	342.6
2	12/03/2013	funding round	Early	3.0	-
2	19/07/2016	funding round	Late	10.5	-
2	01/01/2017	Acquisition	-	-	89.4
3	06/03/2014	funding round	Early	9.1	-

Table B.1: Illustrative dataset

The table shows illustrative data on venture capital funding rounds and exits for three fictitious firms. Amounts raised and valuations should be interpreted in units of million USD.

two deals on the same day, I sum up the respective deal values and record it under the earlier tag. I then label 'Seed' "Early Stage" rounds as "early-stage", and "Expansion" and "Later Stage" rounds as "late-stage". I drop the minority of firms that only have late-stage funding rounds in the data. I also drop any funding rounds that occur after the firm records an exit.

In some cases, the deal values are missing. This occurs in 2,706 of the 33,158 (8%) US funding rounds, and a slightly higher share of UK rounds. Data on amounts raised is used to compute the burn rate and exit multiples, both of which are used in model estimation. Therefore, I follow existing literature and impute amounts raised in these cases. The methodology I use follows Jagannathan et al. (2022) closely. Specifically, I estimate the following regression

$$\log(\text{Amount raised}_{i,r}) = \beta_0 + \beta_1 \log(\text{Amount raised}_{i,r-1}) + \beta_2 X_{i,r} + u_{i,r} \tag{B.1}$$

where Amount raised$_{i,r}$ is the amount raised by firm i in round r (in 2015 Mn USD) and X_i is a list of firm-specific controls and fixed effects. I include the investment stage, industry, year-quarter fixed effects, and fixed effect for the number of investors, which I cap at ten investors (i.e. all rounds with 10 or more investors are lumped together). I run this regression separately for the US and UK. In addition, for some rounds I do not observe the previous funding amount and so run the specification with only the controls, $X_{i,r}$.[66] The results are displayed in Table B.2. I run this on all deals between 2005 and 2022, which is why the observation counts exceed the sample used in the estimation (where data is censored). Using the results of these regressions I impute missing amounts raised using fitted values.

	United States		United Kingdom	
	Log(Amount$_r$)	Log(Amount$_r$)	Log(Amount$_r$)	Log(Amount$_r$)
Log(Amount$_{r-1}$)	-	0.47***	-	0.55***
Constant	0.35	-0.52	0.49	0.54
Industry FE	Y	Y	Y	Y
Stage FE	Y	Y	Y	Y
# Investors FE	Y	Y	Y	Y
YQ FE	Y	Y	Y	Y
Adj. R^2	0.31	0.50	0.26	0.55
Observations	61,004	36,196	7,133	2,816

Table B.2: Imputation model

The table reports regression results from estimation of equation (B.1). ***, p<0.01.

Among exit types, I classify IPOs, reverse takeovers, and secondary sales as successes. I also classify "Mergers" as successes when the acquiror is an investor, whereas trade-sales are classified as acquisitions. This categorisation means that SPACs are included as successes. For exit values, I obtain data from CRSP on the market cap of firms six months after IPO for US IPOs and from Bloomberg for UK IPOs.[67] Furthermore, a non-negligible share of exit values are missing in the Thomson Reuters data. Where possible, I fill in missing exit values with data from Crunchbase. When the deal value remains missing – the majority case – I set it equal to 1.5 times total invested capital, implying an exit multiple of 1.5X, following (Kerr et al., 2014).

The final dataset comprises all firms that raised their first early-stage funding round between 2005-2015 inclusive. Table B.3 reports summary statistics for the main sample.

[66] I run this on all the data, regardless of whether previous funding information is available.
[67] Insiders typically cannot cash out until six months after an IPO (Brav and Gompers, 2000).

	United States		United Kingdom	
	Count	Share	Count	Share
Sample overview				
Firms	11,030	–	1,066	–
Successes	400	3.6%	26	2.4%
Acquisitions	2,034	18.4%	97	9.1%
Funding rounds — total	33,158	–	2,218	–
Early-stage	21,886	66.0%	1,534	69.2%
Late-stage	11,272	34.0%	684	30.8%
Sectors				
Biotechnology	995	9.0%	97	9.1%
Communications and Media	289	2.6%	43	4.0%
Computer Hardware	298	2.7%	31	2.9%
Computer Software and Services	3,607	32.7%	254	23.8%
Consumer Related	328	3.0%	53	5.0%
Industrial/Energy	537	4.9%	99	9.3%
Internet Specific	3,082	28.0%	246	23.1%
Medical/Health	1,063	9.6%	109	10.2%
Other Products	463	4.2%	85	8.0%
Semiconductors/Other Elect.	368	3.3%	49	4.6%
Total	11,030	100.0%	1,066	100.0%

Table B.3: Sample overview

The table reports the number of firms, funding rounds, and outcomes in the estimation sample, and their sectoral composition for the United States and United Kingdom. Funding rounds and outcomes are censored at seven years following a firm's first funding round. Shares are within-country totals of firms (or rounds where applicable).

B.2 Definitions

Thomson Reuters define funding rounds in the following way.

Seed stage This stage is a relatively small amount of capital provided to an inventor or entrepreneur to prove a concept. This involves product development and market research as well as building a management team and developing a business plan, if the initial steps are successful. This is a pre-marketing stage.

Early stage This stage provides financing to companies completing development where products are mostly in testing or pilot production. In some cases, product may have just been made commercially available. Companies may be in the process of organising or they may already be in business for three years or less. Usually such firms will have made market studies, assembled the key management, developed a business plan, and are ready or have already started conducting business.

Expansion stage This stage involves working capital for the initial expansion of a company that is producing and shipping and has growing accounts receivables and inventories. It may or may not be showing a profit. Some of the uses of capital may include further plant expansion, marketing, working capital, or development of an improved product. More institutional investors are more likely to be included along with initial investors from previous rounds. The venture capitalist's role in this stage evolves from a supportive role to a more strategic role.

Later stage Capital in this stage is provided for companies that have reached a fairly stable growth rate; that is, not growing as fast as the rates attained in the expansion stages. Again, these companies may or may not be profitable, but are more likely to be than in previous stages of development. Other financial characteristics of these companies include positive cash flow. This also includes companies considering IPO.

B.3 Additional tables and figures

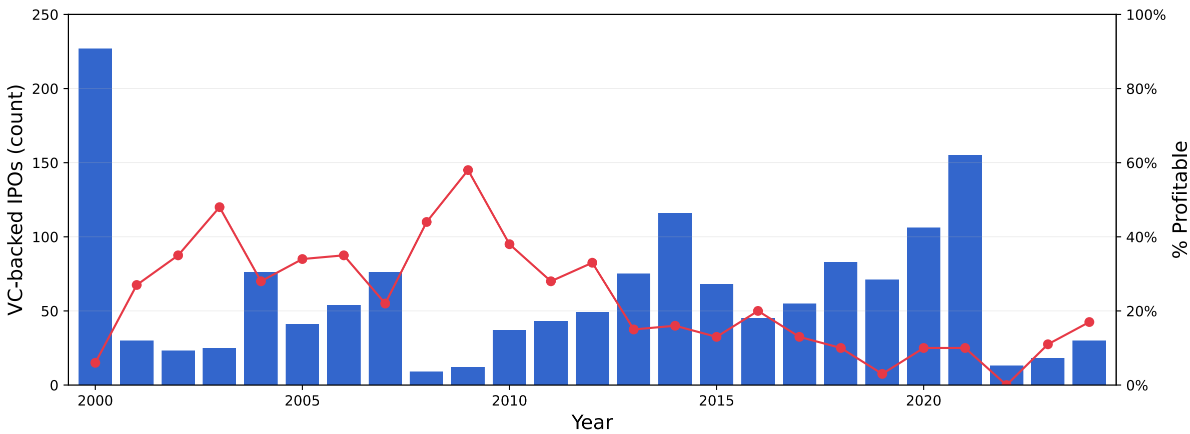

Figure B.1: US-headquartered, VC-backed IPOs

The figure reports the number of VC-backed IPOs of US-headquartered companies from 2000-2024 (bars, left axis), and the share of these companies that were profitable (line, right axis). The data is from Jay Ritter's website (`https://site.warrington.ufl.edu/ritter/`), Table 4d. See source for details.

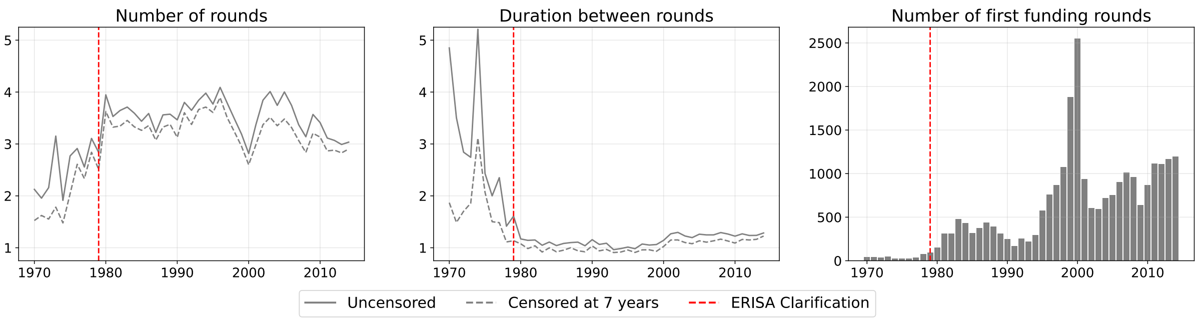

Figure B.2: US historical funding patterns

The figure extends the US sample back to 1970 and, for cohorts of firms defined by their year of first funding, reports the average number of funding rounds, the average duration between funding rounds, and the total number of firms. The dashed line shows the results when data is censored at seven years following each firm's first funding round, as in the main estimation exercise. The solid line applies no censoring. The vertical red line refers to the clarification of ERISA in 1979.

C Quantitative appendix

C.1 Additional details on US estimation

This section provides additional details on the estimation procedure.

Moment computation The estimation procedure involves two steps. First, all parameters except for $\tilde{\sigma}$ are estimated by minimising equation (21). Then, $\tilde{\sigma}$ is recovered from the model condition for the number of first funding rounds. The moments included in equation (21) are reported in Table 2. Before computing moments, I drop observations occurring more than seven years after a firm's first funding round, ensuring uniform censoring.

The complication lies in estimating $\lambda + \nu_e$ and $\lambda + \nu_l + \hat{\phi}_s$ in way consistent with the model's structure. In the model, the within-stage distribution for the time between funding rounds is hypoexponential: for two successive early-stage rounds, $T_{br}^{e \to e} \sim \text{Hypo}(\kappa + \omega_e, \lambda + \nu_e)$, and for two successive late-stage rounds, $T_{br}^{l \to l} \sim \text{Hypo}(\kappa + \omega_l + \hat{\phi}_d, \lambda + \nu_l + \hat{\phi}_s)$. There are two problems. First, the parameters of the hypoexponential distribution are interchangeable within stage, so the two estimated hazards are identified only up to ordering. Second, the data used to estimate the model is censored, meaning these distributions no longer apply.

First, consider the censoring issue. To state the issue more formally, denote by $t^s_{i,n}$ the time of the nth funding round for firm i in state $s \in \{e, l\}$ and the duration between round n and round $n+1$ for firm i in stage s by $\tau^s_{i,n} = t^s_{i,n+1} - t^s_{i,n}$. Firm i completes $N^e_i \in \{1, 2, 3, ...\}$ early-stage rounds and $N^l_i \in \{0, 1, 2, ...\}$ late-stage rounds, but N^e_i and N^l_i are unobserved because the data is censored.[68] Without censoring, the duration between funding rounds is drawn from the unconditional distribution with CDF $\Pr(\tau^s_{i,n} \leq \boldsymbol{\tau})$ for $s \in \{e, l\}$. However, when data is censored, observations are instead drawn from the distribution

$$\Pr(\tau^s_{i,n} \leq \boldsymbol{\tau} | t^s_{i,n} + \tau^s_{i,n} \leq T)$$

where T is the censoring date, set at seven years after the firm's first funding round in the baseline estimation.

In the model, the assumption of *ex ante* homogeneous firms implies that the relevant unit of observation within each stage is the funding round, j, not the firm-round number tuple, (i, n). Furthermore, the duration to the next round, τ^s_j, is independent of the time of the current round, t^s_j, owing to the model's Markov property. Therefore,

$$\Pr(\tau^s_{i,n} \leq \boldsymbol{\tau} | t^s_{i,n} + \tau^s_{i,n} \leq T) = \Pr(\tau^s_j \leq \boldsymbol{\tau} | \tau^s_j \leq T_j)$$

where $T_j = T - t^s_j$ is the censoring date for a given funding round (e.g. with $T = 7$, the duration to the next round for a round at year $t = 4$ has $T_j = 7 - 4 = 3$).

This means that, under censoring T, the observed duration associated with funding round j in stage $s \in \{e, l\}$ is drawn from a truncated hypoexponential distribution. The truncation date, T_j is funding round specific and durations are independent across funding rounds in the model. Therefore, the likelihood for the data is given by

$$L^s = \prod_j \frac{f(\tau^s_j; h^s_1, h^s_2)}{F(T^s_j; h^s_1, h^s_2)}$$

[68] N^e_i starts at 1, not 0, because the sample is firms that have at least one funding round, which is necessarily an early-stage round.

where $f(\cdot)$ and $F(\cdot)$ are, respectively, the unconditional PDF and CDF of the hypoexponential distribution and (h_1^s, h_2^s) are the parameters of the hypoexponential distribution. The estimates are shown in Table C.1.

	UK			US		
	Point estimate	95% CI	N	Point estimate	95% CI	N
h_1^e	1.60	(0.91, 2.29)	468	1.54	(1.47, 1.61)	10,856
h_2^e	2.13	(0.93, 3.32)	468	3.30	(3.02, 3.57)	10,856
h_1^l	0.80	(0.58, 1.02)	331	0.89	(0.84, 0.93)	6,688
h_2^l	2.99	(1.34, 4.65)	331	5.94	(5.05, 6.83)	6,688

Table C.1: Estimates of hypoexponential distribution

The table reports parameter estimates for MLE estimates of hypoexponential distribution parameters, estimated on the duration between funding rounds within stages under round-specific censoring.

The second issue is to assign the estimated parameter values to $\lambda+\nu_e$ and $\lambda+\nu_l+\hat{\phi}_s$. I set the estimates for $\lambda + \nu_e = \max\{h_1^e, h_2^e\}$ and $\lambda + \nu_l + \hat{\phi}_s = \max\{h_1^l, h_2^l\}$. Choosing the larger value is conservative: it implies that firms spend relatively less time in search, biasing against finding large frictions. For instance, in the late-stage, setting the estimate for $\lambda + \nu_l + \hat{\phi}_s = 5.94$ implies a search duration of approximately two months, rather than 13 months for $\lambda + \nu_l + \hat{\phi}_s = 0.89$. Furthermore, the value for $\kappa + \omega_l + \hat{\phi}_d$ should be close to the other parameter value and, in practice, VCs seek to provide funding for one to two years, not several months.

Finally, computing the parameters by MLE is computationally costly. Therefore, for the simulated data, I simply take the values for $\lambda + \nu_e$ and $\lambda + \nu_l + \hat{\phi}_s$ from the model, rather than estimating them. With $S = 20$ (see below), this is innocuous.

Implementation I simulate a sample of 220,600 firms, $S = 20$ times the size of the empirical 2005-2015 sample. Model parameters excluding $\tilde{\sigma}$, Θ, are jointly estimated by minimising the criterion

$$C(\Theta) = g(\Theta)' g(\Theta) \tag{C.1}$$

where the j-th element of $g(\Theta)$ is given by

$$g_j = \frac{\tilde{m}_j - m_j(\Theta)}{\frac{1}{2}|\tilde{m}_j| + \frac{1}{2}|m_j(\Theta)|}$$

and where $\tilde{m}_j$ is the empirical moment and $m_j(\Theta)$ is the model moment, computed under Θ. Standard errors are computed as

$$\widehat{\mathrm{Var}}(\hat{\Theta}) = (G^\top W G)^{-1} G^\top W \, \Omega \, W G (G^\top W G)^{-1}$$

where $W = I$ is the identity matrix, Ω is the variance-covariance matrix of $g(\Theta)$, G is the Jacobian of moment conditions, $g(\Theta)$. I compute Ω via bootstrap. I draw $N = 11,030$ firms with replacement from my sample, compute the moments, and compute $g(\Theta)$ at my estimated Θ for that draw. I repeat this $B = 1000$ times. I then compute Ω as

$$\hat{\Omega} = \frac{1}{B-1} \big(g_b - \bar{g}_b\big)' \big(g_b - \bar{g}_b\big)$$

where g_b is an $B \times J$ matrix of moment conditions at each bootstrap, and $\bar{g}_b$ is a $1 \times J$ vector that averages across bootstrapped samples.

The entry cost parameter, $\tilde{\sigma}$, is estimated from the model condition for the number of first funding rounds. To compute its standard error, I sample parameter Θ from $N(\hat{\Theta}, Var(\hat{\Theta}))$, $B = 1000$ times, and compute $\tilde{\sigma}$ at each iteration. The standard error is then the standard deviation of the bootstrapped sample.

Parameter	Mean # early rounds	Mean # late rounds | to late	Mean duration b/w rounds	Estimate of $\lambda + \nu_e$	Estimate of $\lambda + \nu_l + \hat{\phi}_s$	Mean burn rate | early round	Mean burn rate | late round
Early-stage dev. level, p_e	-1.65	-0.02	0.94	0.00	0.00	1.18	0.02
Late-stage dev. level, p_l	-1.21	-0.75	1.14	0.00	-0.02	1.05	0.83
Early-stage cost, k_e	0.78	-0.01	-0.51	0.00	0.00	0.30	-0.02
Late-stage cost, k_l	0.39	0.65	-0.62	0.00	0.02	-0.32	0.27
R&D arrival rate, κ	-0.75	-0.06	0.04	0.00	-0.01	0.47	0.05
Success payoff, π	-1.19	-0.65	1.07	0.00	-0.02	1.04	0.78
Failure rate search, λ	-0.09	-0.44	0.13	0.20	0.11	-0.03	0.40
Acq. offer rate, ϕ	-0.47	-0.24	0.41	0.00	0.03	0.40	0.10
Mean acq. 'synergy', ξ	-0.82	-0.29	0.67	0.00	-0.00	0.70	0.32
Early-stage rate, ν_e	0.30	0.02	-0.17	0.80	0.00	0.06	0.02
Late-stage rate, ν_l	-0.06	0.35	-0.15	0.00	0.84	0.18	0.17

continued ...

Parameter	Mean time-to-exit	Acq-to-success ratio	Mean success multiple	Share one round	Share funded after year 5	Share to late-stage	Share acq multiple > 10X
Early-stage dev. level, p_e	0.15	-0.31	-0.55	0.02	0.53	1.99	-2.12
Late-stage dev. level, p_l	0.11	-2.24	-1.14	0.70	0.01	0.69	-0.50
Early-stage cost, k_e	-0.02	0.20	-0.17	-0.50	0.02	-0.38	-0.01
Late-stage cost, k_l	-0.07	0.77	0.18	-0.23	-0.06	-0.18	0.54
R&D arrival rate, κ	-0.23	-1.12	0.01	0.40	-0.69	0.71	0.02
Success payoff, π	0.10	-1.19	-0.09	0.69	0.00	0.68	-0.40
Failure rate search, λ	-0.08	-0.02	-0.21	0.42	-0.52	-0.27	-0.39
Acq. offer rate, ϕ	-0.04	0.52	-0.24	0.31	-0.13	0.19	-0.13
Mean acq. 'synergy', ξ	-0.00	-0.30	-0.48	0.49	-0.03	0.41	0.29
Early-stage rate, ν_e	0.03	0.07	-0.02	-0.46	0.31	0.12	0.17
Late-stage rate, ν_l	0.02	-0.17	-0.06	-0.09	0.10	0.18	-0.46

Table C.2: Elasticity of moments with respect to parameters

The table reports the elasticity of each moment (or auxiliary parameter) with respect to each estimated parameter. Starting from the estimated parameter values, each parameter is individually adjusted by $\pm 1\%$, the model is simulated, and moments are recomputed using a centred difference.

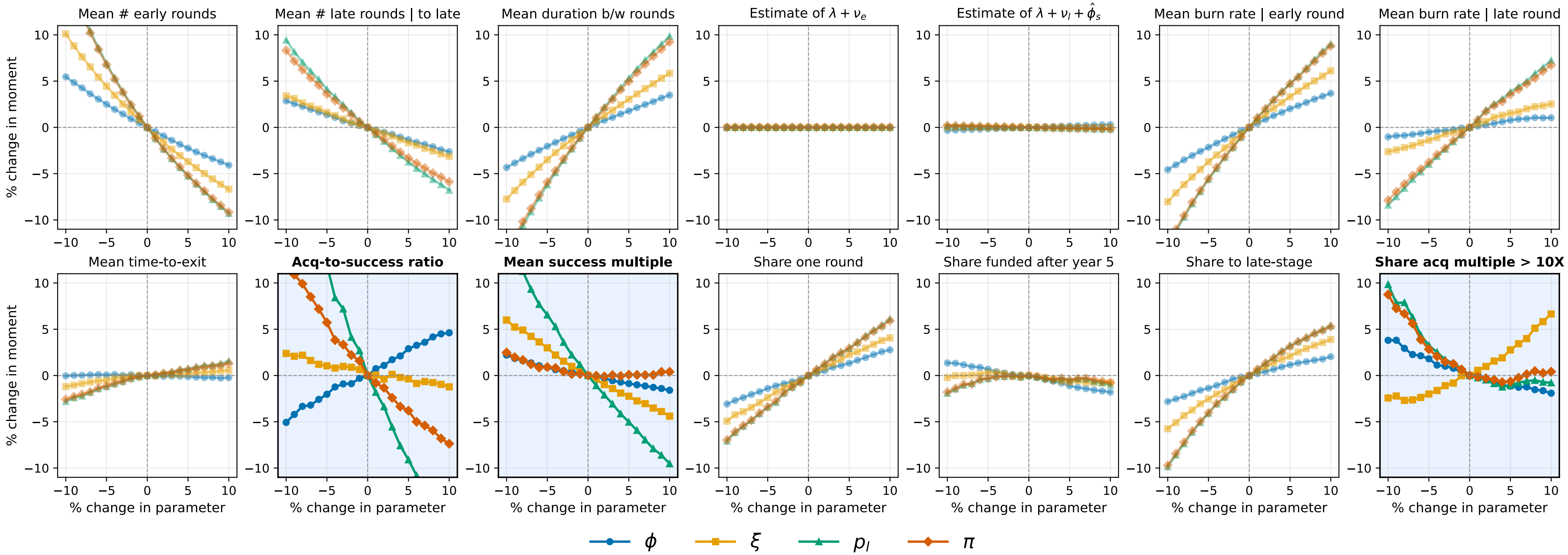

Figure C.1: Comparative statics – moments to parameter changes

Each panel plots the percent change in a moment when varying one parameter by ±10% around the baseline estimate, holding other parameters fixed. Lines correspond to parameters $\{\phi, \xi, p_l, \pi\}$. Baselines are normalized to zero; axes report percent changes. For comparability, y-limits are held at ±10% across panels. Shaded panels (acquisition-to-success ratio, mean success multiple, share of acquisitions with multiple $> 10\times$) are those most informative for separating $\{p_l, \pi, \xi, \phi\}$. Moments are computed from simulated data under the estimated baseline, with 1,000,000 simulated firms.

C.1.1 Robustness and extensions

Parameter	US, 2005-2015 (C7) Estimate	US, 2005-2015 (C7) Std. Error	US, 2005-2010 (C7) Estimate	US, 2005-2010 (C7) Std. Error	US, 2005-2010 (C12) Estimate	US, 2005-2010 (C12) Std. Error
ρ	0.08	–	0.08	–	0.08	–
κ	0.26	0.01	0.25	0.03	0.26	0.02
λ	0.62	0.04	0.71	0.10	0.64	0.15
p_e	0.23	0.01	0.18	0.01	0.18	0.01
k_e	4.16	0.18	3.81	0.35	3.79	0.23
ν_e	2.54	0.13	2.68	0.25	2.75	0.18
p_l	0.30	0.01	0.21	0.01	0.24	0.01
k_l	16.15	0.25	11.44	0.66	11.81	1.06
ν_l	5.20	0.26	5.87	0.63	6.70	0.66
π	218.18	15.40	223.42	32.07	220.95	29.39
ϕ	0.65	0.07	0.46	0.12	0.44	0.10
ξ	0.74	0.08	0.98	0.13	1.01	0.20
$\tilde{\sigma}$	2.68	0.37	1.60	1.00	2.80	1.11

Table C.3: Parameter estimates for the US under alternative sample periods and censoring

The table reports parameter estimates from the model in section 2.4. "US, 2005-2015 (C7)" replicates the US parameter estimates under the baseline sample period and seven-year censoring (see Table 1). "US, 2005-2010 (C7)" and "US, 2005-2010 (C12)" report estimates for the 2005-2010 subsample (firms first funded between 2005-2010), estimated under seven-year and twelve-year censoring, respectively. To maintain consistency, all estimations use a simulated sample of 220,600 firms ($S = 20$ times the size of the 2005-2015 empirical sample). Parameters (except $\tilde{\sigma}$) are estimated by minimising (21) (equal weighting of centred percentage deviations; see footnote 43) and I report asymptotic standard errors, where the covariance matrix of moment conditions is computed from 1,000 bootstrap samples, clustered at the firm level. With $S \geq 20$, simulation noise is minuscule, so I report standard errors that simply reflect sampling variability. To avoid local minima in the estimation, I initialise a simulated annealing algorithm with 200 (randomly selected) starting points for each estimation. With the resulting candidate parameter estimates, I run a local minimisation (Nelder-Mead) using the results from the first step as initial guesses. With these estimates, the entry cost parameter, $\tilde{\sigma}$, is recovered from the condition determining the flow rate of first-time funding rounds. To compute its standard error, I take 1,000 samples from the asymptotic distribution of parameter estimates, solve the model, and recover the implied value for $\tilde{\sigma}$. The standard error is the standard deviation of $\tilde{\sigma}$ among this sample. All amounts are in 2015 USD.

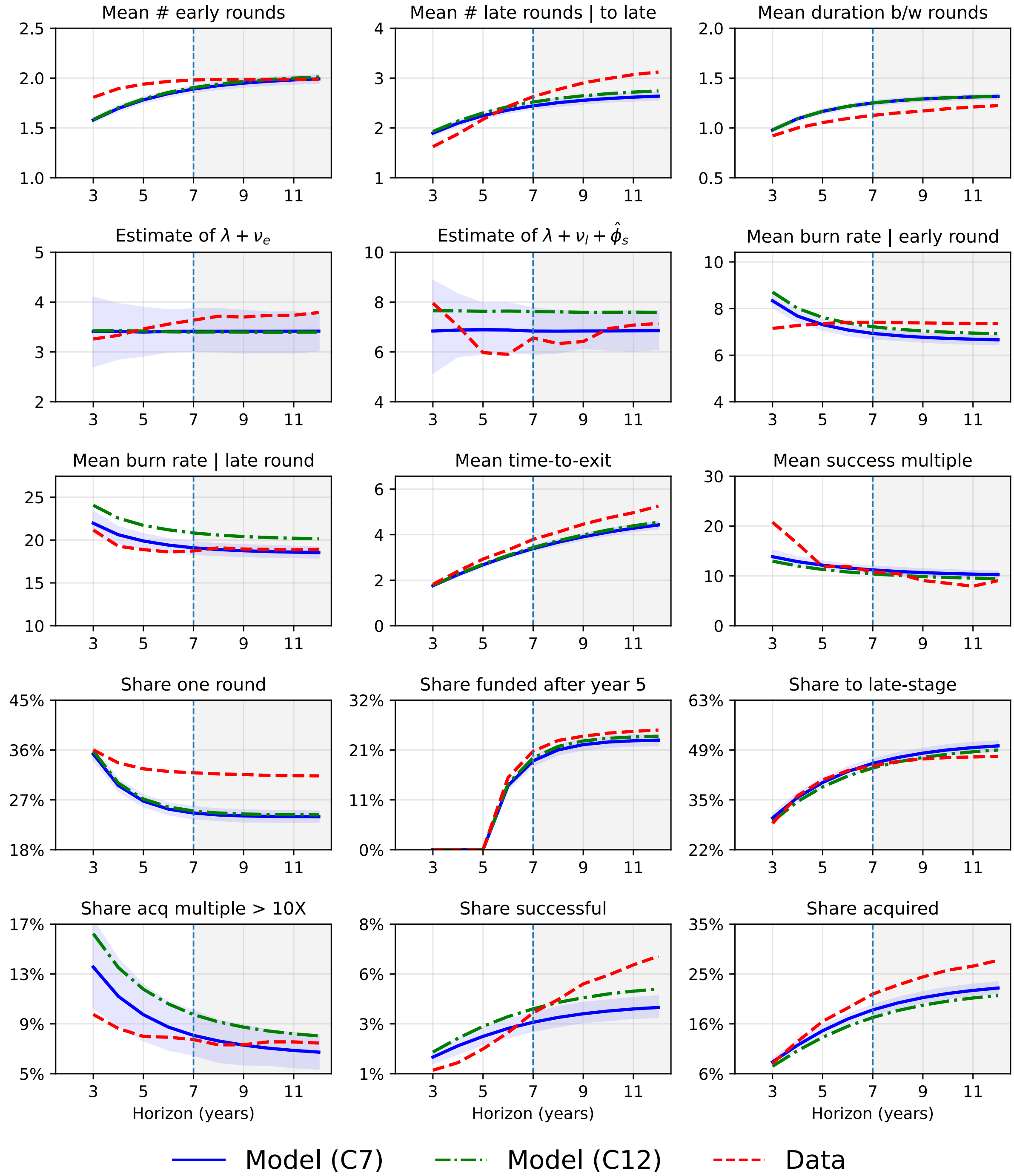

Figure C.2: Out-of-horizon censoring diagnostics (2005-2010 sample)

The figure plots moments computed on data censored at horizon h (x–axis) using the 2005–2010 sample (red, “Data”). Targeted moments and the share of acquisitions and successes are shown; the targeted acquisition-to-success ratio – the ratio of these two measure – is redundant and so excluded. Blue (“Model (C7)”) and green (“Model (C12)”) show model–implied moments when parameters are fixed at the estimates from the 7-year and 12-year censoring specifications, respectively; see Table C.3. For the empirical series, the data *are* re–censored at each h and moments recomputed. For the model series, I simulate 200 samples of the same size as the 2005–2010 data ($N = 5137$) and compute moments after censoring at each h. The blue band is a pointwise 95% simulation interval (2.5–97.5 percent quantiles) and reflects finite–sample variability under the model; parameter uncertainty is not included. The vertical line marks $h = 7$; the shaded region indicates out of sample outcomes with respect to the C7 specification.

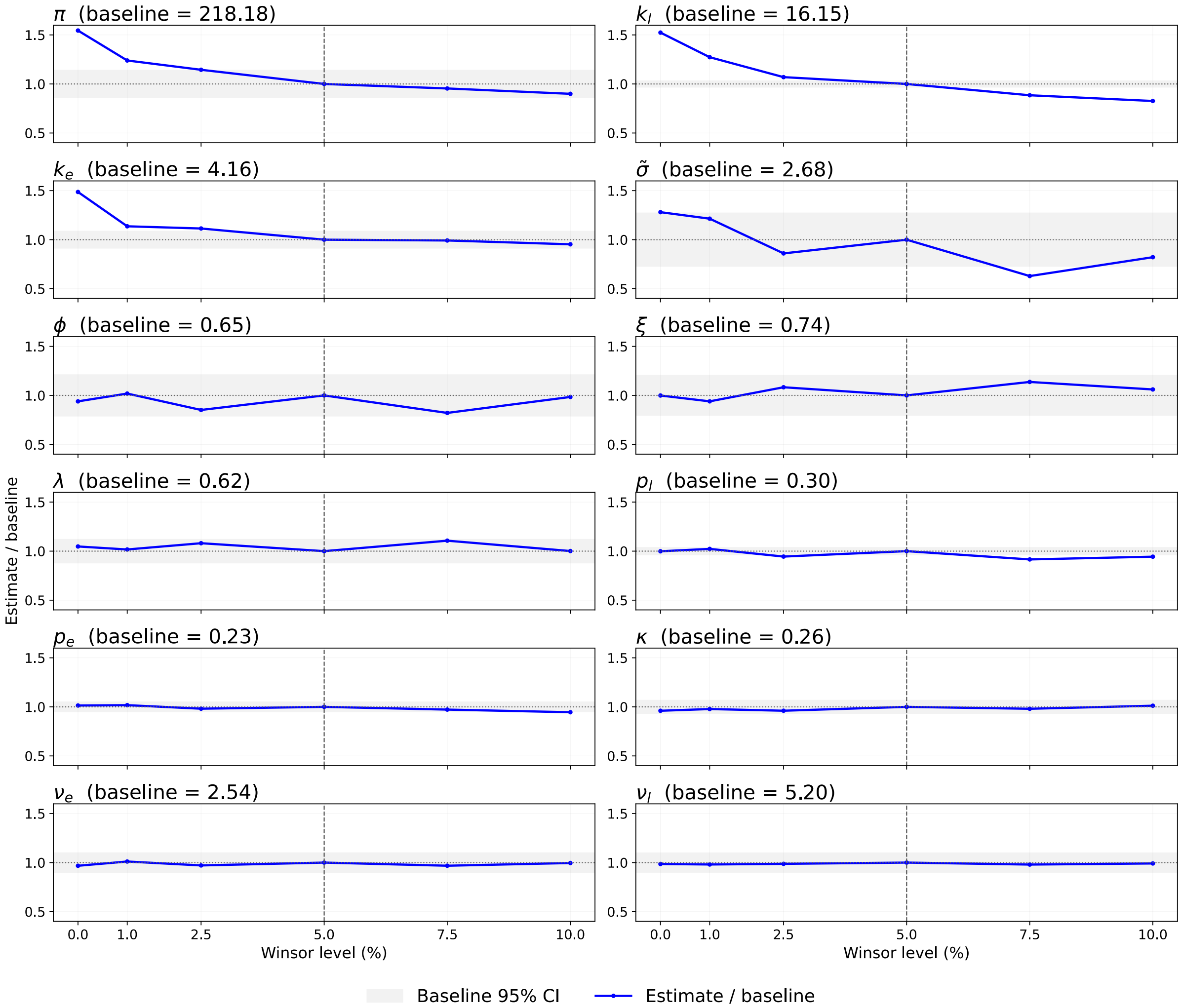

Figure C.3: Parameter estimates under different burn rate winsor thresholds

The figure plots parameter estimates normalised to 1.00 at $w = 5\%$ (baseline), under winsorising thresholds $w = 0\%, 1\%, 2.5\%, 5\%, 7.5\%, 10\%$. Gray bands show 95% confidence intervals for baseline parameter estimates; standard errors are reported in Table 1. Parameters are ordered by maximum percentage deviation from baseline. In each estimation, the mean burn rate is computed after winsorising the burn rate distribution (within stage); the same procedure is applied when computing burn rates from simulated data used in the estimation.

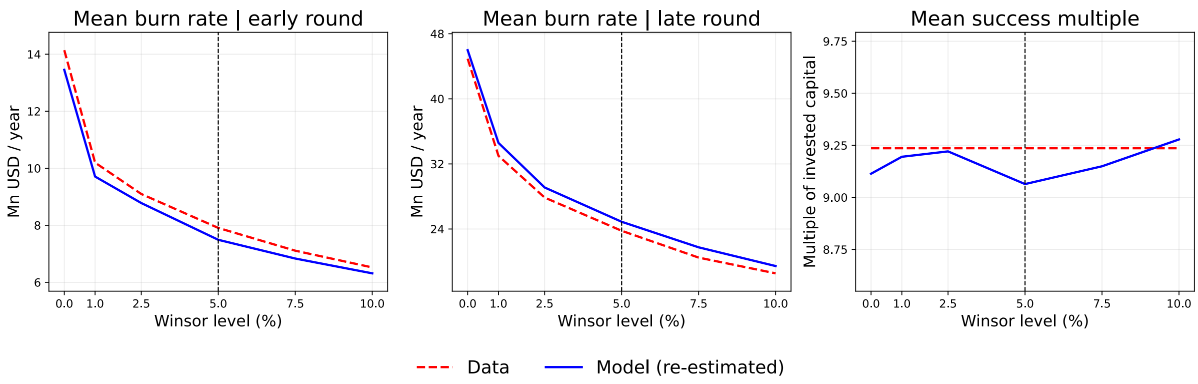

Figure C.4: Model fit under different burn rate winsor thresholds

The figure reports empirical and model moments under different burn rate winsor thresholds. The model fits well across thresholds; the focus on mean burn rates and the success multiple reflects that movements in k_e, k_l, and π (particularly relevant to these moments) are the largest among all parameters when the winsor threshold is varied.

C.2 Additional details on UK estimation

	US		UK		
Parameter	Estimate	Std. Error	Estimate	Std. Error	Δ(US-UK)
ρ	0.08	–	0.08	–	0.00
κ	0.26	0.01	0.25	0.07	0.01
λ	0.62	0.04	0.81	0.17	-0.19
p_e	0.23	0.01	0.26	0.07	-0.03
k_e	4.16	0.18	3.46	1.02	0.70
ν_e	2.54	0.13	1.09	0.19	1.45***
p_l	0.30	0.01	0.30	0.04	0.00
k_l	16.15	0.25	12.04	2.53	4.11
ν_l	5.20	0.26	2.28	0.41	2.92***
π	218.18	15.40	205.26	66.56	12.92
ϕ	0.65	0.07	0.33	0.07	0.32***
ξ	0.74	0.08	0.81	0.72	-0.07
$\tilde{\sigma}$	2.68	0.37	0.98	2.32	1.70

Table C.4: US-UK Parameter Estimates

The table reports parameter estimates from the model in section 2.4 for the US and UK. Estimates for the US are reproduced; see Table 1. The estimation procedure for the UK mirrors the US: I again use a simulated sample of 220,600 firms, although this now corresponds to $S \approx 207$ times the empirical sample size. To avoid local minima in the estimation, I initialise a simulated annealing algorithm with 200 (randomly selected) starting points for each estimation. With the resulting candidate parameter estimates, I run a local minimisation (Nelder-Mead) using the results from the first step as initial guesses. Significance levels for differences in parameter estimates across countries (except $\tilde{\sigma}$) are computed from the asymptotic parameter estimate distributions, assuming independence. For $\tilde{\sigma}$, I take 1,000 samples from the asymptotic distribution of parameter estimates for each country, solve the model, recover the implied value for $\tilde{\sigma}$, and compute the difference $\Delta_{\tilde{\sigma}} = \tilde{\sigma}_{US} - \tilde{\sigma}_{UK}$. The significance level is determined by the share of cases with $\Delta_{\tilde{\sigma}} < 0$. Significance levels for Δ(US-UK) are: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	Share successful (%)	Share acquired (%)	Entry value ($Mn)
UK: baseline	4.4	15.8	0.8
US: baseline	4.9	24.5	3.3
US: $\nu_e^{UK},\ \nu_l^{UK}$	3.0	19.7	1.6
US: ν_e^{UK}	4.5	22.3	2.2
US: ν_l^{UK}	3.4	22.2	2.5

Table C.5: US-UK decomposition

The table reports uncensored outcomes and the entry value across different counterfactual scenarios. Rows one and two show the UK and US baselines. Rows three to five compare different counterfactual scenarios, where the US financing market meeting rates are replaced with their respective UK estimates.

Table C.6 reports targeted and untargeted moments for the UK estimation; the UK parallel to Table 2. Figure C.5 shows the fit of the model to the data across a wider range of outcomes; the UK parallel to Figure 5.

Panel A: Targeted moments		
Moment	Data	Model
Mean # early rounds	1.44	1.44
Mean # late rounds | progressed to late	1.94	1.69
Share one round (%)	52.72	43.15
Mean duration b/w rounds (years)	1.46	1.72
Estimate of $\lambda + \nu_e$	2.13	1.90
Estimate of $\lambda + \nu_l + \hat{\phi}_s$	2.99	3.31
Share receiving funding after year 5 (%)	11.73	11.52
Mean burn rate | early round ($Mn/year)	5.80	5.39
Mean burn rate | late round ($Mn/year)	18.79	19.43
Share to late-stage (%)	33.11	34.03
Mean time-to-exit (years)	3.82	3.32
Acquisition-to-success ratio	3.73	3.68
Mean exit multiple | success	8.88	9.01
Share acquisition multiples > 10X (%)	11.34	11.54
# first funding rounds	0.48	0.48
Panel B: Untargeted moments		
Share successful (%)	2.44	3.46
Share acquired (%)	9.10	12.73
Mean time-to-success (years)	3.96	3.48
Mean time-to-acquisition (years)	3.78	3.27
Mean exit multiple	5.37	6.10

Table C.6: Targeted and untargeted moments

Panel A shows 15 moments (including two auxiliary parameter estimates) used in the estimation procedure. Empirical moments are computed from the sample of 1,066 firms. Other than for the estimates of $\lambda+\nu_e$ and $\lambda+\nu_l+\hat{\phi}_s$, simulated moments are computed using the parameter values given in Table C.4 to simulate a sample of 220,600 firms; preserving the same simulated sample size as for the US estimation, for consistency. Note that For $\lambda+\nu_e$ and $\lambda+\nu_l+\hat{\phi}_s$, the parameter values are taken directly to reduce the computational burden of the estimation procedure. In computing the empirical and simulated moments, data on a given firm is censored seven years after its first funding round. Panel B shows a set of untargeted moments.

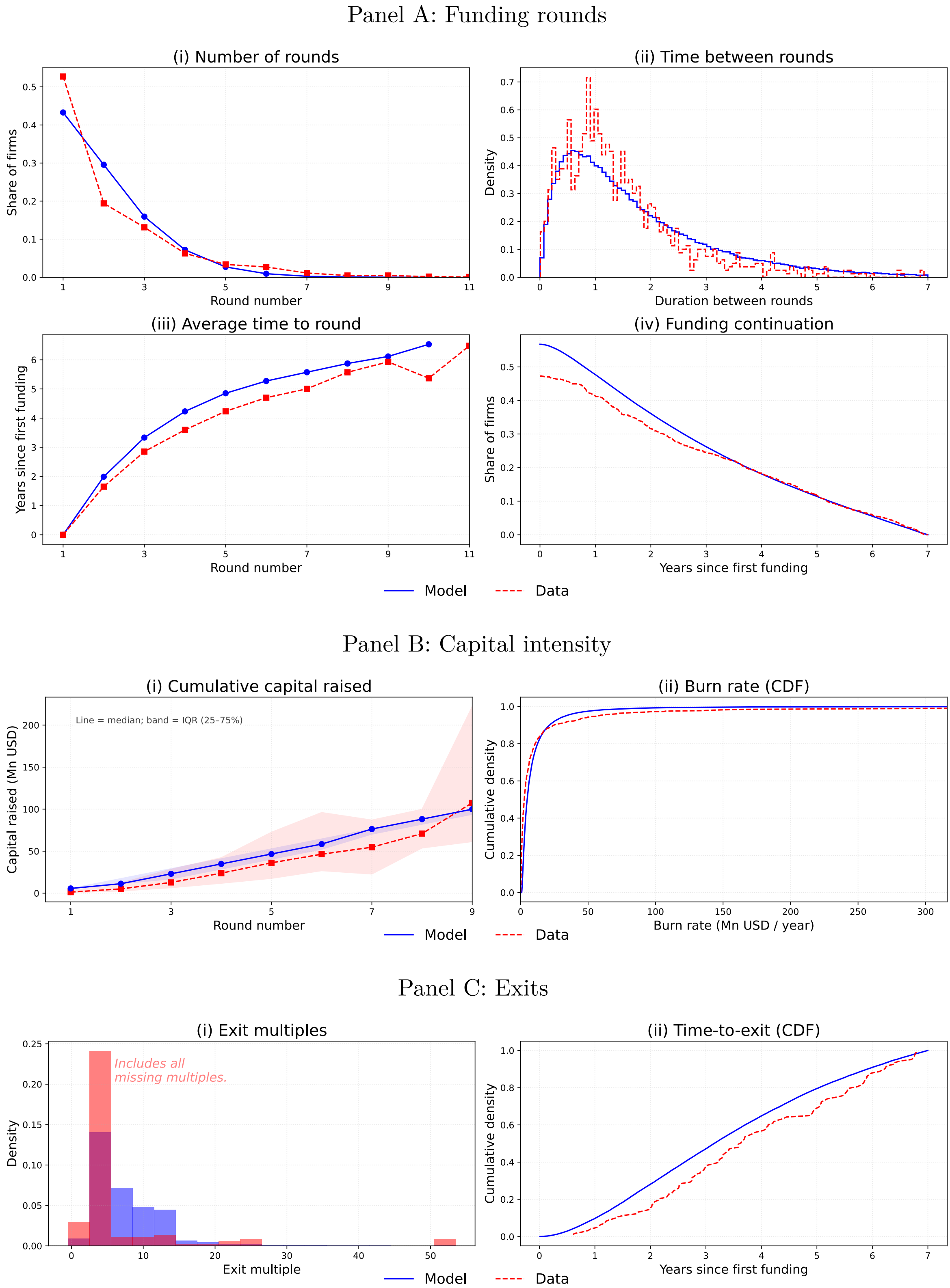

Figure C.5: UK Model validation

The figure shows outcomes in the model and data across funding rounds (panel A), capital intensity (panel B) and exits (panel C). All empirical and simulated data is censored at seven years following a firm's first funding round. "Funding continuation" in Panel A (iv) reports the share of firms that raise capital more than x years after their first funding round. To aid visualisation, cumulative capital raised is shown up to the 9th round; the burn rate CDF is reported up to the 99th percentile; and exit multiples are winsorised at 50X. Missing empirical exit multiples are set to 1.5X, following Kerr et al. (2014).

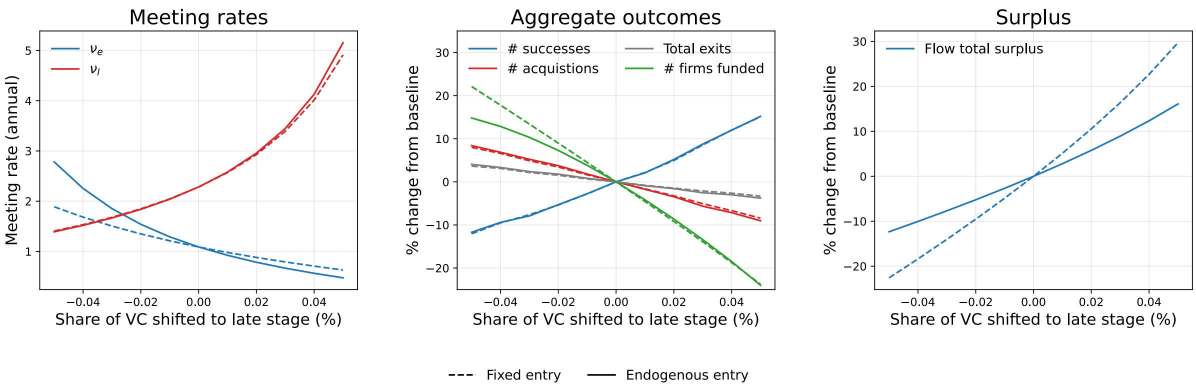

Figure C.6: UK policy counterfactual

The figure reports the results of a budget-neutral policy intervention that shifts capital from the early to late-stage. The intervention is implemented by adjusting the meeting rates, (ν_e, ν_l), while holding total (flow) investment, K^{total} fixed, $K^{\text{total}} = K^{\text{e}} + K^{\text{l}}$ and $K^{\text{i}} = k_i(\mu^p_{d,i} + \mu^u_{d,i})$ for $i \in \{e, l\}$. The policy intervention is conducted under two assumptions on start-up creation: (i) holding start-up creation constant at its initial value (dashed); (ii) allowing start-up creation to adjust endogenously, according to $V_{s,e}/\tilde{\sigma}$ (solid). The left-panel shows the mapping between the meeting rates (y-axis) and the share of capital reallocated from the early to late-stage. The share reallocated is defined as $(K^l_{\text{new}} - K^l_{\text{init}})/K^{\text{total}}$. The centre-panel shows the change in the number of funded firms (green), the number of successful firms (number funded $\times$ share successful; blue), the number of acquired firms (number funded $\times$ share acquired; red) and the number of total exits (i.e. success + acquisition; gray) as capital is reallocated between stages. The right-panel shows the change in total (flow) surplus. Flow surplus is defined analogously to equation (15), but augmented to account for acquisitions and two stages. Specifically, it is given by $\kappa p_l \pi \times \mu^p_{d,l} + \hat{\phi}_d p_l \, \pi E[\epsilon | \epsilon > V_{d,l}/p_l \pi] \times \mu^p_{d,l} + \hat{\phi}_s p_l \, \pi E[\epsilon | \epsilon > V_{s,l}/p_l \pi] \times \mu_{s,l} - K^{\text{total}} - (\tilde{\sigma}/2) \times \Lambda^2$, where Λ is the flow rate of start-up creation. In case (i), this is held constant at the initial value of $V_{s,e}/\tilde{\sigma}$. In case (ii), it equals $V_{s,e}/\tilde{\sigma}$ as $V_{s,e}$ adjusts endogenously to changes in (ν_e, ν_l).